Pricing Mistakes and the Non-Neutrality of Money

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Abstract

Standard sticky-price models assume firms reset prices without error. Using UK and Argentine price microdata, I document that consecutive regular price changes are negatively correlated, more strongly when prices adjust more frequently, and with negligible correlation beyond one spell. A large class of standard models cannot jointly match these facts, but models with idiosyncratic pricing mistakes can. I ask how mistakes affect monetary policy's real effects and their measurement using price microdata. First, even though firms perfectly observe monetary shocks, adding mistakes to a standard model can increase cumulative output effects by changing which firms adjust prices. Second, once mistakes are possible, these real effects are completely unidentified from standard micro moments (frequency and kurtosis of price changes) because mistakes distort them. I restore identification with a sufficient statistics result that incorporates regression coefficients measuring the documented serial correlation of price changes. Accounting for mistakes increases inferred real effects from the UK microdata by around 50%.

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1 Introduction

Sluggish price adjustment plays a central role in macroeconomics: it is an important reason why business cycles are inefficient, and a key channel by which monetary policy affects real quantities and can address those inefficiencies. One hypothesis is that prices take time to adjust because price-setters underreact to aggregate shocks due to imperfect information (Lucas, 1972). These models feature choices that can be viewed, ex post, as mistakes.

In contrast, the current dominant paradigm emphasizes nominal rigidities. In these models there are no mistakes, but price adjustment may be subject to adjustment costs, prespecified contracts or other frictions (Sheshinski and Weiss, 1977; Taylor, 1979; Calvo, 1983). A popular way to quantify their importance is to build a structural model, discipline it using price microdata, and study the aggregate response to a monetary shock. This response depends on *how often* prices change and *which* prices do so after a shock: the composition of adjusting firms (Golosov and Lucas, 2007; Caballero and Engel, 2007; Midrigan, 2011). A central finding is that key price microdata moments directly identify both margins and thus the aggregate response to monetary shocks (Alvarez et al., 2016a).

In this paper, I first show that a large class of standard models cannot match three new facts that I document using price microdata from the UK and Argentina. I trace this failure to the seemingly innocuous assumption that firms *perfectly* reset prices to their value-maximizing level, i.e., firms do not make mistakes. I relax this assumption to allow for mean-zero, idiosyncratic mistakes or *imperfections* (Costain and Nakov, 2015, 2019). Models with imperfections can successfully match the new facts. Crucially, and in contrast to Lucas (1972) and the subsequent literature, firms in my model have perfect information about aggregate variables. Yet I show that this alternative notion of mistakes also has important consequences for aggregate price adjustment, due to its interaction with nominal rigidities.

I do so by focusing on two separate exercises. First, a comparative-static exercise: I introduce reset imperfections into a simple model with nominal rigidities, holding other structural parameters fixed. I show that imperfections can increase the cumulative output response to a monetary shock even as they increase how often prices change on average. This occurs because imperfections can change which firms adjust after the shock, slowing down the aggregate price response. Second, a measurement exercise: how does accounting for imperfections change the aggregate adjustment we infer from price microdata? Extending the sufficient statistics result in Alvarez et al. (2016a), I prove that the micro moments I introduce provide additional statistics needed to discipline monetary non-neutrality. Furthermore,

ignoring them would render real effects completely unidentified from standard micro moments alone. Quantitatively, accounting for imperfections raises the inferred cumulative output effect of a monetary shock by around 50% in the UK data. Through the lens of a structural model, the increase reflects a change in the inferred composition of adjusting firms.

Identification of Reset Imperfections with Price Microdata. I argue that predictability patterns in consecutive price changes of a given product are useful for discriminating among different sticky-price models. I focus on three moments: first, the regression coefficient when predicting the price change for a given product with the previous price change for that same product and appropriate fixed effects; second, how that coefficient comoves with average price adjustment frequency; and third, the coefficient on the price change from two spells ago in a similar regression.

I systematically analyze what different models predict for these moments. I first show that a large class of models implies no predictability. This class includes standard formulations of Calvo (1983) and Taylor (1979), multiproduct models (Alvarez and Lippi, 2014), random menu cost models (Alvarez et al., 2021) and models with imperfect or infrequent information (Reis, 2006; Woodford, 2009; Alvarez et al., 2011). This is because these models share a key property: with price adjustments, all idiosyncratic state variables are reset to the same value for all firms. Price resets erase any existing idiosyncratic differences, so variables determined before reset have zero predictive power for variables after it. Since that property hinges on the assumption that idiosyncratic shocks follow random walks, I also study models where idiosyncratic shocks mean-revert. In that case, transitory shocks to demand or costs generate a negative coefficient, so that price changes feature *reversals*. Those reversals get weaker with frequency: as prices change more often, there is less time for the shocks to revert. Since mean reversion takes time to occur, reversals last for more than one spell.

I also consider models with imperfect reset (Costain and Nakov, 2015, 2019; Stevens and Morales-Jiménez, 2025), which have different implications. In these models, firms incur costs of choosing prices precisely, so at the time of reset new prices feature some idiosyncratic and serially uncorrelated *imperfections* or "mistakes". These models also predict reversals, because firms partially correct past reset imperfections. These reversals get stronger with frequency because imperfections occur at each price change and thus become more important relative to other sources of variation when adjustment is more frequent. Since these imperfections are uncorrelated with each other, reversals only last for one spell. These three predictions jointly separate imperfect reset from the alternatives discussed above.

I take these predictions to the data using the price microdata underlying the construction of the UK Consumer Price Index, as well as a novel dataset from Argentina, which provides evidence from the large inflationary spike and subsequent disinflation at the start of the Milei presidency. After controlling for key confounders (such as sales and product substitutions) and using appropriate fixed effects, I find that (i) there is a consistent pattern of reversals; (ii) reversals get stronger with higher frequency of price changes, both when variation reflects differences in adjustment frequency across product categories (in the UK) and when it reflects changes in aggregate adjustment frequency due to inflation (in Argentina); and (iii) there is little predictability after one spell. The evidence thus lines up with imperfect reset and rejects the discussed alternatives. I also review other plausible stories (measurement error in prices, active experimentation and unobserved heterogeneity, among others) and argue they are unlikely to explain the evidence. Given this empirical success, I study the implications of these imperfections with two exercises.

Introducing Reset Imperfections: A Comparative Static. I consider an economy populated by firms that adjust prices subject to nominal rigidities. At the time of price adjustment, firms are subject to a control cost, which is a cost of choosing the reset price precisely—a stand-in for the attention and internal communication required to determine and implement the exact right price. Firms optimally choose a reset price that is *random*, with its precision influenced by the magnitude of the control cost. At the moment of adjustment, firms set the right price *on average* but not *in each realization*.

In this setup, I perform the following comparative static: take an economy with perfect precision and increase the cost of precision while leaving all other structural parameters unchanged. I am interested in the effects on the average adjustment frequency, as well as monetary non-neutrality. Following Alvarez et al. (2016a), I measure monetary non-neutrality by the cumulative impulse response of output (CIR) to a small permanent monetary shock.

I first assume that the timing of price changes is exogenous, as in Calvo (1983). In that case, imperfections are irrelevant for my objects of interest: adding them alters neither the average adjustment frequency nor the aggregate response to monetary shocks. The reason is that both how often firms change prices and which firms do so are fixed by assumption. Since adjusting firms fully incorporate the shock, imperfections wash out among them.

Given that the exogenous timing assumption is key, I relax it by considering a model where firms can choose when to change prices, but doing so entails a fixed cost (as in standard menu cost models) as well as a control cost. In this case, the price adjustment decision is

characterized by an inaction region: firms reset when their price gaps reach its boundaries but choose the new price imprecisely. This captures the idea that deciding *when* to change prices and *which* price to set are separate decisions: firms can precisely determine when adjustment is warranted while remaining imprecise about the price they choose.

Once adjustment timing is endogenous, I show that moving from perfect to imperfect reset by making precision costly increases both the average frequency of price changes and the cumulative output effect of monetary shocks. Intuitively, adding imprecision generates an additional source of variation in prices. Firms optimally adjust to it by increasing the average frequency of price changes and widening their inaction bands. Due to these forces, imperfections also change the composition of firms that adjust. After a monetary shock, many price changes are driven by imperfections and subsequent corrections. Since the aggregate shock is fully incorporated on the first price change, price changes after the initial one do not contribute to aggregate flexibility. In contrast, wider bands imply that some firms take longer to adjust, which slows down aggregate adjustment. I show that although the initial response to the monetary shock is faster, the subsequent response is more persistent, and the cumulative output effect is larger. This result overturns the conventional wisdom that a higher frequency of price changes implies greater aggregate price flexibility.

Accounting for Reset Imperfections in Measurement. I then turn to studying a different but related question: how do imperfections change the way we measure monetary non-neutrality from the microdata? Given the documented quantitative importance of the composition of adjusting firms, I consider the general class of models in Alvarez et al. (2016a). This class can generate wide variation in the degree of *state dependence* in price-setting, that is, how strongly the probability of changing prices depends on how far a price is from its frictionless level. High state dependence implies that after an expansionary shock, there are significantly more price increases and fewer price cuts. By changing *which* prices adjust, high state dependence speeds up aggregate price adjustment and thus lowers non-neutrality. Within this class, Alvarez et al. (2016a) prove that monetary non-neutrality is *point identified* by two statistics: the frequency and kurtosis of price changes. Intuitively, a thick tail of large price changes implies that some firms need to wait a long time before they can change prices for the first time, which signals low state dependence and higher inferred non-neutrality.

I extend this class to allow for reset imperfections: I maintain the assumption that firms set the right prices *on average* by assuming reset gaps are drawn from a mean-zero, symmetric (but otherwise unrestricted) distribution. Once we allow for reset imperfections,

the two statistics mentioned above are no longer sufficient. In fact, the opposite is true: for any given frequency and any kurtosis above 2, one can find examples such that non-neutrality can be made arbitrarily large or small while still matching those two observables. A notable corollary is that Calvo (1983) pricing no longer represents the theoretical upper bound on inferred non-neutrality. In terms of measurement, imperfections matter because they alter the distribution of price changes and thus distort our measurement of the degree of state dependence from the microdata in potentially arbitrary ways.

I show how to bring in discipline via a *generalized sufficient statistics result*, which restores set identification by adding specific predictability-regression coefficients as new sufficient statistics. Intuitively, these new statistics allow us to discipline the size of imperfections and restore the relation between the state dependence in price-setting and moments of the microdata. Using the derived formula, I can measure implied non-neutrality directly from statistics estimated using the UK data and compare the resulting implications with those obtained using the Alvarez et al. (2016a) formula, which assumes perfect reset. Under the baseline specification, the lower bound on inferred non-neutrality is at least 50% higher than the perfect-reset benchmark, although it is imprecisely estimated.

While the generalized sufficient statistics result does not require solving a specific structural model, it is completely silent about what features are needed to match the data and how those change when we account for imperfections. I estimate a structural model to tackle these issues. If we ignore the new moments and just target standard micro moments with a perfect-reset model, matching the data requires an intermediate degree of state dependence. Once we add imperfections and discipline them by matching an extended set of moments which includes predictability coefficients, we obtain essentially no state dependence, as in a Calvo model. Consistent with the sufficient statistics result, this model delivers around 50% higher non-neutrality than the one that ignores imperfections. Note that while adding imperfections to a Calvo model does not change non-neutrality, imperfections still matter because they change our assessment of whether a Calvo model fits the data.

The class of models covered by the sufficient statistics result assumes that idiosyncratic shocks follow random walks. The quantitative conclusion on non-neutrality does not depend on this. I also study what happens if I match the observed size of reversals using mean-reverting idiosyncratic shocks. The same conclusion applies: once I calibrate shocks to match the observed size of reversals, a Calvo model allows me to roughly match the frequency and kurtosis of price changes, and thus measured non-neutrality increases. Thus, reversals matter for non-neutrality even outside the class of models covered by the sufficient statistics result.

Related Literature. This paper contributes to the literature that quantifies monetary non-neutrality using price microdata, with important contributions including Bils and Klenow (2004); Golosov and Lucas (2007); Klenow and Kryvtsov (2008); Nakamura and Steinsson (2010); Midrigan (2011); Vavra (2014); Alvarez et al. (2016a, 2021); Gagliardone et al. (2025a). I introduce a new set of moments and show they are informative about imperfections in reset. I also prove that these moments are new sufficient statistics for monetary non-neutrality and show their quantitative importance.

The paper also relates to the literature on stochastic choice due to control costs (van Damme, 1991; Mattsson and Weibull, 2002; Flynn and Sastry, 2023, 2024), in particular to a small literature applying stochastic choice to price-setting models: the pioneering work of Costain and Nakov (2015, 2019); Costain et al. (2022), and Stevens and Morales-Jiménez (2025). Relative to these papers, I present empirical evidence on a new moment that is directly informative about the prevalence of reset imperfections. I also provide analytical versions of these models and a sufficient statistics result to measure non-neutrality, complementing the numerical results in those papers. Furthermore, I reconcile their conflicting findings on the importance of reset imperfections for inferred monetary non-neutrality.

The paper is also related to a strand of literature that uses imperfect information to model infrequent price adjustment and monetary non-neutrality (Phelps, 1969; Mankiw and Reis, 2002; Woodford, 2003, 2009; Maćkowiak and Wiederholt, 2009). While the conceptual justification for control costs is similar to that for information-processing costs in the rational inattention literature, I show that models with imperfect reset have distinct empirical implications: standard imperfect information models do not generate reversals, while models with imperfect reset do. Furthermore, the increase in non-neutrality comes from a different source: changes in the composition of adjusters. This last point relates the paper to Alvarez et al. (2016b); Baley and Blanco (2019); Argente and Yeh (2022), which also raise real effects by altering the composition of adjusting firms through different mechanisms.

Quantitative sticky-price models sometimes feature transitory idiosyncratic shocks, but the degree of mean reversion is usually treated as an irrelevant parameter. Jenuwine and Tielens (2025) is a notable exception: it introduces vintage-conditional moments and argues that matching those moments requires large and very transitory shocks to desired prices. These findings are consistent with the presence of a large transitory component in prices (see also Klenow and Malin (2010)). However, I argue that highly transitory shocks to costs or productivity are unlikely to fully explain the evidence, since they are inconsistent with reversals getting stronger with adjustment frequency.

The Argentine evidence connects the paper to a literature that uses high-inflation episodes to learn about pricing frictions (Gagnon, 2009; Alvarez et al., 2018; Alvarez and Neumeyer, 2020; Nakamura et al., 2018; Blanco et al., 2024a,b,c; Gagliardone et al., 2025b; Miyahara et al., 2025). The paper is particularly related to Miyahara et al. (2025) in that both consider informational frictions in settings of high inflation. However, I model and identify them differently, and focus on non-neutrality at low inflation.

ROADMAP. Section 2 introduces the key empirical moment and gives a preview of what different models predict. Section 3 presents the empirical evidence. Section 4 studies the effects of adding imperfections to models with nominal rigidities. Section 5 focuses on the effects of imperfections on our measurement of non-neutrality. Section 6 concludes. The Appendix contains the main supporting results and proofs, while the Online Appendix provides additional empirical results, theoretical extensions, and technical details.

2 Price Reversals in Different Models

2.1 Moments of Interest

I argue that the temporal correlation of price changes is a useful moment to distinguish across different models with sticky prices, which in turn have distinct implications for non-neutrality. Assume that we have a panel dataset of the prices charged by several different firms for comparable products. We can think of each unit i as a specific product at a specific store. Each firm changes prices at times $\tau_{i,n}$, $n = 1, 2, \ldots$, which will in general differ across firms. Let $\Delta p_{i,n} = p_{i,\tau_{i,n}} - p_{i,\tau_{i,n-1}}$ be the size of a non-zero (log) price change that occurs at time $\tau_{i,n}$. I am interested in the coefficient β_1 in the following panel regression:

$$\Delta p_{i,n} = \sum_t \alpha_t \mathbf{1}\{\tau_{i,n-1} = t\} + \beta_1 \Delta p_{i,n-1} + \text{Error}, \tag{2.1}$$

that is, we are trying to predict the size of the price change for unit i, $\Delta p_{i,n}$, using the size of its last price change, $\Delta p_{i,n-1}$, and fixed effects by the time of the $\Delta p_{i,n-1}$ price change. In order to get a more intuitive grasp of (2.1), consider the following thought experiment: assume that, in our sample, we see two units, $i = 1, 2$, changing prices at the same time (which is captured by α_t). One of the units increases prices by $\Delta p_{1,n-1} = +5\%$ and the other one by $\Delta p_{2,n-1} = +20\%$. The regression (2.1) is asking: does that predict anything about the relative size of their *next* non-zero price changes, whenever those occur?

We can also extend this logic to see how predictability depends on the horizon. In particular, I also consider the following specification:

$$\Delta p_{i,n} \quad = \quad \sum_t \alpha_t \mathbf{1}\{\tau_{i,n-2} = t\} + \beta_{1,2}\Delta p_{i,n-2} + \text{Error}, \tag{2.2}$$

where the key difference is that we are using the price change from two spells ago ($\Delta p_{i,n-2}$).

2.2 What Do Standard Models Predict?

Having defined the moments of interest, I now study what different models predict for these moments. Consider an economy populated by a continuum of firms. Each firm has a frictionless optimal price (in logs) of $z_{i,t}$. The actual (log) price it charges is $p_{i,t}$, which may differ from $z_{i,t}$ due to pricing frictions, so we let $x_{i,t} = p_{i,t} - z_{i,t}$ be the price gap. Assume that a firm's profits are given by $\pi(x) = -Bx^2$. Online Appendix OA.D expands on the general equilibrium microfoundations and corresponding approximations being made. These are standard and follow from Alvarez and Lippi (2014). Under those microfoundations, shocks to $z_{i,t}$ reflect joint shocks to idiosyncratic firm productivity and preferences. In what follows, we consider several assumptions about how $z_{i,t}$ evolves, and how $p_{i,t}$ adjusts to it.

2.2.1 Idiosyncratic Random Walks and Perfect Reset

First, assume that $z_{i,t} = \sigma W_{i,t}$, where $W_{i,t}$ is a standard Brownian motion, independent across firms. In terms of the timing of price adjustment, assume that prices adjust at some (random) stopping times $\{\tau_{i,n}\}_{n=1}^{\infty}$, with $\mathbb{E}[\tau] < +\infty$. Assume furthermore that $\tau_{i,n}$ can be deterministic (once every T months, as in Taylor (1979)), random but independent of $z_{i,t}$ (for example, as in Calvo (1983)), or endogenous and depend on $x_{i,t}$, as in menu cost models.

Since the only relevant state is x, we can write the value of the firm as $V(x) = \mathbb{E}[\int_0^{\infty} e^{-\rho s}\pi(x_{t+s})ds|x_t = x]$. At reset, the firm sets new prices such that the price gap is $x^* = \arg\max_x V(x)$. Thus, firms *always* choose the same price gaps upon adjustment, which implies a new price of $p^*_{i,t} = z_{i,t} + x^*$. Given that $z_{i,t}$ is a martingale and $\tau_{i,n}$ is a stopping time, $p_{i,\tau_{i,n}}$ is also a martingale. Therefore, its increments are uncorrelated over time, so:

$$\beta_1 \quad = \quad \frac{\text{Cov}(\Delta p_{i,n}, \Delta p_{i,n-1})}{\text{Var}(\Delta p_{i,n-1})} = 0,$$

in this model.[1] The economic intuition is simple, and in fact, more general: since all firms go back to x^* after changing prices, this effectively eliminates all memory from the process, and therefore what happens after reset cannot be predicted using what happened before.

In order to make that general intuition concrete, consider all models that share a key property: all firms resetting prices at time t fully reset their idiosyncratic states to the same point in the state space. This class contains standard formulations of time-dependent models (see Taylor (1979); Calvo (1983) among others), random menu cost models (Dotsey et al., 1999; Alvarez et al., 2021), and models with multiproduct firms (Alvarez and Lippi, 2014). Several models with imperfect or infrequent arrival of information such as Reis (2006); Woodford (2009); Afrouzi et al. (2026), and the version with no price plans in Alvarez et al. (2011), are also included. This assumption also allows for aggregate states, such as stochastic trend inflation, sectoral shocks, the price index of competitors in models with strategic complementarities (Alvarez et al., 2023) or time-varying volatility (Vavra, 2014).

To describe this class of models, assume that time is discrete. There is a large number of firms. Each firm i has an observable type, $\theta_i \in \Theta$, where Θ is finite. Denote idiosyncratic states by $x_{i,t} \in \mathbb{R}^n$ and aggregate states by $X_t \in \mathbb{R}^m$. Whenever firm i changes prices, its idiosyncratic state moves to $x^*_{i,t+1} = H^*(\theta_i, X_t, x_{it})$. Otherwise idiosyncratic states evolve according to some law of motion. The key condition satisfied by these models is:

Condition 1. *All firms of type θ with aggregate state X_t reset to the same value of the idiosyncratic state, i.e., $x^*_{i,t+1} = H^*(\theta_i, X_t)$ does not depend on x_{it}.*

If Condition 1 is satisfied, it follows from the Markov property that, conditional on X_t and θ_i, the timing and sizes of all future price changes (realized at time $t+1$ or after) are independent of the value of the idiosyncratic state right before reset. Therefore, any pre-reset idiosyncratic differences must have no predictive power for variables after reset.

In order to see that formally, let $y_{i,s} = \mathcal{Y}(X_s, x_{i,s}, \theta_i)$ be any variable for firm i (of type θ_i) determined at time s *after* the last reset time $\tau_{i,n-1}$ (i.e., $s > \tau_{i,n-1}$) that is a function of the aggregate and idiosyncratic states at that date. Let $w_{i,\tau_{i,n-1}} = \mathcal{W}(X_{\tau_{i,n-1}}, x_{i,\tau_{i,n-1}}, \theta_i)$ be a variable determined at or *before* the last reset that depends on the relevant states, and $\lambda_{\theta,t}$ be a fixed effect for type interacted with the time of the last reset. Assume we observe a sample with a finite time horizon T. The following proposition shows that, once we include the proper fixed effects, variables determined at reset have no predictive power for variables determined purely after reset.

[1] In this economy aggregates are always at steady state, so the time fixed effects in (2.1) play no role.

Proposition 1. ***No Predictability.*** *If Condition 1 holds, then the estimand of the following regression is* $\beta = 0$*:*

$$y_{i,s} = \sum_{\theta \in \Theta} \sum_{t} \lambda_{\theta,t} \mathbf{1}\{\tau_{i,n-1} = t, \theta_i = \theta\} + \beta w_{i,\tau_{i,n-1}} + Error. \tag{2.3}$$

Proof. See Appendix C.1. □

The intuition is simple: once we add type-by-time fixed effects, we are effectively comparing firms of the same type that reset prices at the same time t. All remaining variation in $y_{i,s}$ comes from shocks (both idiosyncratic and aggregate) *after* t. Since all firms being compared started at the same point in the state space upon reset, any differences these firms may have had are completely erased after reset, so there is no predictability. Therefore, models in this class predict $\beta_1 = \beta_{1,2} = 0$ for the regressions in (2.1)-(2.2).

There are two main assumptions that deliver Condition 1: (i) idiosyncratic components follow random walks; and (ii) reset is perfect. I analyze what happens once we relax those assumptions below. I discuss other ways to break this property and their implications in Section 3.3.

2.2.2 Mean Reversion in Desired Prices

Assume now that the frictionless optimal price follows $dz_{i,t} = -\eta z_{i,t} dt + \sigma dW_{i,t}$ with $\eta > 0$, i.e., it features mean reversion. Assume the profit function is quadratic and that stopping times $\tau_{i,n}$ are as in Calvo: price changes occur randomly with arrival rate κ, and the timing is independent of $z_{i,t}$. Assuming firms choose their reset prices to maximize the present value of profits, it can be shown that price changes are $\Delta p_{i,n} = \bar{c}(z_{i,\tau_{i,n}} - z_{i,\tau_{i,n-1}})$, where $\bar{c}$ is a known constant.

This model has three key implications. First, one can show that $\beta_1 = -\frac{\eta}{2(\kappa+\eta)} < 0$ (see Appendix B.1 for a proof) so this model generates reversals. Since shocks to the desired price are mean-reverting, price changes in one direction are usually followed by price changes in the opposite direction, which generates reversals. Second, $\partial|\beta_1|/\partial\kappa < 0$: the strength of reversals (measured by the magnitude of β_1) is *decreasing* in the frequency of price changes, κ. When prices adjust more quickly, there is less time for mean reversion in z to kick in, so the measured predictability is smaller. Finally, Appendix B.1 also shows that $\beta_{1,2} = \beta_1(1 + 2\beta_1) < 0$, so predictability lasts more than one spell. Intuitively, since $z_{i,t}$ takes time to revert to the mean, mean reversion takes more than one spell to occur, and thus price

changes from two or more spells away still have predictive power. I show in Appendix B.1 that similar patterns hold in the Taylor (1979) model (analytically) and in a menu cost model (numerically).[2]

2.2.3 Imperfect Reset

Go back to the setup with $\eta = 0$ and Calvo timing, but alter what happens at reset: assume that, due to the difficulty of the pricing process, the firm has imperfect control over where the price gap will land after reset.[3] I model this by assuming that the reset gap is drawn from a distribution F that is chosen by the firm, similarly to Costain and Nakov (2015, 2019); Stevens and Morales-Jiménez (2025). The value of resetting prices is:

$$V^* = \sup_F \int V(x)dF - \mathcal{C}(F), \text{s.t} \int dF = 1,$$

that is, the firm chooses a *distribution* with CDF F for the new price gap, subject to a *control cost* $\mathcal{C}(F)$. The basic idea behind control costs is that perfectly choosing the optimal price is costly, so managers optimally decide to make some "mistakes" rather than eliminate them entirely. One natural interpretation is managerial costs: the time spent in meetings discussing new prices, or the number of analysts hired to evaluate the pricing decision. Spending more resources will lead to a price closer to the profit-maximizing one, and firms choose precision by balancing that trade-off. It is worth emphasizing that this notion of "mistakes" does not rest on irrationality but quite the opposite: agents are optimally deciding how much to invest in precision.

In order to get a closed-form solution, assume that, as is standard in the literature, $\mathcal{C}(F) = \theta \mathcal{I}(F||\bar{F}) = \theta \int \log\left(dF/d\bar{F}\right) dF$, i.e., costs are proportional to the relative entropy between the chosen distribution F and a baseline distribution $\bar{F}$, which I take to be $\mathcal{N}(0, \bar{\sigma}^2)$. Under these assumptions, it can be shown that $x^* \sim \mathcal{N}(0, (\bar{\sigma}^{-2} + \frac{2B}{\theta(\rho+\kappa)})^{-1})$. Thus, in contrast

[2]The baseline menu cost calibration features reversals ($\beta_1 \approx -0.2$) that last more than one spell ($\beta_{1,2} \approx -0.12$). Increasing the frequency of price adjustment is conceptually less clear in a menu cost model, since it is endogenous and driven by several structural parameters. I consider changes in three parameters that increase the steady-state frequency: i) higher trend inflation, ii) smaller menu cost, iii) higher volatility of idiosyncratic shocks. For the first two parameter changes, $|\beta_1|$ decreases, whereas for the last one it increases. Thus, except for changes in idiosyncratic volatility, menu cost models tend to produce a similar comovement between average adjustment frequency and $|\beta_1|$. However, I argue that variation in idiosyncratic volatility is unlikely to explain the variation in the frequency I exploit in Section 3. See Appendix B.1.1 for details.

[3]The assumption of Calvo timing under imperfect reset can be justified by using a multiproduct model, see Section 5.6 and Online Appendix OA.C.4 for details.

Model Features		β_1		$\beta_{1,2}$
Mean Reversion?	Imperfect Reset?	Sign?	Higher Freq.?	After Two Spells?
No	No	Zero	Still Zero	Zero
Yes	No	Negative	Smaller $\lvert\beta_1\rvert$	$\beta_{1,2} < 0$
No	Yes	Negative	Larger $\lvert\beta_1\rvert$	Zero

Table 1: Summary of implications from different models.

to the previous model, price gaps after reset are *stochastic:* Firms choose the right price *on average*, but make "mistakes" *in each realization.* Thus, prices satisfy:

$$p_{i,\tau_{i,n}} = z_{i,\tau_{i,n}} + x^*_{i,n}, \tag{2.4}$$

where $x^*_{i,n}$ are independent and identically distributed (i.i.d.) across firms and spells, independent of $z_{i,t}$, and have mean zero.

Given (2.4), the estimand of regression (2.1) is $\beta_1 = -\frac{\text{Var}(x^*)}{\text{Var}(\Delta p)} < 0$ so this model generates reversals. Intuitively, larger-than-average price changes are partially explained by imperfections, which are expected to be *reversed* at the next price change. Using that $\text{Var}(\Delta p) = \sigma^2/\kappa + 2\text{Var}(x^*)$ (see Lemma 2 in the Appendix) one can show that $\partial|\beta_1|/\partial\kappa > 0$, i.e., the strength of the reversals *increases* with frequency. This is due to two channels. First, mechanically, higher frequency reduces σ^2/κ, which is the variance component coming from $z_{i,t}$. Thus, the $x^*_{i,n}$ component in prices becomes more important, and reversals become stronger. Second, if κ is higher, firms endogenously choose more imprecisely: since any given price is expected to last less time, it is optimal to save on precision costs, which are paid at each price change. The model also implies that $\beta_{1,2} = 0$: since imperfections are corrected at the next price change, predictability only lasts for one spell.

Table 1 summarizes the implications of different models. Both mean-reverting shocks to desired prices and imperfect reset can generate $\beta_1 < 0$, but they differ in their comovement with frequency, and in whether predictability lasts for multiple spells or not. With this in mind, we turn to the data to measure β_1, its comovement with frequency, and $\beta_{1,2}$.

3 Empirical Evidence

I measure these moments using two datasets: the price microdata underlying the construction of the CPI in the United Kingdom, and a new dataset from a recent high-inflation episode in Argentina. In both cases, the predictions align with models with imperfect reset. I also

discuss some alternative stories and why I view them as unlikely candidates to explain all the evidence.

3.1 Evidence from the United Kingdom

I use the microdata underlying the construction of the UK Consumer Price Index (CPI), collected by the United Kingdom Office for National Statistics (ONS). It contains monthly price quotes for the vast majority of goods and services in the CPI. This database is publicly available, and has been used extensively in the pricing literature (Adam and Weber (2023); Blanco et al. (2024c); Jenuwine and Tielens (2025) among others). The baseline sample is 1996M01 to 2019M12. The data is structured as a panel, where units are referred to as *product*s. A product is defined as a particular good or service at a given store. An *item* is a narrowly defined category of products. Examples include "Cinema Admission" or "Men's Formal Shirt, Long Sleeve". Thus, several different products will map to a given item: for example, two formal shirts of different brands or two identical formal shirts sold in different stores would be classified as two different products under the same item. Finally, a *cell* is the finest level of disaggregation used by the ONS. Its definition varies by item, but generally it is defined as item by region by type of shop (small or large). For items that have just a handful of quotes, the ONS may define the cell at a coarser level, such as item by region, item by type of shop, or just item.

I perform several basic data cleaning steps, which mostly follow Adam and Weber (2023).[4] First, given that some stores are visited more than once per month, some products have two different price quotes. I drop those products. Second, if the same exact product is not found in the store, or if the store leaves the sample, it can be substituted by a similar product. I treat substitutes as completely different products. Thus, in any event that triggers a product substitution, I treat that as an exit of one product and entry of a new product with a potentially new price. I drop prices below one penny and price quotes that are flagged as invalid, and I restrict the sample to products that have at least 2 consecutive price changes. In order to account for temporary sales, I use a 3-month V-shaped filter as my baseline, but results are robust to using other popular filters (Nakamura and Steinsson, 2008; Kehoe and Midrigan, 2015). Another typical source of measurement error is the presence of spurious small or large price changes. For small price changes, I follow Alvarez et al.

[4]My code is partially based on the code provided by Gado et al. (2026), available at https://www.ukmicrocpi.com

(2016a) and drop price changes smaller than 0.1% in absolute value. For large price changes, I first drop all price changes that are larger than log(3) in absolute value. Second, I compute standardized price changes as $\tilde{\Delta}p_{i,t} = (\Delta p_{i,t} - \mu_c)/\sigma_c$ where $\Delta p_{i,t}$ is the observed price change (in logs) for product i that belongs to a cell c at time t, and μ_c and σ_c are the mean and standard deviation of all price changes for cell c, respectively. I further control for outliers by dropping the top and bottom 1% of observations in terms of their standardized price changes. Appendix A shows that results are robust to these choices.

Regression Specifications and Results. Following the preceding discussion, I estimate the following regressions:

$$\Delta p_{i,n} = \text{Fixed Effects} + \beta_1 \Delta p_{i,n-1} + \text{Error}, \tag{3.1}$$

$$\Delta p_{i,n} = \text{Fixed Effects} + \beta_{1,2} \Delta p_{i,n-2} + \text{Error}. \tag{3.2}$$

The notation follows that of the previous section.[5] As before, each firm has a type $\theta_i \in \Theta$. In my baseline specification, I define types by the firm's cell. Consistent with the discussion surrounding Proposition 1, I include a set of type-by-time fixed effects of the form $\lambda_{\theta,t}\mathbf{1}\{\tau_{i,n-1} = t, \theta_i = \theta\}$ for the first regression (and a similar one with $\tau_{i,n-2}$ instead of $\tau_{i,n-1}$ for the second one). I also include fixed effects by the time of the price change being predicted, $\gamma_{\theta,t}\mathbf{1}\{\tau_{i,n} = t, \theta_i = \theta\}$. Appendix A shows that results are similar for several different configurations of fixed effects (including no fixed effects at all).

In order to measure the comovement of $|\beta_1|$ with frequency, I split the sample in two: items with "low" frequency of price changes, defined as those with frequency lower than the average (which is around one price change every 9 months), and those with "high" frequency, i.e., higher than average. It has been documented that UK CPI data feature large variation in frequencies across items (Adam and Weber, 2023; Blanco et al., 2024c).

Figure 3.1 presents the results. There are three key patterns. First, focusing on the left-hand estimate, β_1 is negative and large in magnitude. Even after we filter out sales, prices feature strong reversals in consecutive spells. Second, the center part shows what happens when we compare the values of β_1 across categories with high and low frequency of price changes. The figure shows that items with higher frequency of price changes also

[5] In Online Appendix OA.A.1 I also present results for the timing of price changes (i.e., the extensive margin) since they might be of interest to some readers. The overall predictability patterns are consistent with models of imperfect reset.

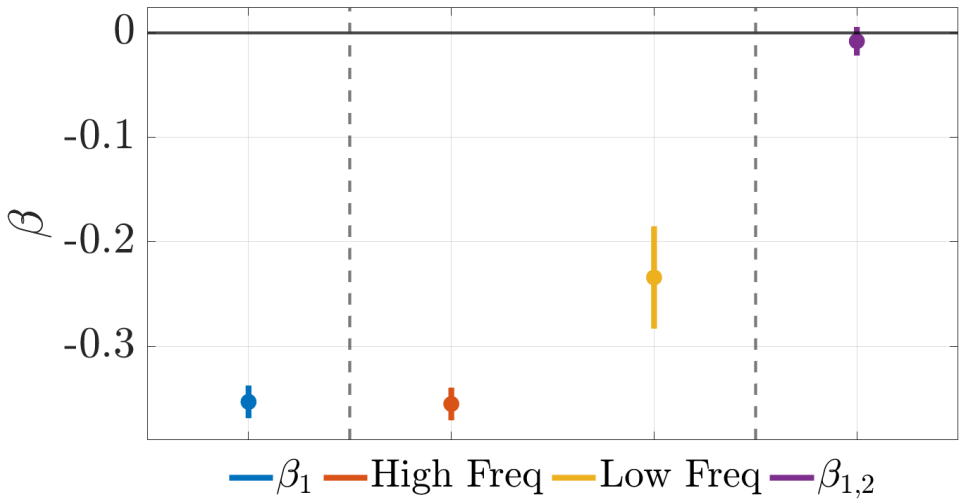


Figure 3.1: Empirical Estimates for equation (3.1), for the full sample (left) and the frequency subsamples (middle). High (low) Freq. refers to items with frequency of price changes above (below) the average. The right-hand estimate is for (3.2). Bars reflect 95% confidence intervals. Standard errors are clustered by item.

feature larger β_1 in magnitude. Third, focusing on the right part, predictability using price changes from two spells ago ($\beta_{1,2}$) is essentially zero.[6]

Appendix A.1.2 shows that I obtain the same patterns using other sales filters. However, if we did not filter out sales, results would meaningfully change: β_1 would go down significantly, and $\beta_{1,2}$ would be large and significantly *positive.* This last finding is consistent with models of price plans (Eichenbaum et al., 2011; Kehoe and Midrigan, 2015; Alvarez and Lippi, 2020): if firms often alternate between two fixed price points p^H and p^L, it would often happen that $\Delta p_{i,n} = \Delta p_{i,n-2}$, leading to a positive $\beta_{1,2}$. The fact that we find $\beta_{1,2} \approx 0$ in all three cases provides reassurance that the filters remove temporary sales.

Overall, these patterns are consistent with models of imperfect reset, and inconsistent with the other models considered above.[7] However, at least regarding the comovement

[6]The finding that $\beta_{1,2}$ is not significantly different from zero is not robust to different sales treatments and fixed effects: the point estimate of $\beta_{1,2}$ changes little, but it may become statistically significant. Thus, I formally test the null that $\beta_{1,2} = \beta_1(1 + 2\beta_1)$. At conventional levels, I reject it for all filters.

[7]The reader may also wonder: what are the stochastic properties needed in $z_{i,t}$ to generate the magnitude of β_1 observed in the data? I study this question in Section 5.7. There, I show that matching my evidence without imperfections would require very large and transitory shocks: the estimated standard deviation is around 30% *per month* and the half-life of these shocks is around 3 months. In terms of the empirical

with frequency, an alternative explanation of this fact is that heterogeneity in frequency is correlated with heterogeneity in the degree of mean reversion: a world where sectors with higher frequency also had faster mean reversion would also explain this pattern. It could also be the case that transitory shocks follow a more general ARMA process and just last for a few months, which would explain the absence of predictability after the first price change. Given these concerns, I study the same patterns in a dataset from Argentina. There, the *aggregate* frequency of price changes varies substantially over time. Furthermore, I observe a subsample where prices change every few weeks due to very high inflation. If reversals are purely due to transitory shocks, they should be much weaker when trend inflation is high.

3.2 Evidence from Argentina

In this subsection, I supplement the above evidence with data from Argentina. The Argentine case is exceptional in that it has large variation in the frequency of price changes, driven by variation in aggregate inflation. Furthermore, the database I present below allows me to track products by barcode across different stores, which allows me to use appropriate fixed effects to purge more potential sources of unobserved heterogeneity across products.

Figure 3.2 shows the monthly inflation rate for the sample April 2023 - January 2025, obtained from INDEC. As we can see, monthly inflation hovers around 8-10% at the beginning of the sample. A few months before Milei's presidency starts, around the time elections are taking place, inflation begins to increase. It reaches a peak of 25.5% per month in December 2023, when Milei's presidency starts. While the exact causes for this require further scrutiny, one possible narrative attributes this to expectations about an imminent depreciation of the peso, due to the perception that Milei would lift currency controls and even possibly dollarize the economy (Caravello et al., 2023b,a).[8] After the initial shock, inflation remains elevated for a few months, then decreases to around 4% by June 2024, and continues a slow downward trend, settling around 2% by December 2024. The red and blue lines in the figure depict the frequency of price changes, for both price increases (red) and decreases (blue). As expected, price hikes become more frequent when inflation is high, and less frequent when it is low. The opposite is true for price cuts. The frequency of price

evidence regarding costs, Gagliardone and Tielens (2026) estimate the idiosyncratic component in marginal costs to be far more persistent than what matching my evidence requires, although this evidence is disputed (see Jenuwine and Tielens (2025)).

[8]The currency controls were partially lifted during December 2023, with the official exchange rate almost tripling its value. On the other hand, dollarization plans were discarded.

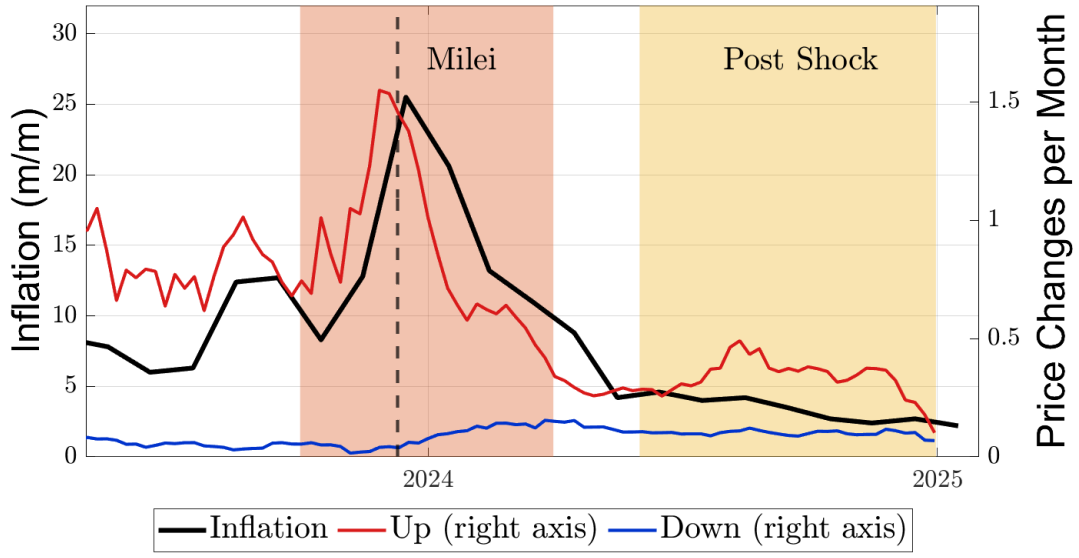


Figure 3.2: Black: monthly inflation rate. Red: frequency of price increases. Blue: frequency of price decreases. Both frequencies are measured weekly and converted to monthly by multiplying by 52/12. The dashed vertical line indicates the start of the Milei presidency (December 10, 2023). Sample: April 2023 to January 2025. "Milei" subsample runs from October 1, 2023 to March 31, 2024. The "Post Shock" subsample runs from June 1, 2024 to December 31, 2024.

increases during the inflation spike is quite high: prices change more than once a month.

For my regressions, I split the sample into two windows. First, I consider the high-inflation period around the inflationary shock. I label this the "Milei" subsample, and it corresponds to the time frame shaded in red, which runs from October 1, 2023 to March 31, 2024. Second, I consider a "Post Shock" sample (June 1, 2024 to December 31, 2024), where the main inflationary shock had dissipated and inflation sits around 2-4% per month.

Data Description. I use data from Sistema Electrónico de Precios Argentinos (SEPA). This system was created by the Argentine government in 2016 with the idea of fostering competition across retailers. In particular, all medium and large retailers in the country are required to submit daily data on all their posted prices in each store. The government would then post those prices on a user-friendly website, where consumers could compare the price of a given barcode-level product in all stores that sold that product around a given physical address. This dataset is new: to my knowledge, the only paper that uses this data source

is Daruich and Kozlowski (2023), whose authors obtain the data for an earlier subsample. My sample covers a period where the underlying inflation rate varies significantly, which provides useful variation.

My sample ranges from July 2021 to December 2024. However, due to the prevalence of price controls during 2022 and early 2023, I only focus on the two subsamples described above, which provide large variation in aggregate frequency. Although I have daily data, I aggregate to weekly by considering prices on the same weekday on different weeks. The database contains between 37 and 55 unique retail chains (depending on the year), between 70 and 100 thousand different products (barcodes), and between 3 and 5 thousand unique physical stores. The data contains information on the posted price for each barcode at each store. Relative to the UK CPI data, one downside of this dataset is that I do not observe product categories.

Regarding sales, promotions or other types of discounts, the dataset records these as a separate price column. Therefore, my dataset always includes the baseline regular price, which does not change with sales or discounts unless they are recorded as changes to the regular price. Due to institutional reasons, a large number of temporary sales in Argentina are of the form "buy one, get one X% Off", which my data filters automatically. Furthermore, typical sales tend to be less frequent when inflation is high (Alvarez and Lippi, 2020). Nevertheless, I also apply Filter A in Nakamura and Steinsson (2008) to the observed series to remove any potential sale-like behavior.[9]

Results. As before, I focus on the specifications (3.1)-(3.2). There are two main differences relative to the specification for the UK data. First, given that I observe barcodes, I now let j be a given barcode, and consider barcode-by-time fixed effects. Notice that this eliminates an important source of variation: we are now comparing the same exact barcode across different stores, so assuming that the cost of sourcing that barcode changes for all retailers at the same time, all remaining variation in prices must be due to costs or desired markups at the retailer level. Second, due to the large synchronization in repricing across stores of the same retailer, I now cluster standard errors at the time-by-retailer level. This meaningfully increases standard errors relative to other clustering specifications that ignore this.

Figure 3.3 reports the estimated coefficients of running each specification on different subsamples. Although the setting is completely different, we see that the same patterns

[9] I use this filter instead of a simple V-shaped filter because, due to high inflation, it is more likely that the price increases above its pre-sale value when the sale ends.

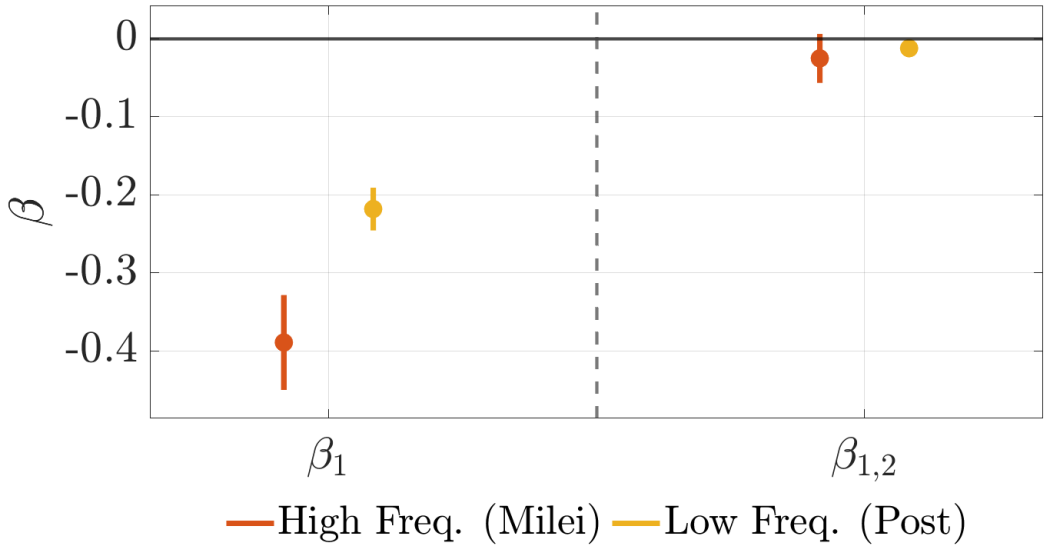


Figure 3.3: Regression coefficients for specifications (3.1) and (3.2), for different subsamples. Red: "Milei" subsample. Yellow: "Post Shock" subsample. See Figure 3.2 or the text for definitions of the subsamples. Bars reflect 95% confidence intervals. All regressions include barcode-by-time fixed effects. Standard errors are clustered at the retailer-by-time level.

emerge. First, β_1 is significantly negative. Second, $|\beta_1|$ is much larger during the Milei subsample, which has higher frequency of price changes. The magnitude of the coefficient is smaller in the Post Shock sample, with lower frequency. Finally, in both subsamples, predictability only lasts one spell: $\beta_{1,2}$ is not statistically significant at the 5% level in the "Milei" subsample, and it is very small in magnitude for the Post Shock period. These results present significant evidence against the hypothesis that these reversals are driven by mean-reverting shocks. In particular, we find strong relative reversals during the high-inflation period, when prices tend to remain unchanged for only a few weeks. If reversals were driven by a structural mean-reverting process with a fixed reversion speed, one would expect little reversion to occur in such short periods of time.

3.3 Main Empirical Facts and Alternative Explanations

While the macroeconomic context is quite different between the UK and Argentina, three facts are common across both datasets.

Fact 1. ***Idiosyncratic Price Reversals.*** *The β_1 coefficient in regression (2.1) is negative.*

This means that if a firm changes prices more than average at t, it is expected that it will change prices less than average at the next price change.

Fact 2. ***Higher Adjustment Frequency Implies Stronger Reversals.*** *The β_1 coefficient in regression (2.1) is, in absolute value, increasing in adjustment frequency.*

Thus, higher adjustment frequency correlates with stronger reversals.

Fact 3. ***Reversals Last One Spell.*** *The $\beta_{1,2}$ coefficient in (3.2) is approximately zero.*

Finally, predictability only lasts one spell: sizes of price changes from two spells ago have essentially no predictive power for the sizes of current price changes.

Going back to the predictions of different models in Table 1, we see that these three facts are consistent with models of imperfect reset, and inconsistent with standard models. Before studying the consequences of reset imperfections, I briefly review some additional evidence, and other a priori plausible explanations for the above patterns, and why I view them as unlikely candidates to fully explain my empirical evidence.

Measurement Error. The reader may worry that the apparent error component in (2.4) is just an artifact of improper measurement. In this regard, several things should be noted: First, in the UK, price quotes used for the CPI undergo extensive checks, both at the time of collection and afterwards, to ensure that there are no recording errors. In Argentina, the data is self-reported digitally by firms every day, minimizing the potential for unintentional human error. As the previous discussion implies, data collection procedures are different, and yet we find the same patterns. Eichenbaum et al. (2014) argue that many of the small price changes recorded in the data are actually due to measurement error. Appendix A.1.1 shows that the estimates are essentially unchanged if we exclude small price changes.

Standard Noisy Information. If firms do not know their frictionless optimal prices and learn them over time by observing noisy signals, it may seem like this would generate "mistakes" that firms later correct. Online Appendix OA.B.1 shows that with zero trend inflation and random walks in idiosyncratic fundamentals, this cannot generate reversals. Intuitively, noisy information introduces two offsetting forces: i) the firm responds partially to noise, which is later corrected, generating negative correlation; ii) the firm initially underreacts to fundamental shocks, which generates *positive* correlation between consecutive price-change sizes. If the true process follows a random walk, optimal inference is such that both forces cancel, the perceived price gap also follows a random walk, and we obtain zero predictability. This result relies on agents updating optimally: if instead agents overreact by

overweighting new signals relative to their prior (departing from the Bayesian benchmark), predictability can arise. I discuss the connection to my notion of imperfections in Online Appendix OA.B.2.

Active Experimentation. If firms try to learn about demand by experimenting with their prices, this will generate "mistakes" that are then corrected when the firm gains more information about the demand curve. Argente and Yeh (2022) document evidence for experimentation upon the introduction of a new product. Since I can track products at the barcode level in the Argentine data, I test this by running the same regression as the one reported in Figure 3.3, but restricting the subsample to a set of products available at the beginning of the sample in 2021. These products have been in the market for a few years prior to the start of the sample, so any idiosyncratic learning effects should have dissipated. Appendix A.2 shows that results are unchanged. While this does not directly rule out active experimentation in response to constantly renewed sources of uncertainty, this empirical finding suggests that an explanation centered on initial learning is insufficient. Furthermore, it is unclear whether this story can explain the observed comovement with adjustment frequency.

Unobserved Heterogeneity. Fixed effects may not capture all relevant heterogeneity among firms.[10] Appendix B.2 argues that this is unlikely to explain my findings. I consider the model in Alvarez et al. (2021), and allow for heterogeneity in menu costs and volatility of idiosyncratic shocks. With low trend inflation, I show that the model predicts $\beta_1 = 0$ even if there is unobserved heterogeneity. For the Argentine evidence, one may worry that different sources of heterogeneity (such as different menu costs, or heterogeneous beliefs across firms about future trend inflation) may interact non-trivially with the large inflationary spike. Consider this issue in a simple menu cost model. In that case, some units will have wider adjustment bands than the others. In the presence of trend inflation, this will generate a positive correlation in consecutive sizes, which is the opposite of what we observe.

Convex Adjustment Costs. If, in addition to fixed costs, firms were subject to convex adjustment costs as in Rotemberg (1982), firms would implement large price increases through several smaller price increases. This would generate a positive correlation in the size of price changes, which is counterfactual.

[10] Online Appendix OA.A.2 presents results by size (chains with at least ten stores versus fewer than ten). There is no clear evidence of differences in predictability. This goes against heterogeneous cost pass-through by size (Amiti et al., 2019) driving predictability.

Customer Markets. In terms of adding state variables to the firm's problem, one natural option would be to extend the model to add the customer base of the firm as a state variable that evolves endogenously. However, it is unclear that this must meaningfully interact with price-setting. For example, Afrouzi et al. (2025) builds a model that matches a key empirical fact: markups correlate with the sales per customer, *not* with the total size of their customer base. Consequently, in their model, pricing decisions are independent of the number of customers that the firm has. Thus, I deem this potential explanation implausible.

4 Comparative-Static Exercise

The above discussion shows that while standard models have trouble matching the empirical evidence, models with imperfect reset can do it. For that reason, I study their implications in more detail for the remainder of the paper. In this section, I consider economies populated by firms that may be subject to these imperfections and study their price adjustment behavior at the *micro* level (i.e., the steady-state average adjustment frequency), as well as their response to a *macro* shock: an aggregate increase in the quantity of money.

I evaluate the effects of imperfections by means of a comparative-static exercise: I start with an economy with perfect reset, and make reset imprecise by changing the cost of precision, while leaving all other parameters unchanged. The main result is that imperfections can matter because they may change the composition of firms that adjust after an aggregate shock. In particular, if the timing of price adjustment is endogenous, imperfections can simultaneously increase the *micro* adjustment frequency and slow down the *macro* response to a monetary shock.

Objects of Interest. I define my objects of interest following Alvarez et al. (2016a); Alvarez and Lippi (2022). The first one is the steady-state frequency of price changes, denoted by N. I also study the impulse response of output to a permanent monetary shock of size δ. In the economies I study, firms have a single state variable $x_{i,t}$. Using the same approximations as Alvarez et al. (2016a), the output response is $Y(t,\delta) = -\mathbb{E}[x_{it}]$. In order to summarize this in a single number, I use the Cumulative Impulse Response (CIR), defined as $\mathcal{M}(\delta) = \int_0^\infty Y(t,\delta)dt$. Since it is easier to characterize analytically, I focus on small shocks. Thus, I can write $Y(t,\delta) = \delta Y(t) + o(\delta)$ and $\mathcal{M}(\delta) = \mathcal{M}'(0)\delta + o(\delta)$, so my objects of interest are $Y(t)$ and its integral $\mathcal{M}'(0) = \int_0^\infty Y(t)dt$. Note that in this set-up, the first-order response of the aggregate price level is $P(t) = 1 - Y(t)$. Thus, studying the adjustment in $Y(t)$ is

directly informative about aggregate price adjustment.

4.1 Imperfect Reset in Calvo: An Irrelevance Result

I first study the comparative static of interest in the economy in Section 2.2.3. The main result is that the average frequency and output response to a monetary shock do not depend on the presence of imperfections so these are irrelevant in this case.

Proposition 2. ***Irrelevance of Imperfect Reset under Calvo.*** *Assume that the timing is Calvo with arrival rate κ. Then $Y(t) = e^{-\kappa t}$, $\mathcal{M}'(0) = 1/\kappa$ and $N = \kappa$.*

Proof. Let $I_{i,t} = 1$ if firm i has adjusted at least once since the shock, zero otherwise. Then:

$$\mathbb{E}[x_t] = \underbrace{\mathbb{E}_t[x_t|I_{i,t}=0]\text{Prob}(I_{i,t}=0)}_{\text{Have Not Adjusted}} + \underbrace{\mathbb{E}_t[x_t|I_{i,t}=1]\text{Prob}(I_{i,t}=1)}_{\text{Already Adjusted}}.$$

Since firms that adjust are selected randomly and x evolves as a random walk, we have that $\mathbb{E}_t[x_t|I_{i,t}=0] = -\delta$. Conversely, $\mathbb{E}_t[x_t|I_{i,t}=1] = 0$: since $\mathbb{E}[x^*] = 0$, while individual firms will not close their price gaps, resetting firms do so *on average.* Given that after reset x evolves as a random walk, the expectation of interest equals the initial average position of firms. Thus, $Y(t,\delta) = \delta\text{Prob}(I_{i,t}=0) = \delta e^{-\kappa t}$, where the last step uses that timing is as in Calvo. From here, the result follows directly. The expression for the frequency follows directly from the Calvo assumption. □

The above result shows that, in Calvo, imperfect reset is irrelevant for both the *micro* frequency and the *macro* adjustment to an aggregate monetary shock. The intuition is simple. First, note that the aggregate shock gets fully incorporated *on average* the first time firms reset their prices. Imperfections do "wash out" in aggregation when considering this term. This implies that output effects are generated by firms that have not adjusted since the shock. Given that reset timing is exogenous and only depends on κ, non-systematic imperfections are irrelevant. This intuition also shows that the same result would obtain in any purely time-dependent model, as long as firms do not make mistakes on average. Note that this is very different from what a standard imperfect-information model would generate (Lucas, 1972; Mankiw and Reis, 2002; Sims, 2003; Afrouzi et al., 2026): there, firms do not fully incorporate the shock after the first price adjustment because they learn about it gradually, and thus this influences monetary non-neutrality.

The above derivation shows that, if the timing of price adjustment is exogenous, adding

mean-zero imperfections does not change aggregate responses. For that reason, I turn to studying models where the timing of price changes is endogenous. In that case, imperfections change the composition of firms that adjust, and therefore have an impact on price adjustment both at the micro and macro levels.

4.2 Optimal Adjustment Policies under Endogenous Timing

The core of the setup is the same as in Section 2.2.3: if prices are not adjusted, gaps follow $dx_{i,t} = \sigma dW_{i,t}$. $V(x)$ is the value of the firm, which has profits $\pi(x) = -Bx^2$ and discount rate ρ. The main difference is that the manager now chooses *when* to change prices. Assume that if prices change, the manager is expected to get a value V^*, which is net of all costs of price changes. On the other hand, the net benefit of waiting an additional instant is $(\pi(x) - \rho V)dt + \mathbb{E}[dV] = (\pi(x) + \frac{\sigma^2}{2}V_{xx} - \rho V)dt$. Given that the manager optimally chooses when to adjust, the value function solves:

$$0 = \max\left\{\pi(x) + \frac{\sigma^2}{2}V_{xx} - \rho V, V^* - V\right\}. \tag{4.1}$$

Note that if $V^* - V < 0$, it is optimal to wait, so the maximum is attained by the left argument. On the other hand, if $V^* - V \geqslant 0$, it is optimal to adjust, and the maximum is attained on the right. Thus, (4.1) fully summarizes the decision on *when* to adjust prices.

The key difference with respect to a standard menu cost model is what happens at the moment when prices are reset, i.e., how V^* is determined. Now, a new price gap is drawn from a distribution F which is chosen by the firm subject to a cost $\mathcal{C}(F)$. Thus, letting $\mathcal{D}$ be the space of CDFs, V^* is determined by:

$$V^* = \max_{F\in\mathcal{D}} \int V(x)dF(x) - \mathcal{C}(F). \tag{4.2}$$

The standard menu cost model is nested as a particular case: if $\mathcal{C}(F) = c$ is independent of F, then $\sup_{F\in\mathcal{D}} \int V(x)dF(x) - c = \max_{x^*} V(x^*) - c$. In contrast, if $\mathcal{C}(F)$ varies with F, this encodes that choosing a new price precisely is costly, and the value of adjusting reflects it.

Note that x is perfectly observed by the agent choosing *when* to change prices. However, the agent has imperfect control over *what* price is set conditional on changing prices. Online Appendix OA.B.2 shows that this model is equivalent to one in which we explicitly separate decisions between two agents: a manager, who decides *when* to reset prices, and a pricer who decides *what* the new price is. While this model makes this separation quite stark for

analytical simplicity, in practice the decisions of *when* to change prices and *what* prices to set may be informed by different variables. For example, a manager may see her profits going down over time and interpret that as a clear signal of the need to change prices, but she might be unsure about the exact price that maximizes said profits.

It is also important to emphasize that, as explained in Woodford (2009), most of the price-setting literature has focused on mechanisms that effectively reduce "precision" in the *timing* of price adjustments, primarily using random menu costs or multiproduct firms. Both of these ingredients partially decouple the probability of adjusting from the relevant state (price gaps), thus making timing more "imprecise", i.e., less state-dependent. However, these models retain full precision conditional on adjustment. In contrast, the model in this section features high precision in *when* to adjust, but potentially lower precision in the decision of what price to set. I generalize this in Section 5, where I allow less state dependence in the timing of price adjustment.

Exogenous Reset Distribution. Assume that the total cost is given by $\mathcal{C}(F) = \theta\mathcal{I}(F||\bar{F}) + c$, where c is a standard menu cost. Take the limit where $\theta \to +\infty$. In that case, the reset distribution is exogenous and given by the baseline $\bar{F}$, and the value upon adjustment is $V^* = \int V(x)d\bar{F}(x) - c$. In order to have a well-defined limit when $c \to 0$, assume that the reset gap is distributed as $x^* = \bar{x}\tilde{x}$, where $\tilde{x}$ has a CDF $\tilde{F}$, with a symmetric distribution with support $[-1, 1]$. Thus, the distribution scales with $\bar{x}$, which in turn scales with c in the limit. The next proposition characterizes the optimal inaction region.

Proposition 3. ***Inaction Region.*** *As $c \to 0$, there is a non-trivial inaction region $(-\bar{x}, \bar{x})$ with:*

$$\bar{x} = \left(\frac{6\sigma^2}{B}[\mathbb{E}[(1-\tilde{x}^2)^2]]^{-1} \times c\right)^{1/4}(1 + o(1)), \tag{4.3}$$

where $[\mathbb{E}[(1-\tilde{x}^2)^2]]^{-1} \geqslant 1$ (strict unless $\tilde{x} = 0$ almost surely).

Proof. See Appendix C.2. The key is to show that imperfect reset creates an additional term in the value-matching condition; the rest of the proof follows Dixit (1991). □

The proposition shows that, when uncertainty over the reset is large, it has a first-order impact on the width of the inaction bands. Intuitively, a firm that is unsure about what price it will charge has wider inaction bands. The result also shows that we can get any desired size of bands by lowering the size of the menu costs. This shows that the key distinct implication of control costs is reset imperfections.

Endogenous Reset Distribution. In Online Appendix OA.B.3, I study a different limit where $c, \theta \to 0$ at the correct rate. That limit shows that control costs act similarly to menu costs and lead to wider adjustment thresholds. I also analytically characterize the endogenous reset distribution and how imperfections affect the value of changing prices. An important takeaway is that large mistakes in price space can arise even for very small information costs (as indexed by θ): since the value function is locally flat close to 0, deviations from the optimal price that are "large" in the policy space lead to "small" losses in profits (see also Akerlof and Yellen (1985)). This provides a theoretical argument for why relatively large ex-post imperfections could arise even if control costs were small in practice.

4.3 Characterization of Objects of Interest

I am interested in comparing two economies: one with no imperfections (where F is degenerate at $x = 0$), and another one with the same structural parameters (ρ, B, c, σ) but imperfect reset, captured by a non-degenerate but otherwise arbitrary F. In the model described above, all that matters for aggregation and aggregate responses are the values of $(\sigma^2, \bar{x}, F)$. The nature of the monetary shock is such that firms change their $x_{i,t}$ on impact, but not their optimal adjustment thresholds and reset distributions.

The following proposition derives the stationary distribution of price gaps, $g(x)$, and the objects of interest as a function of $(\sigma^2, \bar{x}, F)$. In particular, one can characterize the output impulse response function (IRF) $Y(t)$ and cumulative impulse response using the results in Alvarez and Lippi (2022).

Proposition 4. ***Aggregation and Response to Monetary Shocks.*** *The frequency of price changes is $N = \sigma^2/(\bar{x}^2 - Var(x^*))$ and the stationary distribution of price gaps is $g(x) = (x + \bar{x} - 2\int_{-\bar{x}}^{x} F(z)dz)/(\bar{x}^2 - Var(x^*))$. The response of output to a monetary shock is*

$$\begin{aligned} Y(t) &= \sum_{j=1}^{\infty} B_j \exp(\lambda_j t), \quad \textit{where } \lambda_j = -\frac{\sigma^2}{\bar{x}^2}\frac{(j\pi)^2}{2} \textit{ and} \\ B_j &= \frac{1}{1 - \mathbb{E}[\tilde{x}^2]}\left((1 - \int_{-1}^{1} cos\left(j\pi(u+1)\right) d\tilde{F}(u)\right)(\frac{2}{j\pi})^2. \end{aligned}$$

The cumulative impulse response is $\mathcal{M}'(0) = \mathbb{E}_F[(\tau(x^*))^2]/(6\mathbb{E}_F[\tau(x^*)])$, where $\tau(x) = \mathbb{E}[\tau | x(0) = x] = \left(\frac{\bar{x}^2 - x^2}{\sigma^2}\right) 1_{\{|x| \leqslant \bar{x}\}}$ is the expected time until the next price adjustment for a firm that starts at x and $\mathbb{E}_F[k(x^*)] = \int k(x) dF$.

Proof. See Appendix C.3. For $Y(t)$, since the key symmetry properties at reset continue to

hold in this model, we can apply the formulas in Alvarez and Lippi (2022) directly to get the first part. By integrating this expression one can obtain $\mathcal{M}'(0)$. See Appendix C.3.2. □

Having characterized the objects of interest in terms of $(\sigma^2, \bar{x}, F)$, I now proceed to study the comparative static of interest. In order to predict how $\bar{x}$ would change when adding imperfections, I make use of the analytical characterization in Proposition 3.

4.4 Comparative Static

Assume that we have a standard (perfect reset) menu cost model with some structural parameters (ρ, c, B, σ^2). Fixing those, I now add imperfections by changing the cost functional to induce a non-degenerate reset distribution F. This captures that precision may vary over time: price-setting may become harder, for example due to broad shifts in consumer patterns that require effort to understand; or easier, for example due to the introduction of new technologies that simplify data analysis.

First, Proposition 5 shows that both the size and the frequency of price changes generally increase when we move to imperfect reset.

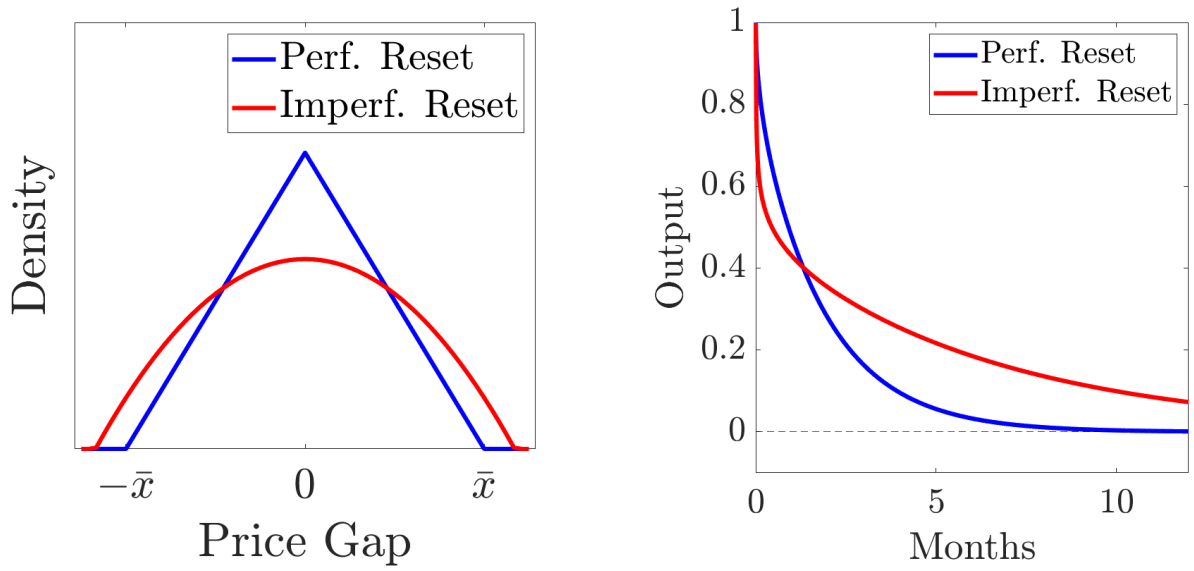
Proposition 5. ***Steady-State.*** *Fix structural parameters and assume c is sufficiently small. Moving from an economy with perfect reset to imperfect reset has the following effects:*

i) The average size of price changes $(\bar{x})$ strictly increases.

ii) The average frequency of price changes (N) weakly increases. This effect is strict unless $\exists a \in \mathbb{R}^+ : Prob(|x^*| = a) = 1$.

Proof. See Appendix C.5. □

Thus, imperfect reset generates two changes in observables. First, Proposition 3 directly implies that firms will expand their inaction bands, for the intuitive reasons explained above. However, after reset, firms will tend to land closer to the adjustment bands, which will tend to increase the frequency. This can be seen from Proposition 4: given $(\sigma^2, \bar{x})$, higher variance of x^* increases the frequency. In principle, one may think that the final effect on the frequency is ambiguous. However, the proposition shows that the frequency weakly increases when the costs are sufficiently small. Intuitively, given the new source of variation in prices, firms optimally choose to adjust both on the extensive margin (i.e., adjust more frequently) and the intensive margin (have wider inaction bands). Thus, adding imperfections looks similar to an increase in idiosyncratic volatility (Vavra, 2014).



(a) Stationary distribution of price gaps.

(b) Impulse response to a small monetary shock.

Figure 4.1

In order to see how this affects non-neutrality, Figure 4.1a compares the stationary distribution with perfect and imperfect reset. There are two effects: First, with imperfect reset there is less mass close to $x^* = 0$, and more mass closer to the inaction bands. Intuitively, since firms do not land exactly at $x = 0$, they are closer to adjusting again. However, the inaction bands also widen, so firms that do land close to $x = 0$ now take *longer* to adjust than before. Thus, while adding imperfect reset increases the average frequency of price changes, it also causes some firms to take longer to adjust than before.

Figure 4.1b compares the IRFs with perfect and imperfect reset. As we can see, both forces discussed above play a role. Initially, the economy with imperfect reset adjusts faster, so the first effect (firms are closer to adjustment bands) dominates. However, as time passes, the output response is given by firms that were close to $x = 0$ at the time of the shock. Since those firms take longer to reach the adjustment thresholds, eventually both IRFs cross, and the decay is slower in the economy with imperfect reset. The proposition below formalizes these observations.

Proposition 6. ***Faster Initial Adjustment, Slower Decay.*** *Fix structural parameters and assume c is sufficiently small. Let $Y^{imp}(t)$ be the IRF with imperfect reset and $Y^{per}(t)$ the one with perfect reset. Then:*

i) $\exists t_1 > 0$ *such that* $Y^{imp}(t) < Y^{per}(t)$ *for all* $t \in (0, t_1)$.

ii) $\exists t_2 \geqslant t_1$ *such that* $Y^{imp}(t) > Y^{per}(t)$ *for all* $t > t_2$*, and* $\lim_{t\to+\infty} \frac{Y^{per}(t)}{Y^{imp}(t)} = 0$.

Proof. See Appendix C.4. □

This result also shows that inferring aggregate adjustment speed by looking at the initial response can be misleading, since imperfections decouple the two. This is important since several studies use the Flexibility Index in Caballero and Engel (2007), which is derived from the initial response, as a measure of overall non-neutrality.

While this result characterizes the shape of the IRF, it does not explain what effect dominates in terms of the cumulative IRF. The proposition below shows that, even though the average frequency of price changes rises, the cumulative impulse response of output increases as a consequence of incorporating imperfections.

Proposition 7. ***Cumulative Output Response.*** *Fix structural parameters and assume* c *is sufficiently small. Moving from an economy with perfect reset to imperfect reset weakly increases* $\mathcal{M}'(0)$*. The effect is strict unless* $\exists a \in \mathbb{R}^+ : Prob(|x^*| = a) = 1$.

Proof. See Appendix C.5. □

Strikingly, the proposition shows that even though prices look more flexible at the *micro* level due to higher average frequency, *macro* price flexibility *decreases*: monetary shocks have larger effects on output. This implies that, out of the two effects described in Proposition 5, the fact that bands widen dominates, and non-neutrality increases.

Intuitively, while adding imperfections increases the total number of price changes, it also changes *which* prices change after an aggregate shock, i.e., the composition of adjusters. To see how composition changes, focus first on the economy with perfect reset. There, once a firm changes prices, it resets to $x = 0$. This leaves the firm as far as possible from the adjustment bands, making it less likely to reset again. This means that, after a monetary shock, most price changes are being used by firms that have not yet adjusted prices, and thus contribute to aggregate price flexibility. In contrast, when reset is imperfect, firms that reset after the shock may land close to the adjustment bands, and reset again. This implies that a smaller share of price changes is being used by firms that have not adjusted. Since the adjustment bands are wider, firms that are close to $x = 0$ when the shock hits actually take longer to adjust. It turns out that this second effect dominates, and the cumulative effect on output increases as a result. Intuitively, the economy with imperfect reset allocates price changes "worse" across firms in terms of aggregate adjustment.

A crucial implication is that, if the reset distribution F were to change over time, this could simultaneously lead to higher frequency of price adjustment at the micro level and slower price adjustment at the macro level. In contrast, looking at the frequency of price changes through the lens of perfect-reset models would lead to the completely opposite conclusion: higher micro frequency means lower non-neutrality. Overall, once we entertain the possibility of imperfect reset, higher average frequency does not automatically imply higher aggregate price flexibility: the reason behind the increase in frequency matters.

5 Measurement

The preceding section shows that adding imperfect reset can weaken the connection between individual price adjustment and aggregate price flexibility. In this section, I turn to a different but related question: how does the potential presence of imperfections change our *measurement* of monetary non-neutrality from price microdata? The main result is a generalization of the sufficient statistics result in Alvarez et al. (2016a): I show that regression coefficients similar to the ones studied in Section 2 become new sufficient statistics. Applying the result to the UK data, I find that accounting for imperfections raises measured non-neutrality by around 50%.

5.1 Measurement in the Simple Endogenous Timing Model

To start, consider the endogenous timing model in Section 4. Assume now that we want to infer $\mathcal{M}'(0)$ from the microdata. In particular, we calibrate the structural parameters to match a given frequency N and average (absolute) size of price changes $\bar{x}$, and then compute the objects of interest. I compare these objects in two scenarios: a standard menu cost model, where F is a point mass at 0, and a general version with imperfect reset, where F is non-degenerate. I show below that, if the distribution of reset gaps is arbitrary, we can obtain $\mathcal{M}'(0) \to +\infty$ while still matching the same average frequency and sizes of price changes. The next proposition formalizes that point.

Proposition 8. ***Unbounded Non-Neutrality.*** *Fix an average frequency $N^{data} > 0$ and a size of price changes $\bar{x} > 0$. For any bound $C \in \mathbb{R}$, there exists a symmetric distribution F with $supp(F) \subset (-\bar{x}, \bar{x})$, and a variance of shocks σ^2 such that: i) the average frequency is $N = N^{data}$, and ii) $\mathcal{M}'(0) > C$.*

Proof. Take F to be discrete with $\text{Prob}(x^* = \alpha\bar{x}) = \text{Prob}(x^* = -\alpha\bar{x}) = (1-\varepsilon)/2$ with

$\alpha \in (0,1)$ and $\text{Prob}(x^* = 0) = \varepsilon$. Using the formulas in Proposition 4 we have:

$$\mathcal{M}'(0) = \frac{1}{6N} \frac{(1-\varepsilon)((1-\alpha^2))^2 + \varepsilon}{[(1-\varepsilon)((1-\alpha^2)) + \varepsilon]^2}.$$

Taking $\alpha \to 1$ and $\varepsilon \to 0$ in a way such that $1 - \alpha^2 = o(\varepsilon)$, $\mathcal{M}'(0) = \frac{1}{6N}\frac{1}{\varepsilon} + O(\varepsilon) \to +\infty$. Note also that $\sigma^2 = N\bar{x}^2((1-\varepsilon)((1-\alpha^2)) + \varepsilon) \to 0$. □

The intuition is that, by choosing F appropriately, we can have $\sigma^2 \to 0$ and still match the average frequency of price changes. As $\sigma^2 \to 0$, the *average* frequency of price changes is entirely driven by firms that reset, land very close to the bands, and reset again. However, in this construction, a small mass of firms (ε) does land at $x^* = 0$. Since $\sigma^2 \to 0$, these firms will take an arbitrarily long time to adjust after an aggregate shock. In the limit, it takes them infinitely long, so they never adjust, and thus the CIR diverges. This proposition is important because it shows that when reset is imperfect, measured monetary non-neutrality is no longer bounded by that of a Calvo model, $1/N$.

While this exercise reveals that imperfect reset can materially change inference, the result was derived under very particular assumptions about the timing of price changes and the distributions of reset gaps. For this reason, I extend the model to allow for more generality along those two margins, and also consider how additional price microdata moments can help to identify monetary non-neutrality.

5.2 Random Menu Costs and Multiproduct Firms

Since monetary non-neutrality crucially depends on assumptions about how the timing of price changes works, I describe a more general model that nests several popular pricing models in the spirit of Alvarez et al. (2016a), with or without imperfections. The control cost model extends naturally to multiproduct firms with random menu costs, following Alvarez and Lippi (2014) and Alvarez et al. (2021). A firm sells n products, with total losses $By = B\sum_{i=1}^{n} x_i^2$. When prices are not adjusted, gaps evolve as $dx_i = \sigma dW_{i,t}$, where $W_{i,t}$ are Brownian motions, independent across i. The firm always has the option to readjust the prices of all its products by paying a fixed cost $\Psi \leqslant +\infty$. Additionally, as in a random menu cost model, opportunities to change prices at lower costs arrive with arrival rate κ. Once an opportunity arrives, a cost ψ is drawn from a distribution $G(\psi)$ with support $[0, \Psi)$. When the firm decides to change prices, it *must* change prices for all products. It can be shown that the problem of *when* to reset depends only on y. At reset, the firm chooses a

control-cost-optimal distribution for each product's gap. Appendix B.3 provides full details.

Several observations are in order. First, the solution to this model implies a generalized hazard function $\Lambda : [0, \bar{y}] \to \mathbb{R}^+$, an inaction threshold $\bar{y} \in \mathbb{R}^{++}$ and a reset distribution F, nesting a broad class of models. As I show below, these objects are all that matters: their underlying microfoundations are irrelevant. As a corollary, this implies that one could have microfounded $(\Lambda, \bar{y})$ in other ways, such as using information costs (Woodford, 2009; Costain and Nakov, 2019; Stevens and Morales-Jiménez, 2025). Second, multiproduct firms with random menu costs can generate flat adjustment hazards, which I show in Section 5.6 are required to match the data within the class of models I consider.

5.3 Object of Interest and Assumptions

Let $m(x_1, \ldots, x_n) = -\mathbb{E}\left[\int_0^\tau \sum_{i=1}^n x_i(t)dt | x(0) = x\right]$. Following Alvarez et al. (2016a), the cumulative output for a shock of size δ is:

$$\mathcal{M}(\delta) = \frac{1}{n} \int \cdots \int m(x_1 - \delta, \ldots, x_n - \delta) g(x_1, \ldots, x_n) dx_1 \cdots dx_n.$$

where $g(x_1, \ldots, x_n)$ denotes the density of price gaps. I focus on small shocks, so the object of interest is $\mathcal{M}'(0)$, which I refer to as the CIR. I describe the assumptions about the behavior of price gaps below.

Assumption 1. ***Random Walk and Adjustment Policy.*** *The price gap of each product evolves as $dx_{i,t} = \sigma dW_{i,t}$ unless adjusted. Firms adjust with hazard $\Lambda(y)$, or whenever $y = \bar{y}$. The Brownian processes $W_{i,t}$ are independent across all (product,firm) pairs.*

This assumption states that, if not reset, price gaps follow a Brownian motion. Furthermore, the timing of adjustment is fully described by a hazard Λ and barrier $\bar{y} > 0$. The next assumption imposes some structure on Λ.

Assumption 2. ***Non-Decreasing Hazard.*** *$\Lambda(y)$ is non-decreasing and either $\bar{y} < +\infty$ or $\sup_y \Lambda(y) > 0$.*

This assumption states that the probability of changing prices must be weakly higher if prices are more misaligned, at least on average. This is a standard implication of state-dependent models. The second part only states that the probability of changing prices at some point is positive. Assumptions 1 and 2 are essentially equivalent to the ones in Alvarez et al. (2016a) and Alvarez et al. (2021). As the next assumption shows, the key difference is what happens at reset.

Assumption 3. ***Independent Reset Gaps.*** *Reset gaps are given by realizations of a random variable X_i^* with CDF $F(x)$. The draws of X_i^* are independent across products, firms, and time, and independent of the Brownian shocks of all firms.*

This assumption states explicitly how resetting prices works: the firm resets by getting n independent draws from a random variable with CDF F. These draws are completely independent of each other, and of any other random variable. This is the key generalization relative to the literature: this assumption allows for reset gaps to be stochastic. The next assumption puts some restrictions on F.

Assumption 4. ***Non-Systematic Imperfections and Regularity.*** *F is a symmetric distribution around 0, with bounded fourth moments, $\int x^4 dF(x) < +\infty$.*

The first part essentially says that mistakes are non-systematic. Note that it automatically implies that $\mathbb{E}[x_i^*] = 0$, i.e., firms close their price gaps *on average, but generally not in each realization.* The distribution is symmetric, so erring on the positive or negative side is equally likely. This assumption is satisfied in control cost models with no trend inflation and symmetric profit functions. The second part is imposed for regularity: bounded fourth moments are required to ensure that some of the objects that show up in the formulas actually exist. Given these assumptions I define a set of *admissible objects* as:

Definition 1. ***Admissible Objects.*** *A set of admissible objects is a tuple $(\sigma^2, n, \bar{y}, \Lambda, F)$ such that $\sigma^2 \in \mathbb{R}^+$, $n \in \mathbb{N}$, $\bar{y} \in \mathbb{R}^{++} \cup \{+\infty\}$,$\Lambda : \mathbb{R}^+ \to \mathbb{R}^+$, F is a CDF and (Λ, F) satisfy Assumptions 2-4.*

Scope. Note that under our definition, all models in Alvarez et al. (2016a, 2021) are included, which are particular cases of the model introduced in Section 5.2. These papers consider a quite general class of models, which includes the time-dependent models in Calvo (1983), Taylor (1979) and Reis (2006) (as limit cases), random menu cost models, and multiproduct models. All these correspond to different specifications of $(\sigma^2, n, \bar{y}, \Lambda)$, but they all share one thing in common: F is a degenerate distribution at 0, so firms perfectly close their price gaps in each realization every time they reset prices. Our definition of objects crucially relaxes this assumption, by allowing for randomness in the point of reset. Note also that the control cost models introduced in Section 5.2 imply additional restrictions between Λ and F beyond what Assumptions 2-4 impose. My sufficient statistics result does not rely on this additional structure. Thus, any other pricing model that satisfies Assumptions 1-4 is also covered.

5.4 Only Match Standard Moments? Anything Goes

Under perfect reset, Alvarez et al. (2016a) prove that $\mathcal{M}'(0)$ is exactly pinned down by two observable statistics: the frequency of price changes N, and the kurtosis of the distribution of price changes, Kurt(Δp) as $\mathcal{M}'(0) = \text{Kurt}(\Delta p)/(6N)$. Since Proposition 8 showed that monetary non-neutrality can be unbounded for any given frequency, one may wonder if using additional information can restore identification. Proposition 9 shows that the answer is negative. In fact, allowing for imperfections renders non-neutrality completely unidentified if we consider the frequency and kurtosis alone.

Proposition 9. ***Anything Goes.*** *Let $A_i = (\sigma_i^2, n_i, \bar{y}_i, \Lambda_i, F_i)$ be a set of admissible objects. Denote the implied frequency and kurtosis of price changes by N_i and $Kurt_i(\Delta p)$, and denote the CIR to a marginal shock by CIR_i. Then for any $Kurt(\Delta p) > 2$, $N > 0$ and for any bound $\bar{B} > 0$ there exist A_1 and A_2 such that: (i)$N_1 = N_2 = N$ and $Kurt_1(\Delta p) = Kurt_2(\Delta p) = Kurt(\Delta p)$; and (ii) $CIR_1 < -\bar{B} < 0 < \bar{B} < CIR_2$.*

Proof. See Appendix C.6 for the details. Intuitively, we have enough degrees of freedom to set $\sigma^2 = 0$, alter the CIR by varying (Λ, F), but still match any desired frequency and kurtosis (above 2). □

The proposition says that we can always find admissible objects such that we match any frequency N and kurtosis of price changes (as long as it is above 2), and yet the non-neutrality can be arbitrarily large or small, even negative. This shows that the assumption of perfect reset in Alvarez et al. (2016a) is crucial. Intuitively, with imperfect reset, we could observe a thick or thin tail of price changes due to these imperfections, and thus the tight link between kurtosis and non-neutrality completely breaks down. The main message of Proposition 9 is that the restrictions in Assumption 4 are too mild to make any progress on the question of non-neutrality: The result underscores the need to further discipline the distribution of reset gaps. In what follows, I show how to do that using additional measurable statistics.

5.5 Generalized Sufficient Statistics

I propose restricting F by means of measurable statistics. In particular, the following proposition shows that we can measure key moments of the reset gap distribution using predictability regressions, such as the ones discussed in Section 2.

Proposition 10. ***Predictability.*** *Consider the following pooled OLS regressions:*

$$\Delta p_{i,n} = \alpha_1 + \beta_1 \Delta p_{i,n-1} + u_{1,i,\tau_{i,n}}, \quad (\Delta p_{i,n})^2 = \alpha_2 + \tilde{\beta}_2 (\Delta p_{i,n-1})^2 + u_{2,i,\tau_{i,n}},$$

$$\textit{and} \quad \Delta \tau_{i,n} = \alpha_3 + \tilde{\beta}_3 (\Delta p_{i,n-1})^2 + u_{3,i,\tau_{i,n}}.$$

Then letting $\beta_2 = \tilde{\beta}_2 - \tilde{\beta}_3 N\, Var(\Delta p)(1 + 2\beta_1)$,

$$\beta_1 = -\frac{Var(x^*)}{Var(\Delta p)}, \quad \beta_1 \in [-\frac{1}{2}, 0], \quad \textit{and} \quad \beta_2 = (\beta_1)^2 \frac{[Kurt(x^*) - 1]}{[Kurt(\Delta p) - 1]}. \tag{5.1}$$

Proof. See Appendix C.7. □

The proposition shows that predictability regressions are *directly* informative about the size of imperfections: if they are prevalent, firms will tend to correct them in the next price change. This generates $\beta_1 < 0$, as explained in Section 2. Large imperfections tend to be followed by large corrections, so squared price changes are positively correlated, and how often these extreme realizations occur depends on $\text{Kurt}(x^*)$. Note that Assumptions 1 - 3 are crucial for this: under those, the only source of predictability is randomness in x^*, and thus these regressions exactly identify the key moments of the reset gap distribution.

Once we measure the key moments of F, the following proposition shows that we can restore partial identification of non-neutrality.

Proposition 11. ***Generalized Sufficient Statistics Result.*** *Assume* $\sigma^2 > 0$. *The CIR is bounded as* $\frac{L}{N} \leqslant CIR \leqslant \frac{U}{N}$ *with:*

$$U = \frac{(1 - 2\beta_2) Kurt(\Delta p) + 2\beta_1(2\beta_1 + 3) + 2\beta_2}{6(1 + 2\beta_1)^2} \textit{ and } L = \frac{\beta_1}{(1 + 2\beta_1)} + U.$$

Proof. See Appendix C.7. □

Given the negative result in Proposition 9, Proposition 11 is one of the central results of the paper. It shows how to map the moments in Proposition 10 directly to monetary non-neutrality. Note that if there is no predictability, $\beta_1 = \beta_2 = 0$, and that would imply $U = L = \text{Kurt}(\Delta p)/6$, so we revert exactly to the result in Alvarez et al. (2016a). Once we add imperfect reset, an extra term appears, which can be bounded.

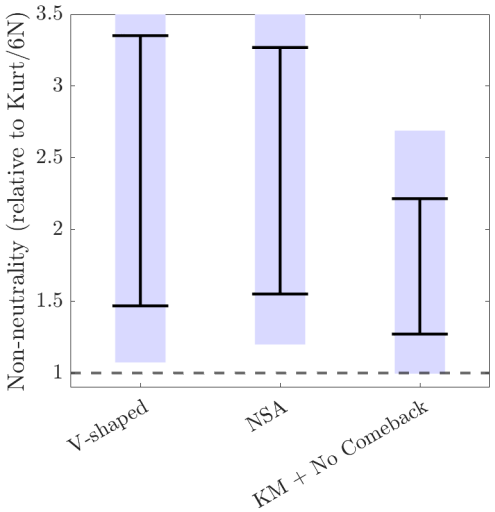
It is important to note that the lack of point identification is mainly a consequence of my focus on a small set of moments. Online Appendix OA.B.5 shows that, under some regularity assumptions, it is possible to identify F using higher-order moments of the joint distribution

of consecutive price changes and durations. Furthermore, for the single-product case, I show that given F, Λ is point-identified from the distribution of price changes, using a similar logic to Alvarez et al. (2021). However, implementing this construction in practice requires estimating higher-order moments and recovering entire functions, which can be quite sensitive to measurement assumptions and sampling error. The generalized sufficient statistics result directly bounds monetary non-neutrality using a small set of observable moments and applies to a broader class of models, so I use it as the basis for my quantification.

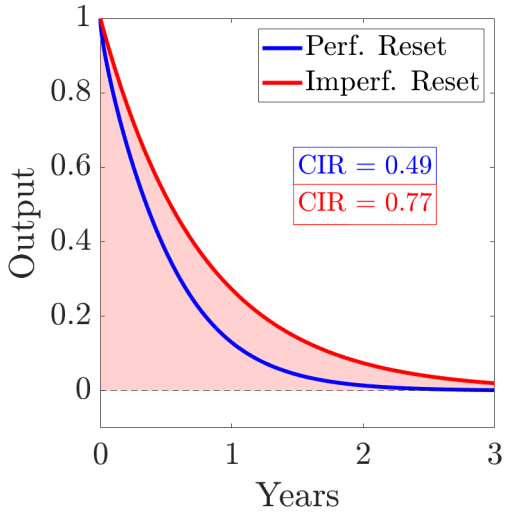
Quantification. The sufficient statistics result requires matching $N, \text{Kurt}(\Delta p)$,β_1 and β_2. I estimate them using the UK data, for which inflation is relatively low and stable. For the frequency, I use the CPI-weighted average frequency of price changes, which yields $N = 1.3$ price changes per year. I compute the kurtosis using standardized price changes (discussed in Section 3). This is due to the well-known bias that heterogeneity introduces in the kurtosis (Alvarez et al., 2016a). I obtain $\text{Kurt}(\Delta p) = 3.81$. The point estimates of predictability coefficients are $\beta_1 = -0.353$, $\beta_2 = 0.193$. Using these values, the lower bound implies that the cumulative impulse response is around 0.72. This is around 50% higher than what we would obtain using the formula that ignores imperfections, $\text{Kurt}(\Delta p)/(6N) = 0.49$.

In order to quantify uncertainty, I estimate joint confidence sets via bootstrap. Appendix A.3 contains the implementation details. For each draw, I compute L and U, and then divide them by $\text{Kurt}(\Delta p)/6$ to obtain the factor by which measured non-neutrality changes. Figure 5.1a presents the resulting bounds for different sales filters with their corresponding confidence intervals. Using the estimates corresponding to the V-shaped filter implies around 50% higher non-neutrality than the standard formula, and this is statistically significant at 90% confidence. However, we also see that the estimate is relatively noisy. The bounds are virtually unchanged when I use Filter A in Nakamura and Steinsson (2008). For the Kehoe and Midrigan (2015) filter, even after applying it, there is still a nontrivial number of prices that return exactly to the same level after a price change, which distorts the estimated coefficients and thus the bounds. Once we eliminate all comeback prices from the filtered series, the figure shows we still get non-neutrality that is around 27% larger than the benchmark, although still with significant uncertainty.

It is also important to note that inference based on the kurtosis relies on the exact distributional assumptions for underlying shocks, as well as on measuring the kurtosis precisely, which is a notoriously difficult task. I show in Appendix B.4 robustness to different measurement assumptions, such as treatment of outliers and trimming of small price changes.



(a) Estimated bounds for different sales filters, reported relative to $\text{Kurt}(\Delta p)/(6N)$. Shaded areas are 90% confidence intervals.



(b) Blue: Calvo+ with perfect reset. Red: baseline model in Table 2.

Figure 5.1

Inferred non-neutrality always increases, but the point estimates for the bounds are sensitive to these measurement assumptions: for the lower bound, the increase relative to their perfect-reset counterparts ranges from +20% to +100%. Overall, the finding that inferred non-neutrality increases is robust, but the exact magnitude is affected by the standard measurement issues in this literature.

The reader may be also concerned about how heterogeneity across categories may bias my conclusions. Indeed, Appendix B.5 documents significant heterogeneity in the key moments across product categories, and extends the setup to account for it. In this extension, $CIR = \sum_{j=1}^{J} \alpha_j CIR_j$, where CIR_j are the item-level cumulative IRFs, and α_j are expenditure shares. We can bound this alternative formula directly from the data. The main results are unchanged by heterogeneity: accounting for imperfect reset still significantly increases estimated non-neutrality under this formula relative to the one that ignores imperfections.

5.6 Structural Models

The sufficient statistics result in Section 5.5 shows that, at the point estimate of the lower bound, monetary non-neutrality is around 50% higher once imperfections are accounted for. However, the analysis so far leaves several questions open. First, it is unclear why the non-

neutrality is larger. What properties should a structural model have to jointly match those statistics? How do these properties differ from those of a structural model with perfect reset that only matches frequency and kurtosis? How costly are the mistakes required to fit the data? Furthermore, how robust are the conclusions to the assumption that all reversals come from imperfections? In this section, I answer these questions by estimating different versions of the multiproduct model in Section 5.2. Online Appendix OA.B.6 contains additional discussion.

5.6.1 Estimation Details

I use the values for $N, \text{Kurt}(\Delta p)$, β_1 and β_2 described in Section 5.5 as empirical targets. I also add $\text{Var}(\Delta p)$, which is around $(0.147)^2$. Given that these models can be rescaled to match any desired frequency and variance leaving the other three moments unaltered (Alvarez and Lippi, 2014; Alvarez et al., 2021), in all cases I make sure that the estimated models perfectly match N and $\text{Var}(\Delta p)$.

	Data	Calvo ($n \to +\infty$, imperfect)	Calvo+ ($n = 1$, perfect)	Calvo+ ($n = 1$, imperfect)
N	1.300	1.300	1.300	1.300
Std. Dev.	0.147	0.147	0.147	0.147
Kurtosis	3.809	3.811	3.809	3.715
β_1	-0.353	-0.354	0.000	-0.489
β_2	0.193	0.188	0.000	0.318
GMM loss		0.280	2.194×10^3	328.875
Bounds	[0.712,1.634]	[0.769,1.704]	[0.488,0.488]	[-14.040,3.478]
CIR		0.769	0.488	0.377
CIR (relative)		1.574	1	0.791
Cost of imperf.		1.141	0	0.081

Table 2: GMM denotes generalized method of moments. The first set of rows shows the model-implied moments vs. the data. The second set shows the identified set, $\mathcal{M}'(0)$ and $\mathcal{M}'(0)/(\text{Kurt}(\Delta p)/(6N))$, where the frequency and kurtosis are computed from each model. The last row shows the cost of imperfections as a percentage of total revenue.

I consider a multiproduct model with control costs as in Section 5.2. I set the discount rate to $\rho = 0.04$. Under the standard second-order approximation of constant elasticity of substitution (CES) profit functions, the constant is given by $B = \xi(\xi - 1)/2$. I set $\xi = 5$, a standard value in the literature. I parametrize f as a generalized normal distribution, with density function $f(x; b, d) = \frac{d}{2b\Gamma(1/d)} \exp(-(|x|/b)^d)$. By choosing b and d, we can control

the variance and kurtosis of this distribution. I set $\theta \to +\infty$ so that the distribution of X^* equals f.[11] I assume that $G(\psi)$ has a single mass point at $\psi = 0$, as in a Calvo+ model (Nakamura and Steinsson, 2010): firms get free opportunities to reset prices, although there still are reset imperfections.

My baseline calibration uses a limit case of this model ($n, \Psi \to \infty$), which produces a very good fit. I show in Online Appendix OA.C.4 that these assumptions in conjunction with $\theta \to +\infty$ imply that the hazard rate is constant as in a Calvo model $\Lambda(y) = \kappa$ for all y. I set $\kappa = N$ to match the frequency of price changes, and choose parameters $\Omega = (\sigma^2, b, d)$ to minimize a standard generalized method of moments (GMM) quadratic loss: $\ell(\Omega) = \hat{m}(\Omega)' W \hat{m}(\Omega)$ where $\hat{m}(\Omega)$ is a vector that contains the difference between $\mathrm{Var}(\Delta p), \mathrm{Kurt}(\Delta p), \beta_1, \beta_2$ and their model counterparts. For W, I use a block-diagonal matrix: I set the $(1,1)$ element to a large number, to enforce a perfect match of $\mathrm{Var}(\Delta p)$, and the lower 3×3 block is set to the inverse of the covariance matrix of the other three moments, estimated via bootstrap (see Appendix A.3 for implementation details). For comparison, I also present results with $n = 1$, with and without perfect reset.[12]

5.6.2 Results

Table 2 presents the fitted moments, as well as several results of interest. First, the baseline model ($n = +\infty$) fits the data moments quite well. Since we have three parameters but four moments, the reported GMM loss equals the value of the Sargan–Hansen J-statistic, which under the null of correct specification has an asymptotic distribution of χ^2 with one degree of freedom. Its value is well below standard critical values at conventional levels, which means that the null of correct model specification is not statistically rejected.

The baseline model implies around 50% higher non-neutrality compared to the $n = 1$ model that ignores imperfections. The reason is simple: the underlying hazard is constant so non-neutrality is equal to that in the Calvo model, $1/N$. The third column of the table shows what happens when I fit a standard Calvo+ model (Nakamura and Steinsson, 2010) ignoring imperfections. There, I would infer meaningful state dependence in pricing, which reduces non-neutrality. Figure 5.1b presents the output IRFs for both models. It is clear

[11] Note that θ is not identified: three parameters regulate the distribution of reset gaps (b, d, θ), but just two moments $(\mathrm{Var}(X^*), \mathrm{Kurt}(X^*))$ identify them. I prefer fixing $\theta \to +\infty$ to adding more moments because the goal of the exercise is to show what features are *necessary* to match the sufficient statistics.

[12] For $n = 1$ with imperfect reset, the parameter vector is $(\sigma^2, b, d, \kappa, \Psi)$. I add N as a moment to be matched, and ensure its fit is perfect. With perfect reset, parameters are (σ^2, κ, Ψ).

that the baseline model has higher non-neutrality. Thus, this exercise provides a simple explanation of the magnitudes obtained in Section 5.5: the underlying hazard rate can be totally flat, yet we could observe a lower kurtosis due to reset imperfections.

Table 2 also shows what happens when we estimate a version of the model with imperfections and $n = 1$. The minimum loss materially worsens: in order to fit the kurtosis, we need to rely on very large imperfections, which are at odds with the data. When we set $n = 1$, a strict inaction region opens up for small values of $|x|$, which pushes for a bimodal distribution for price changes and lowers the kurtosis. Intuitively, firms that reset and land close to 0 avoid resetting again even if it is free, because potential imperfections may leave the firm worse off. Thus, having large n and flat hazards delivers the best fit for the joint set of moments estimated in the UK data, within the class of models I consider.

This finding seems to conflict with Section 4, which emphasized the interaction between imperfect reset and state dependence as the source of higher non-neutrality. This particular dataset rules out that combination: strong state dependence would generate a distribution of price changes with too little kurtosis, which is not what we observe. However, this apparent contradiction ignores that the question being asked here differs from the one in Section 4: in the present section, imperfections increase *measured* non-neutrality, because they meaningfully change the kurtosis implied by the model. The broader point (under imperfect reset, price changes are allocated "worse" and thus aggregate price adjustment is slower) still applies, but through a different channel: imperfect reset makes us *measure* weaker selection in which firms change prices, which generates high non-neutrality.

Cost of Imperfect Reset. The bottom row of Table 2 shows the costs of imperfect reset, measured in terms of revenue as $(V(0) - V^*)/\xi$. The estimated cost is around 1% of revenue. This cost is in line with direct estimates of the costs of changing prices by Levy et al. (1997) and Zbaracki et al. (2004), which find values of 0.7% and 1.2% respectively. Overall, this shows that imperfections may impose only moderate costs on firms, as in Section 4.2.

Comparison to the Literature on Pricing with Control Costs. It is useful to compare these results with the existing literature that uses control costs in pricing models (Costain and Nakov, 2015, 2019; Stevens and Morales-Jiménez, 2025). These papers do *not* discipline the magnitudes by matching the sufficient statistics; they instead match different moments of the price change distribution. Costain and Nakov (2015) develops a model that is quite close to that in Section 4.2. The non-neutrality implied by that model is almost equal

to that of Golosov and Lucas (2007), even though the model can generate higher kurtosis than it. Therefore, accounting for imperfections would *lower* measured non-neutrality in this case. This conclusion rests on the particular kurtosis of price changes generated by this model (2.3), which is below what most studies find in CPI datasets. Costain and Nakov (2019) develops a generalized version of the model that allows for a smooth hazard function. In their calibrated model, imperfect reset does not matter much because the calibrated size of imperfections is small. On the other hand, Stevens and Morales-Jiménez (2025) find that control costs are a key determinant of price stickiness: the baseline model has a degree of non-neutrality close to that of the Calvo model, but if they eliminate control costs, money is essentially neutral. Furthermore, their model is able to produce very high values of the kurtosis: their benchmark calibration matches a value of 11. This disparity in findings is a practical illustration of the result in Section 5.4: if imperfections are not independently disciplined, measured non-neutrality can be higher or lower, and significantly so.

My analysis reconciles both results. The models in Costain and Nakov (2015, 2019) generate a kurtosis that is below the empirically relevant values for CPI data. Furthermore, Costain and Nakov (2019) implies small imperfections, which would be inconsistent with the observed magnitude of reversals. On the other hand, the model in Stevens and Morales-Jiménez (2025) generates a very large kurtosis and high state dependence by having very large imperfections. In my case, generating high kurtosis in this way is not possible, since such large imperfections would generate implausible values of β_1 and β_2. Overall, this shows that the predictability coefficients introduced in this paper tightly discipline the magnitude of imperfections, and thus clarify their implications for measured monetary non-neutrality.

Trend Inflation. Under some symmetry assumptions for the control cost functional $\mathcal{C}$, changes in trend inflation have only second-order effects in the model, exactly as in Alvarez and Lippi (2014) (see Online Appendix OA.C.3). Thus my analysis also hold for sufficiently low trend inflation. However, the baseline model cannot reproduce the extent to which the frequency of price changes rises with trend inflation once we consider moderate changes. This is an issue that state-dependent models with perfect reset also face: it is hard to write down a model that can *quantitatively* match both (i) kurtosis at low inflation and (ii) the extent to which the frequency rises with trend inflation (Blanco et al., 2024c,b). While I show that a model with flat hazards fits the micro moments quite well, it is unclear whether a more general model with imperfections and meaningful state-dependence can fit the data similarly well. I leave this question for future work.

5.7 Mean Reversion in Desired Prices

While the hypothesis that reversals are purely driven by mean reversion is rejected empirically due to the arguments in Sections 2 and 3, it is still of interest to see how the results on measured non-neutrality would differ under this alternative source of reversals. Assume that desired prices are driven by a variable z, which follows $dz_{i,t} = -\eta z_{i,t}dt + \sigma dW_{i,t}$, with $\eta > 0$. The timing of price changes is assumed to be Calvo: prices get reset with a constant hazard κ. This assumption helps match the values of kurtosis seen in the data and keeps the model tractable. When prices are reset, the size of price changes is given by:

$$\Delta p_{i,n} = \frac{\rho + \kappa}{\rho + \kappa + \eta}(z_{i,\tau_{i,n}} - z_{i,\tau_{i,n-1}}) + \varepsilon^*_{i,n} - \varepsilon^*_{i,n-1}, \tag{5.2}$$

where $\varepsilon^*_{i,n}$ are mean-zero, i.i.d. with some distribution, and independent of all other shocks.[13] In this simple setup, it is possible to solve analytically for all moments of interest, see Online Appendix OA.C.5.

Calibration. I maintain $\rho = 0.04$, and target the same values of data moments described in Section 5.6.1. Given η, I choose parameters to match $N, \text{Var}(\Delta p)$, β_1 and β_2 exactly, leaving $\text{Kurt}(\Delta p)$ as an untargeted moment. I do this instead of full GMM estimation to keep the comparison as clean as possible. The key parameter here is the degree of mean reversion, η. I will consider two approaches: external and internal calibration.

For external calibration, I will follow most of the literature and assume that it comes from variations in marginal cost.[14] I use estimates of the persistence of the process for idiosyncratic marginal costs from Gagliardone and Tielens (2026) (Table 4), who measure a proxy for marginal costs directly from the data. They estimate a monthly autoregressive coefficient of 0.942, which implies a half-life of cost shocks of around 12 months.[15]

For internal calibration, I choose η to match β_1 under perfect reset, ignoring β_2. This yields an implied monthly AR(1) coefficient around 0.77. This exercise is useful for several reasons. First, marginal costs may not be the only source of variation in desired prices. This exercise allows us to assess what properties desired prices must have to rationalize the

[13] I use this notation because, in the presence of mean reversion, the optimal price gap upon reset is not zero. Thus, ε^* captures the pure imperfection component at reset, which is taken to be mean zero.

[14] The microfoundations in Online Appendix OA.D assume CES preferences. This implies that desired markups are constant, and thus all variation in desired prices comes from marginal costs.

[15] For a monthly AR coefficient $\hat{\rho}_m$, we can convert it to η (which is in annual terms) as $\hat{\eta} = -12\log(\hat{\rho}_m)$.

ρ_m	1	0.942	0.771
η	0	0.717	3.118
Half-life (months)	$+\infty$	11.601	2.668
Monthly innovation to z (%)	2.623	6.247	29.656
Std(ε^*) (%)	8.727	7.658	0
Kurt(ε^*)	5.405	6.873	-
Kurt(Δp)	3.859	3.869	3.518
Bounds	$[0.769, 1.692]$	$[0.779, 1.701]$	$[0.875, 1.797]$
CIR	0.769	0.769	0.769
CIR (relative to Kurt(Δp)/(6N))	1.555	1.551	1.706

Table 3: Model-implied moments for different degrees of mean reversion. Monthly innovation in z is computed as $\sigma/\sqrt{12}$, and reported in percentage points. The last column does not match β_2.

observed degree of predictability, regardless of whether desired prices vary due to costs or demand. Second, I can see how far off my sufficient statistics result would be in the extreme case in which predictability is solely driven by mean reversion in desired prices.

Results. Table 3 presents the results. The first column shows what happens if z is assumed to follow a random walk.[16] The second column shows the calibration matching the persistence of measured marginal costs taken from Gagliardone and Tielens (2026). This meaningfully increases the size of monthly shocks required, to around 6% *per month*. The implied standard deviation of imperfections decreases, although it remains nontrivial at around 7.6%. The kurtosis of imperfections meaningfully increases, which is needed to match β_2. The last column shows what happens if we assume perfect reset and use the observed β_1 to calibrate the degree of mean reversion. This implies that shocks have a half-life of around 2.7 months. In order to match the observed variance of price changes, we need very large shocks to desired prices, with a standard deviation of around 30% *per month*. The kurtosis does not change much.

Since the model-implied Kurt(Δp) changes little even if we match β_1 only with mean reversion in desired prices, the takeaway remains: we could see a Calvo model with lower kurtosis than expected, and thus the canonical formula would substantially understate the true non-neutrality. This indicates that β_1 may remain a relevant statistic for non-neutrality

[16]The results are slightly different from Table 2 because in this case, I choose parameters to perfectly match β_1 and β_2 by leaving the kurtosis untargeted.

even in a model with perfect reset. Furthermore, the sufficient statistics result correctly predicts that inferred non-neutrality is significantly higher than under standard inference even though this model falls outside the covered class.

6 Conclusion

I present evidence in favor of models where price-setters make non-systematic mistakes. I then study their consequences for monetary non-neutrality with two different exercises. First, I show that adding mistakes to an otherwise standard economy can simultaneously increase the *micro* adjustment frequency and slow down aggregate adjustment to monetary shocks. This is because mistakes can change the composition of firms that adjust after the aggregate shock. Second, I show that when these mistakes are possible, standard price microdata moments are no longer sufficient for measuring monetary non-neutrality. I show how to recover identification using the strength of price reversals. I show that accounting for imperfections increases measured non-neutrality by around 50%, bringing it in line with a Calvo model.

More broadly, my results imply that idiosyncratic mistakes do not necessarily wash out in aggregation: they can have important macroeconomic consequences in environments where decisions are lumpy (such as price- and wage-setting, investment and durable-purchase decisions), because they can affect both the composition of agents that adjust after an aggregate shock and how that composition is measured using microdata. In those environments, my analysis suggests that models where agents are right *on average* will have different aggregate implications than models where agents are right *every single time.*

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A Additional Empirical Results

A.1 Empirical Estimation of Sufficient Statistics

In this subsection, I present additional details on the empirical estimation of sufficient statistics derived in Section 5.5, as well as robustness checks under several alternative measurement assumptions. I estimate the following regression specifications:

$$\begin{aligned} \Delta p_{i,n} &= \text{Fixed Effects} + \beta_1 \Delta p_{i,n-1} + \text{Error} && \text{(A.1)} \\ (\Delta p_{i,n})^2 &= \text{Fixed Effects} + \tilde{\beta}_2 (\Delta p_{i,n-1})^2 + \text{Error} && \text{(A.2)} \\ \Delta \tau_{i,n} &= \text{Fixed Effects} + \tilde{\beta}_3 (\Delta p_{i,n-1})^2 + \text{Error}, && \text{(A.3)} \end{aligned}$$

where $\Delta \tau_{i,n} = \tau_{i,n} - \tau_{i,n-1}$ is the duration of the current spell, for several sets of fixed effects.

Table 4 shows the estimated coefficients for different sets of fixed effects: the first column only includes a constant, the second column includes time fixed effects for both the time of the current and last price change (i.e., $\sum_t (\gamma_t \mathbf{1}\{\tau_{i,n-1} = t\} + \lambda_t \mathbf{1}\{\tau_{i,n} = t\})$), the third column interacts these with item-level fixed effects (i.e., $\sum_{\{t,\theta\}} (\gamma_{\theta,t} \mathbf{1}\{\tau_{i,n} = t, \theta_i =$

	OLS	OLS	OLS	OLS
β_1	-0.309***	-0.316***	-0.355***	-0.353***
	(0.008)	(0.008)	(0.007)	(0.008)
$\tilde{\beta}_2$	0.411***	0.400***	0.203***	0.190***
	(0.011)	(0.012)	(0.011)	(0.013)
$\tilde{\beta}_3$	-0.705***	-0.648***	-0.196***	-0.178***
	(0.116)	(0.105)	(0.026)	(0.035)
$\beta_{1,2}$	-0.018*	-0.021**	-0.017***	-0.008
	(0.010)	(0.009)	(0.005)	(0.007)
Time FEs	no	yes	no	yes
Time-by-item FEs	no	no	yes	no
Time-by-cell FEs	no	no	no	yes

Table 4: Estimation Results for Equations (A.1)-(A.3) and (3.2) for different sets of fixed effects. The number of observations is $n = 789,069$ in all specifications. "Time FEs" means fixed effects by the times of the price changes on both sides of the regression. "Time-by-item FEs" interacts these with item fixed effects, whereas "Time-by-cell FEs" does it with cell fixed effects instead. Standard errors are clustered by item.

$\theta\} + \lambda_{\theta,t}\mathbf{1}\{\tau_{i,n-1} = t, \theta_i = \theta\}))$ and the last one interacts them with cell fixed effects instead (similar to the previous one, but the type is now the cell of the unit). The sign, magnitude and significance are quite consistent across specifications: price changes feature a consistent *reversal* pattern. For $\tilde{\beta}_2$ and $\tilde{\beta}_3$, we see that fixed effects do not alter the signs, but are quantitatively relevant. This likely reflects the well-known heterogeneity of absolute sizes and frequency of price changes across different categories (Alvarez et al., 2021). Once we add more granular fixed effects, we eliminate that variation, and we obtain lower $\tilde{\beta}_2$ and $|\tilde{\beta}_3|$.

Finally, the point estimate of $\beta_{1,2}$ does not vary much with fixed effects, although its significance does. Given this observation, I formally test the prediction that $\beta_{1,2} = \beta_1(1+2\beta_1)$. Letting $h = \beta_1(1+2\beta_1) - \beta_{1,2}$, under the null hypothesis of $h = 0$, the asymptotic distribution of this statistic is normal, with a variance that can be computed using the Delta method. The estimated variance for this statistic is $\hat{\text{Var}}(\hat{h}) = (1 + 4\hat{\beta}_1)^2\hat{\text{Var}}(\beta_1) + \hat{\text{Var}}(\beta_{1,2}) - 2(1 + 4\hat{\beta}_1)\hat{\text{Cov}}(\beta_1, \beta_{1,2})$. Since $1 + 4\hat{\beta}_1 < 0$ in all cases, this variance is maximized when $\hat{\text{Cov}}(\hat{\beta}_1, \hat{\beta_{1,2}}) = \sqrt{\hat{\text{Var}}(\hat{\beta}_1)\hat{\text{Var}}(\hat{\beta_{1,2}})}$. Under this assumption, standard errors are as large as possible, so inference is more conservative. The formula for the standard errors becomes $\hat{\text{SE}}(\hat{h}) = |1 + 4\hat{\beta}_1|\sqrt{\hat{\text{Var}}(\hat{\beta}_1)} + \sqrt{\hat{\text{Var}}(\hat{\beta_{1,2}})}$. Using the point estimates and standard errors reported in Table 4, one can compute the corresponding z-statistics under the null of $h = 0$. The estimated z-statistics are between -10.8 and -8.4, which are far from typical critical values of around -2. This shows that while $\beta_{1,2}$ may become significant under some

Outliers (% dropped)	Kurt(Δp)	β_1	$\tilde{\beta}_2$	$\tilde{\beta}_3$	$\beta_{1,2}$
None	7.59	-0.370	0.228	-0.155	0.002
		(0.007)	(0.013)	(0.023)	(0.007)
1%	4.24	-0.365	0.208	-0.163	0.001
		(0.007)	(0.013)	(0.027)	(0.007)
2%	3.81	-0.353	0.190	-0.178	-0.008
		(0.008)	(0.013)	(0.035)	(0.007)
4%	3.37	-0.334	0.145	-0.194	-0.015
		(0.009)	(0.014)	(0.046)	(0.007)

Table 5: Estimated kurtosis and predictability coefficients β_1, $\tilde{\beta}_2$, $\tilde{\beta}_3$ and $\beta_{1,2}$ with standard errors in parentheses, for different treatments of outliers. Sales are filtered using a three-month V-shaped filter.

specifications, its absolute magnitude is much smaller than what we would infer from β_1 under AR(1) shocks.

A.1.1 Robustness to Outlier Treatment

Table 5 presents point estimates for the kurtosis and the β's for different treatments of outliers. Unsurprisingly, the estimated kurtosis depends strongly on outliers. Regarding predictability, β_1 and $\tilde{\beta}_2$ get smaller in magnitude as we drop more outliers, while $\tilde{\beta}_3$ gets marginally larger in magnitude. The effect is relatively small for β_1 and $\tilde{\beta}_3$, but more pronounced for $\tilde{\beta}_2$. The signs and significance are stable across specifications.

Since Eichenbaum et al. (2014) argue that a large share of the observed small price changes is due to measurement error, Table 6 shows robustness to varying the threshold used to exclude small price changes. The predictability coefficients are essentially invariant to this, showing that my results are not driven by potential measurement error of this kind.

A.1.2 Robustness to Sales Treatment

Figure A.1 presents estimates analogous to those in Figure 3.1 using different sales filters. As we can see, filtering out sales makes an important difference in the estimated coefficients: without filtering, β_1 goes down significantly, and $\beta_{1,2}$ becomes larger. This is consistent with models of price plans (Eichenbaum et al., 2011; Kehoe and Midrigan, 2015; Alvarez and Lippi, 2020): if firms often alternate between two fixed price points p^H and p^L, it would often be the case that the size of the price change two spells ago is equal to the current size,

Small changes dropped	Kurt(Δp)	β_1	$\tilde{\beta}_2$	$\tilde{\beta}_3$	$\beta_{1,2}$
0.1%	3.81	-0.353	0.190	-0.178	-0.008
		(0.008)	(0.013)	(0.035)	(0.007)
1.0%	3.62	-0.356	0.185	-0.174	-0.005
		(0.008)	(0.013)	(0.032)	(0.008)
2.5%	3.21	-0.368	0.179	-0.166	0.001
		(0.009)	(0.014)	(0.029)	(0.009)
5.0%	2.68	-0.373	0.167	-0.136	0.009
		(0.010)	(0.015)	(0.030)	(0.009)

Table 6: Estimated kurtosis and predictability coefficients β_1,$\tilde{\beta}_2$, $\tilde{\beta}_3$ and $\beta_{1,2}$ with standard errors in parentheses, for different exclusion criteria for small price changes. Sales are filtered using a three-month V-shaped filter.

Filter	Kurt(Δp)	β_1	$\tilde{\beta}_2$	$\tilde{\beta}_3$	$\beta_{1,2}$
V-shaped	3.81	-0.353	0.190	-0.178	-0.008
		(0.008)	(0.013)	(0.035)	(0.007)
Nakamura-Steinsson (A)	3.82	-0.344	0.177	-0.168	-0.018
		(0.010)	(0.014)	(0.033)	(0.007)
Kehoe–Midrigan	4.05	-0.350	0.274	-0.306	0.034
		(0.016)	(0.030)	(0.048)	(0.020)

Table 7: Estimated kurtosis and predictability coefficients β_1, $\tilde{\beta}_2$, $\tilde{\beta}_3$, and $\beta_{1,2}$ with standard errors in parentheses, for different sales treatments.

leading to a positive $\beta_{1,2}$. The fact that we find $\beta_{1,2} \approx 0$ in all three cases is reassuring that we are filtering out temporary sales. Table 7 shows the estimated sufficient statistics and corresponding standard errors for different sales filters. As we can see, the results are quite similar across different filters. Appendix B.4 discusses the implications for non-neutrality.

A.2 Continuing Products

Subsample	β_1	$\beta_{1,2}$
Milei	-0.345***	-0.011
	(0.029)	(0.008)
Post Shock	-0.206***	0.000
	(0.018)	(0.001)

Table 8: Regression results by subsample for (3.1)-(3.2) in the Argentine data. The sample is restricted to barcodes available in a categorized sample from July 2021.

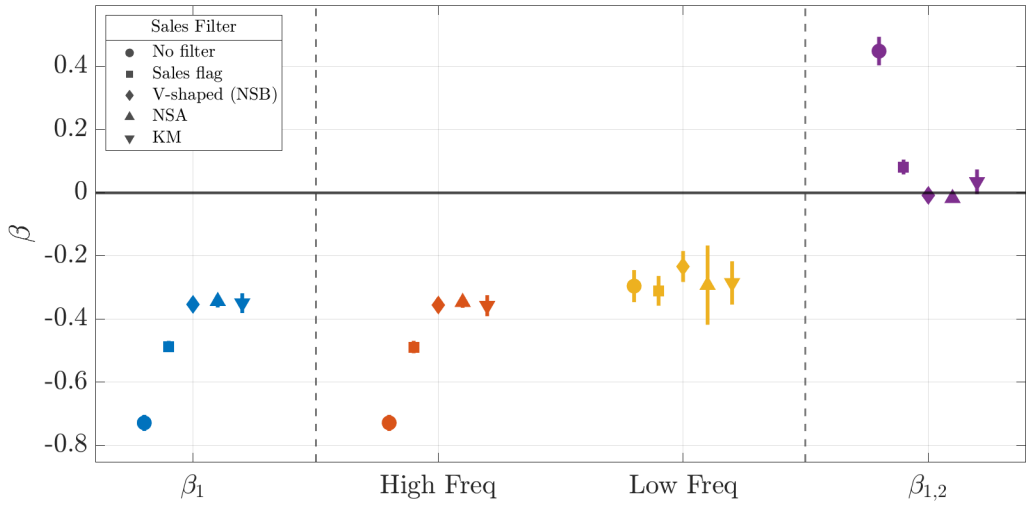


Figure A.1: Estimates of Equation (3.1), for the full sample (left) and frequency subsamples (middle), and of Equation (3.2) for different sales treatments. SEs are clustered by item.

Table 8 shows results if we restrict the sample to a set of products available in a categorized sample in July 2021. These products have thus been on the market for a few years before our main sample starts. Overall, the estimated coefficients show patterns similar to those in the full sample. This implies that our results are not driven by learning upon the introduction of new products to the market, as in Argente and Yeh (2022).

A.3 Bootstrap Implementation Details

I use an item-level bootstrap with 1,000 replications. In each replication, I resample whole items with replacement, where an item consists of the complete set of price paths of all product–store units belonging to that item. I resample by item to be consistent with clustering the standard errors by item. Thus, the procedure preserves arbitrary dependence among observations within an item, including serial dependence within a product-store price path and dependence across products and stores within an item. With this I also avoid having to assume a specific parametric form for the time-series correlation of different products. For each bootstrapped sample, I obtain one estimate for the tuple $(\text{Kurt}(\Delta p), \beta_1, \tilde{\beta}_2, \tilde{\beta}_3)$, which I then use to construct one estimate of each bound. I compute confidence bands using the desired percentiles of the bootstrapped distributions.

B Additional Theoretical and Quantitative Results

B.1 Standard Models with Mean-Reverting Shocks

Take a general time-dependent model with quadratic profits,$\pi = -\frac{1}{2}(p-z)^2$, and assume $z_{i,t}$ follows $dz_{i,t} = -\eta z_{i,t}dt + \sigma dW_{i,t}$. In what follows, I lighten the notation by dropping the i subscripts. As in standard time-dependent models, $\Delta\tau_n = \tau_n - \tau_{n-1}$ has an exogenous distribution, and it is independent for different spells. The optimal reset price solves

$$\max_p \quad -\mathbb{E}_t\left[\int_0^{\Delta\tau} e^{-\rho s}\frac{1}{2}(p - z_{t+s})^2 ds\right],$$

taking the FOC and solving yields $p = \left(\mathbb{E}\left[\int_0^{\Delta\tau} e^{-(\rho+\eta)s}ds\right]/\mathbb{E}\left[\int_0^{\Delta\tau} e^{-\rho s}ds\right]\right) \times z_t = \bar{c}z_t$, so the size of a given price change is $\Delta p_n = \bar{c}(z_{\tau_n} - z_{\tau_{n-1}})$. Letting τ_n denote the adjustment times and $z_n := z_{\tau_n}$:

$$\begin{aligned}
\beta_1 = \frac{\text{Cov}(p_{\tau_{n+1}} - p_{\tau_n}, p_{\tau_n} - p_{\tau_{n-1}})}{\text{Var}(p_{\tau_n} - p_{\tau_{n-1}})} &= \frac{\text{Cov}(z_{n+1} - z_n, z_n - z_{n-1})}{\text{Var}(z_n - z_{n-1})} \\
&= \frac{\mathbb{E}[\text{Cov}(z_{n+1} - z_n, z_n - z_{n-1}|\tau_{n+1}, \tau_n, \tau_{n-1})]}{\mathbb{E}[\text{Var}(z_n - z_{n-1}|\tau_{n+1}, \tau_n, \tau_{n-1})]} \\
&= \frac{\text{Var}(z_t)\mathbb{E}[e^{-\eta\Delta\tau_{n+1}} - e^{-\eta(\Delta\tau_{n+1}+\Delta\tau_n)} - 1 + e^{-\eta\Delta\tau_n}]}{\text{Var}(z_t)\mathbb{E}[2(1 - e^{-\eta\Delta\tau_n})]}.
\end{aligned}$$

Thus $\beta_1 = \frac{2\mathbb{E}[e^{-\eta\Delta\tau_n}] - [1 + \mathbb{E}[(e^{-\eta\Delta\tau_n})]^2]}{2(1 - \mathbb{E}[e^{-\eta\Delta\tau_n}])} = \frac{(\mathbb{E}[e^{-\eta\Delta\tau_n}] - 1)}{2}$. Similarly, for two spells apart we have

$$\beta_{1,2} = \frac{\text{Cov}(z_{n+1} - z_n, z_{n-1} - z_{n-2})}{\text{Var}(z_{n+1} - z_n)}.$$

Proceeding as above for the numerator:

$$\begin{aligned}
\text{Cov}(z_{n+1} - z_n, z_{n-1} - z_{n-2}) &= \mathbb{E}[\text{Cov}(z_{n+1} - z_n, z_{n-1} - z_{n-2}|\tau_{n+1}, \tau_n, \tau_{n-1})] \\
&= -\text{Var}(z)\mathbb{E}\left[e^{-\eta\Delta\tau}\right](1 - \mathbb{E}[e^{-\eta\Delta\tau_n}])^2,
\end{aligned}$$

and since the denominator is the same, we get $\beta_{1,2} = \mathbb{E}\left[e^{-\eta\Delta\tau}\right]\frac{(\mathbb{E}[e^{-\eta\Delta\tau_n}] - 1)}{2} = \beta_1(1+2\beta_1)$ for any time-dependent model. For a Calvo model $\mathbb{E}[e^{-\eta\Delta\tau_n}] = \kappa\int_0^\infty e^{-(\eta+\kappa)t}dt = \frac{\kappa}{\kappa+\eta}$, so $\beta_1 = \frac{\frac{\kappa}{\kappa+\eta}-1}{2} = -\frac{\eta}{2(\kappa+\eta)}$ and $\beta_{1,2} = -\frac{\eta\kappa}{2(\kappa+\eta)^2}$. In the Taylor model, $\mathbb{E}[e^{-\eta\Delta\tau_n}] = e^{-\eta T}$, so $\beta_1 = -(1 - e^{-\eta T})/2$ and $\beta_{1,2} = -e^{-\eta T}(1 - e^{-\eta T})/2$.

B.1.1 Menu Cost Model

In this subsection, I consider a standard menu cost model with mean-reverting desired prices, exactly as in Golosov and Lucas (2007). Let q be the log detrended price, $q = \log(\frac{P}{W\xi/(\xi-1)})$, and z be the target price. The profit function is quadratic, $\pi = -B(q-z)^2$, with $dq = -\mu dt$ and $dz = -\eta z dt + \sigma dW$, where I impose that z has an unconditional mean equal to zero. Firms can adjust prices whenever they want, but they must pay a fixed cost $\Psi = B\tilde{\Psi}$ in order to do so. It can be shown that the value function $V(q,z)$ solves the following HJBQVI:

$$0 = \min\left\{\rho V - (\pi - \mu V_q - \eta z V_z + \frac{\sigma^2}{2}V_{zz}), V(q,z) - (V^*(z,\Psi))\right\}.$$

For given parameter values, I solve this numerically using the procedure detailed in Online Appendix OA.E. The solution yields a value function $V(q,z)$, a hazard function $\Lambda(q,z)$ (which in this case, in the discretized version, implies adjustment probabilities of zero or one), and an optimal reset policy $p^*(z)$. Given these objects, I can directly compute the stationary distribution of (q,z), as well as the frequency of price changes. I also compute regression coefficients β_1 and $\beta_{1,2}$ via simulations.[17]

Here $V^{**}(z) = \max_q V(q,z)$ is the gross reset value, and $V^*(z,\psi) = V^{**}(z) - \psi$ is the value net of the menu cost.

Calibration. I need to calibrate $(\rho, \mu, \eta, \sigma^2, \Psi, B)$. I set $B = 1$, which changes nothing apart from the interpretation of menu costs, which are now measured in units of profits. I calibrate $\rho = 0.04$ and $\eta = -12\log(0.942)$. This parameter follows from the direct estimates of mean reversion in marginal costs in Gagliardone and Tielens (2026). I am left with μ, Ψ and σ^2. I consider a baseline calibration, and then I move each of those parameters one at a time, to see the effects on frequency and predictability coefficients. The baseline calibration features $\mu = 0$, $\Psi = 0.002$ and $\sigma^2 = 0.0325$. This calibration generates a frequency of price changes of around $N = 1.44$, and average sizes of price changes of around 13%. Both are roughly consistent with the UK data. Relative to this baseline calibration, I perform three experiments that increase the frequency of price changes. First, I increase trend inflation to $\mu' = 0.1$, or around 10% per year. Second, I halve the menu cost, i.e., I use $\Psi' = \Psi/2$. Finally, I increase the standard deviation of idiosyncratic shocks by 50%, i.e., $\sigma' = 1.5\sigma$.

[17] In particular, I simulate the joint paths of (p,z) for $N = 4,000$ firms. During the simulations, the time step is one week. I start all firms at $q = z = 0$. I run the simulation for 50 years, and I discard the first 20 years as a burn-in period to ensure convergence.

Name	Frequency	β_1	$\beta_{1,2}$
Base	1.4485	-0.2065	-0.1218
High inflation	1.5959	-0.1513	-0.1017
Low menu cost	2.3519	-0.1377	-0.0987
Higher variance	2.0843	-0.2212	-0.1080

Table 9: Predictability regressions across parametrizations. Regressions include vintage (time of last reset) fixed effects.

Results and Takeaways. Table 9 shows the implied frequency and predictability coefficients for the base calibration, as well as the three alternative ones. As we can see, all alternative calibrations increase the frequency of price changes, but for different reasons: trend inflation makes firms change prices faster to keep up with it (Gagnon, 2009; Nakamura et al., 2018; Alvarez et al., 2018), lower menu costs naturally lead to more price adjustments, and higher volatility implies that we hit the adjustment thresholds more frequently (Vavra, 2014; Baley and Blanco, 2019).[18]

Note that the predictability coefficient β_1 depends on the source of the increase in frequency: If I increase inflation or lower menu costs, that decreases its magnitude, whereas if I increase idiosyncratic volatility, that increases the magnitude of β_1. Therefore, in a state-dependent model, we could obtain predictability increasing with frequency, but only if the increase in frequency is driven by an increase in idiosyncratic volatility. However, if that was the main driver of heterogeneity across categories, we should observe a tight positive relation between the average frequency and the size of price changes. Online Appendix OA.A.3 shows that this is not the case in our sample. Note also that $\beta_{1,2}$ remains consistently negative in all cases. Furthermore, the formula derived under time-dependent models, $\beta_{1,2} = \beta_1(1 + 2\beta_1)$, seems to provide a good approximation of the value of $\beta_{1,2}$ given β_1. Overall, the main patterns emphasized in the text ($\beta_1 < 0$, reversals get weaker with frequency, and predictability lasts more than one spell) continue to hold in a menu cost model.

B.2 Unobserved Heterogeneity

Assume the data is generated by the random menu cost model in Alvarez et al. (2021). For each unit, the timing of price adjustment is determined by a hazard $\Lambda_i(x)$, and they face idiosyncratic shocks with variance σ_i^2. Thus, I allow for heterogeneity in both the distribution

[18]The overall effect is in principle not obvious since higher volatility also expands inaction regions. Baley and Blanco (2019) formally prove that, in the context of their model, the optimal change in adjustment thresholds is not enough to overturn the direct effect, and thus frequency increases with σ^2.

of menu costs that different units face, as well as the variance of those shocks. Assume for simplicity that units belong to one of finitely many types, $\theta \in \Theta$, which are unobserved by the econometrician. Assume that trend inflation is zero and profits are symmetric, so upon adjustment all firms set $x^* = 0$. Price changes in this economy are given by $\Delta p_{i,n} = \sigma_i(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})$, where now the distribution of the stopping times $\tau_{i,n}$ is different for each unit, since it is governed by a different hazard function. Consider a pooled OLS version of β_1. The relevant covariance is:

$$\begin{aligned}\text{Cov}(\Delta p_{i,n-1}, \Delta p_{i,n}) &= \mathbb{E}[\Delta p_{i,n-1}\Delta p_{i,n}] - \mathbb{E}[\Delta p_{i,n}]^2 \\ &= \mathbb{E}[\mathbb{E}[\Delta p_{i,n-1}\Delta p_{i,n}|\theta_i = \theta]] - \mathbb{E}[\mathbb{E}[\Delta p_{i,n}|\theta_i = \theta]]^2 = 0\end{aligned}$$

since once we condition on a type, $\mathbb{E}[\Delta p_{i,n-1}\Delta p_{i,n}|\theta_i = \theta] = 0$ by the same argument as in the main text, and $\mathbb{E}[\Delta p_{i,n}|\theta_i = \theta] = 0$ due to the symmetry assumptions imposed. Thus, in this case, $\beta_1 = 0$. Since the effects of trend inflation on even moments are second order in this model, these results continue to hold when there is low trend inflation.

As discussed in the main text, with significant trend inflation, if heterogeneity manifests itself as different sizes of Ss bands, we would tend to find a positive correlation between consecutive sizes: some firms adjust to trend inflation by doing sporadic, large increases, while others make many small increases.

B.3 Multiproduct Model

Assume that the firm sells n products, indexed by i. Total losses are $By = B\sum_{i=1}^{n} x_i^2$ where x_i is the price gap of product i. When prices are not adjusted, gaps evolve as $dx_i = \sigma dW_{i,t}$, where $W_{i,t}$ are Brownian motions, independent across i. Following Alvarez and Lippi (2014), we can rewrite the problem fully in terms of y. The firm always has the option to readjust the prices of all its products by paying a fixed cost $\Psi \leqslant +\infty$. Additionally, as in a random menu cost model, opportunities to change prices at lower costs arrive with arrival rate κ. Once an opportunity arrives, a cost ψ is drawn from a distribution $G(\psi)$ with support $[0, \Psi)$. When the firm decides to change prices, it *must* change prices for all products: it cannot choose to reprice only a set of products and not the rest. Thus its decision to reset is based only on y. If the firm does not adjust prices, y evolves as $dy = n\sigma^2 dt + 2\sigma\sqrt{y}dW$. Letting $V^*(0)$ denote the value of resetting net of control costs when the menu cost is zero, the

HJBQVI of this problem is:

$$0 = \min\left\{\rho V - (-By + n\sigma^2 V_y + 2\sigma^2 y V_{yy} + \kappa \int_0^{\Psi} \max\{V^*(0) - (\psi + V(y)), 0\} dG(\psi)), V(y) - V^*(\Psi)\right\},$$

where the max inside the integral operator indicates that the firm only resets if doing so increases value. The value of resetting $V^*(0)$ is defined as:

$$V^*(\psi) = \max_{\{f_i^y\}} \int_0^{\infty} \dots \int_0^{\infty} V(\sum_{i=1}^{n} y_i) \prod_{i=1}^{n} f_i^y(y_i) dy_i - \sum_{i=1}^{n} \mathcal{C}(f_i^y) - \psi \quad \text{s.t} \int f_i^y(y_i) dy_i = 1 \ \forall i.$$

where f_i^y denotes the density of $y_i = (x_i^*)^2$, and $\bar{f}^y$ is the corresponding transformed baseline density. Here $V^*(\psi)$ deducts both control costs and the realized menu cost ψ. I reserve $V^{**} = \mathbb{E}[V(y^*)]$ for the expected continuation value under the optimal reset distribution before subtracting any adjustment costs. Thus $V^*(\psi) = V^{**} - \sum_i \mathcal{C}(f_i^y) - \psi$. This is the corresponding control cost problem in the multiproduct case, where it is assumed that all products have the same baseline distribution $\bar{f}^y$, and control costs are additive in the number of products. Due to symmetry in how products enter, all f_i^y will be equal. Via a change of variables we can obtain the density as a function of the price gap of a single product as in the case above, $f(x) = |x| f^y(x^2)$. The solution to this problem yields a value function $V(y)$, a threshold $\bar{y}$, a reset gap density $f(x)$, and a generalized hazard function $\Lambda(y) = \kappa G(V^*(0) - V(y))$, that gives the probability per unit of time of a price change occurring. Note that the probability of a given product changing depends on the sum of squared gaps for *all* products sold by the firm.

B.4 Sensitivity Analysis of Estimated Bounds

	Dropped (%)	Kurt(Δp)	Std(Δp)	β_1	β_2	$\tilde{L}$	$\tilde{U}$
Base	3.03	3.807	0.147	-0.353	0.192	1.483	3.376
Base + no comeback	10.28	4.003	0.144	-0.308	0.123	1.843	3.046
Drop $\lvert\Delta p\rvert < 1\%$	8.16	3.616	0.157	-0.356	0.187	1.311	3.356
Outliers to 1%	2.04	4.241	0.154	-0.365	0.210	2.032	3.936
Outliers to 4%	5.01	3.372	0.135	-0.334	0.146	1.226	3.021
Drop $\lvert\Delta p\rvert >25\%$	16.85	4.379	0.078	-0.262	0.137	1.418	2.172
Drop $\lvert\Delta p\rvert >35\%$	9.98	4.959	0.097	-0.291	0.175	1.655	2.495

Table 10: Robustness of sufficient-statistic bounds to trimming assumptions. The bounds $\tilde{L}, \tilde{U}$ are measured relative to the benchmark, $\tilde{L} = 6L/\text{Kurt}(\Delta p), \quad \tilde{U} = 6U/\text{Kurt}(\Delta p)$.

Sensitivity to Measurement Assumptions. Table 10 presents robustness to different measurement assumptions. The first row repeats the results reported in the main text. The second row further drops *all* prices that come back very close to the previous level at the next price change.[19] The third row shows what happens if we drop all price changes below 1 percent (Eichenbaum et al., 2014). The fourth and fifth rows change the definition of outliers dropped. The last two rows replace the rule based on standardized price changes with a fixed threshold for absolute price changes, using the same values as Alvarez et al. (2024).

Overall, while the exact numbers are sensitive, the main patterns in the data are robust to these measurement changes: the kurtosis is above 3, β_1 is negative and large in absolute value, β_2 is positive, and inferred non-neutrality goes up even at the lower bound. However, this also shows that the exact point estimate for the increase in non-neutrality is sensitive to measurement assumptions. Dropping 4 percent of outliers instead of 2 makes the increase 22% instead of around 50%, whereas dropping 1 percent makes non-neutrality at least 100% larger. Using fixed cut-off rules has a relatively minor impact on the point estimate. Perhaps counterintuitively, dropping all comeback prices makes us infer even higher non-neutrality. This is because $\text{Kurt}(\Delta p)$ increases and β_2 falls.

B.5 Accounting for Heterogeneity Across Items

Heterogeneity is quite pervasive in price microdata. For that reason, I extend the model to multiple sectors to account for it, along the lines of Alvarez et al. (2016a), Online Appendix E. In this case, the cumulative impulse response for aggregate output is given by:

$$CIR = \sum_{j=1}^{J} \alpha_j CIR_j, \tag{B.1}$$

where α_j are the expenditure shares on each item and CIR_j is the CIR for that item, which can be bounded using the corresponding moments.

I run the main three regressions by item, keeping the same fixed effects (cell-by-time). The other moments are straightforward to compute. Table 11 presents some distributional statistics for the key moments across items. As extensively documented, there is significant heterogeneity in the frequency, variance, and kurtosis of price changes. When turning to the predictability coefficients, we also see quite large heterogeneity, with all coefficients even

[19] I define these as $|\Delta p_{i,n} + \Delta p_{i,n-1}| < 0.1\%$. The baseline V-shaped filter only eliminates such patterns if the comeback occurs within three months.

Moments	Percentile				
	10	25	50	75	90
Obs	10	49	205	748	1,637
N	0.401	0.539	0.805	1.239	2.213
Var(Δp)	0.003	0.009	0.016	0.037	0.077
Kurt(Δp)	2.284	2.963	3.801	4.844	6.225
β_1	-0.851	-0.490	-0.284	-0.058	0.242
$\tilde{\beta}_2$	-0.300	0.000	0.168	0.406	0.856
$\tilde{\beta}_3$	-6.059	-2.014	-0.205	1.035	5.256
CIR					
Calvo ($1/N$)	0.438	0.788	1.205	1.784	2.397
Base (Kurt(Δp)/($6N$))	0.230	0.472	0.785	1.181	1.749
Lower Bound (L/N)	0.093	0.380	0.963	1.987	5.040
Upper Bound (U/N)	0.339	0.669	1.359	2.825	6.826

Table 11: Distribution of objects of interest across the 1,187 items in the UK CPI dataset. "Obs" refers to the number of observations included in the regression for β_1 for each category. The formulas for the CIR are only evaluated in categories with $\beta_1 \in (-1/2, 0]$.

switching signs for some items. Part of this is likely to be sampling error, since as the top row shows, there are several items with a small number of observations.[20] The second part of the table shows the implied estimates for the CIR. As previously documented, the large heterogeneity across sectors translates into large dispersion in estimated non-neutrality across items when using the Calvo or Baseline formula. Once we correct for imperfect reset, the estimated bounds for the CIR become more dispersed across items.

Table 12 aggregates across items using (B.1) for different weights and numbers of outliers dropped, while the last column shows the inferred non-neutrality based on aggregated moments, as done in the main text. First, focusing on the weighted estimates under Calvo and the baseline formula, heterogeneity across items increases inferred non-neutrality. This is a typical finding in the literature, following Carvalho (2006); Nakamura and Steinsson (2010). The same is true when looking at the upper and lower bounds: the estimated CIR tends to increase relative to the case where we evaluate the formula using aggregated moments. Second, the conclusion in the main text is robust: accounting for reset imperfections increases non-neutrality relative to the Alvarez et al. (2016a) formula, and brings it closer to Calvo. However, the exact number is sensitive to the number of outliers dropped and the weighting

[20] In particular, if we test the null hypothesis that $\beta_1 \in (-1/2, 0)$ for each individual category with 90% confidence, we reject the null in less than 10% of the categories. Furthermore, these categories have a total CPI weight of only 7.6%.

	Outliers Dropped							No Het.
CIR	0%	1%	2%	5%	6%	8%	10%	
CPI Weights								
Calvo $(1/N)$	0.965	0.989	0.990	0.968	0.964	0.962	0.950	0.769
Base (Kurt$(\Delta p)/(6N)$)	0.811	0.834	0.832	0.799	0.792	0.789	0.777	0.488
Lower Bound (L/N)	78.655	1.885	1.883	1.351	0.925	0.987	0.850	0.712
Upper Bound (U/N)	80.585	3.079	3.007	2.131	1.665	1.663	1.473	1.634
Unweighted								
Calvo $(1/N)$	1.034	1.021	1.014	0.994	0.994	0.984	0.974	
Base (Kurt$(\Delta p)/(6N)$)	0.799	0.785	0.777	0.753	0.749	0.738	0.726	
Lower Bound (L/N)	119.461	-0.630	0.111	1.155	0.951	0.895	0.705	
Upper Bound (U/N)	121.997	0.815	1.460	2.058	1.824	1.668	1.427	

Table 12: Aggregate CIR, computed as a weighted average of item-level CIRs, as a function of the total percentage of categories dropped. The first set of rows weights them using CPI weights, whereas the second set weights all categories equally. The formulas are only evaluated in categories with $\beta_1 \in (-1/2, 0]$, and we ensure the weights on the remaining items sum up to one. The last column shows the results using aggregate measures for $N, \text{Kurt}(\Delta p), \beta_1, \beta_2$. These are computed as medians as in the main text.

scheme. If we drop between 5% and 8% of extreme observations, we get numbers very close to Calvo. If we drop fewer outliers, we get extremely large numbers, while if we drop more, we get numbers closer to the baseline formula. This mainly reflects that, as Table 11 shows, item-level estimates are very noisy. Finally, these conclusions are largely unchanged when we use equal weights instead of CPI weights. Overall, the main conclusions in the main text are relatively robust to accounting for heterogeneity across products, although the exact conclusions are sensitive to the number of outlier categories dropped.

C Proofs

C.1 Proposition 1

Before turning to the proof, it is useful to think that the probability of changing prices from t to $t+1$ is $P(\theta_i, X_t, x_{it})$, the law of motion for idiosyncratic states is $x_{i,t+1} = g(\theta_i, X_t, x_{it}, \varepsilon_{i,t+1})$ and for aggregate states is $X_{t+1} = h(X_t, \varepsilon_{X,t+1})$. While this is not used explicitly in the proof, it is useful for intuition. One subtle thing is that, in order to map a standard menu cost model to this notation, I need to split time periods into two. From t to $t+1$, the firm decides to change prices, and no idiosyncratic shocks are realized. Then, between $t+1$ and $t+2$,

idiosyncratic shocks are realized (but the firm cannot update prices). Between $t+2$ and $t+3$ the firm has once again the chance to change prices, and so on. This is just an artifact of my time indexing, which I could dispense with at the cost of introducing additional notation.

Proof. First, note that any sample we observe is conditional on a realized path of the aggregate state.[21] Furthermore, it is well known that the estimand of the panel regression is a weighted average of the $\beta_{t,\theta}$ that come from regressions across i for each given tuple (t,θ). Focusing on those, we know that as the number of units grows to infinity for each tuple:

$$\beta_{t,\theta} \to \frac{\text{Cov}(y_{i,s}, w_{i,\tau_{i,n-1}} | \theta_i = \theta, \tau_{i,n-1} = t, \{X_\ell\}_{-\infty<\ell\leqslant\infty})}{\text{Var}(w_{i,\tau_{i,n-1}} | \theta_i = \theta, \tau_{i,n-1} = t, \{X_\ell\}_{-\infty<\ell\leqslant\infty})}.$$

Letting $\mathbb{E}_i[\bullet]$ denote the cross sectional expectation, we have:

$$\begin{aligned}
\mathbb{E}_i[y_{i,s} w_{i,\tau_{i,n-1}} | \theta_i = \theta, \tau_{i,n-1} = t, \{X_\ell\}_{-\infty<\ell\leqslant\infty}] &= \\
\mathbb{E}_i\left[\mathbb{E}_i[y_{i,s} | \theta_i = \theta, \{X_\ell\}_{-\infty<\ell\leqslant\infty}, \{x_{i,\ell}\}_{-\infty<\ell\leqslant\tau_{i,n-1}}, \tau_{i,n-1} = t] w_{i,\tau_{i,n-1}} | \theta_i = \theta, \{X_\ell\}_{-\infty<\ell\leqslant\infty}, \tau_{i,n-1} = t\right] &= \\
\mathbb{E}_i\left[\mathbb{E}_i[y_{i,s} | \theta_i = \theta, \{X_\ell\}_{-\infty<\ell\leqslant\infty}, \tau_{i,n-1} = t] w_{i,\tau_{i,n-1}} | \theta_i = \theta, \{X_\ell\}_{-\infty<\ell\leqslant\infty}, \tau_{i,n-1} = t\right] &= \\
\mathbb{E}_i[y_{i,s} | \theta_i = \theta, \{X_\ell\}_{-\infty<\ell\leqslant\infty}, \tau_{i,n-1} = t] \mathbb{E}_i\left[w_{i,\tau_{i,n-1}} | \theta_i = \theta, \{X_\ell\}_{-\infty<\ell\leqslant\infty}, \tau_{i,n-1} = t\right], &
\end{aligned}$$

so the conditional covariance is zero. The first line uses the law of iterated expectations, and that $w_{i,\tau_{i,n-1}}$ is perfectly known from the type, time of reset, and the values of idiosyncratic and aggregate states. From the first to the second line I use that, since $y_{i,s}$ only depends on the idiosyncratic state after reset and aggregate states, but this is only a function of θ, X_t, we can drop the conditioning on $x_{i,t}$. This is because, due to our conditioning, all firms changing prices start at the same $x^*_{i,t+1}$, and the subsequent values of the state only depend on past states and unpredictable idiosyncratic shocks. From the second to the third line I just take $\mathbb{E}_i[y_{i,s} | \theta_i = \theta, \{X_\ell\}_{-\infty<\ell\leqslant\infty}, \tau_{i,n-1} = t]$ out of the expectation. The key thing is that under the true mathematical expectation, adding a fixed effect so that $\tau_{i,n-1} = t$ implies conditioning on the fact that the aggregate state is X_t, which is the only remaining source of potential predictability. If we did not condition on that, then $\mathbb{E}_i[y_{i,s}]$ will generally depend on $\tau_{i,n-1}$, which potentially will generate some correlation with $w_{i,\tau_{i,n-1}}$. □

[21] To see this, consider the thought experiment of obtaining several panel samples for many different realizations of the aggregate state, $\{X_t\}$. Then the estimated time fixed-effect $\lambda_{t,\theta}$ will in general be different for each realization of $\{X_t\}$.

C.2 Proposition 3

Solving the PDE, the value function in the continuation region is $V(z) = -\frac{\sigma^2}{\rho^2}B - \frac{B}{\rho}z^2 + 2b\cosh(\lambda z)$, where $\lambda = \sqrt{2\rho/\sigma^2}$. The value upon reset is $\mathbb{E}_F[V] = -\frac{\sigma^2}{\rho^2}B - \frac{B}{\rho}\text{Var}(x^*) + 2b\mathbb{E}_F[\cosh(\lambda z)]$. Assume that x^* has distribution F and that we scale that distribution properly with c. More concretely, assume that $x^* = \bar{x}\tilde{x}$, where $\tilde{x}$ is symmetric and has support in $[-1, 1]$. Mathematically, we are scaling the distribution with c, but it is simpler to write it as doing it in $\bar{x}$. The relevant value matching and smooth pasting conditions are:

$$\begin{aligned} V(\bar{x}) + c &= -\frac{\sigma^2}{\rho^2}B - \frac{B}{\rho}\text{Var}(x^*) + 2b\mathbb{E}_F[\cosh(\lambda z)] \\ 0 &= -2B\frac{\bar{x}}{\rho} + 2b\lambda\sinh(\lambda\bar{x}), \end{aligned}$$

solving out $b = \frac{B\bar{x}}{\rho\lambda\sinh(\lambda\bar{x})}$, expanding $\cosh(\lambda z) = 1 + \frac{\lambda^2 z^2}{2} + \frac{\lambda^4 z^4}{24} + O(z^6)$ and substituting into the equation above we have:

$$\begin{aligned} -\frac{\sigma^2}{\rho^2}B - \frac{B}{\rho}\bar{x}^2 + 2\frac{B\bar{x}}{\rho\lambda\sinh(\lambda\bar{x})}\cosh(\lambda\bar{x}) + c &= \\ -\frac{\sigma^2}{\rho^2}B - \frac{B}{\rho}\bar{x}^2\text{Var}(\tilde{x}) + 2\frac{B\bar{x}}{\rho\lambda\sinh(\lambda\bar{x})}[1 + \frac{\lambda^2\bar{x}^2}{2}\mathbb{E}[\tilde{x}^2] + \frac{\lambda^4\bar{x}^4}{24}\mathbb{E}[\tilde{x}^4] + O(z^6)]. \end{aligned}$$

Approximation. Consider the approximate solution when $c \to 0$. Guess the solution has the form $\bar{x} = (Ac)^{1/4} + o((c)^{1/2})$. Approximating $b(x) = Bx/(\rho\lambda\sinh(\lambda x))$, since $\sinh(\lambda x) = \lambda x + \frac{\lambda^3 x^3}{6} + \frac{\lambda^5 x^5}{120} + O(x^7)$ as $x \to 0$,

$$\begin{aligned} b(x) &= \frac{Bx}{\rho\lambda(\lambda x + \frac{\lambda^3 x^3}{6} + \frac{\lambda^5 x^5}{120} + O(x^7))} = \frac{B}{\rho\lambda^2}\Big(1 - \frac{\lambda^2 x^2}{6} + \frac{7\lambda^4 x^4}{360} + O(x^6)\Big) \\ &= \frac{B}{\rho\lambda^2} - B\frac{x^2}{6\rho} + \frac{B}{\rho}\frac{7\lambda^2 x^4}{360} + O(x^4) \end{aligned}$$

and also $\cosh(\lambda x) = 1 + \lambda^2\frac{x^2}{2} + \lambda^4\frac{x^4}{24} + O(x^6)$. Thus our equality becomes:

$$\begin{aligned} -\frac{B}{\rho}x^2 + 2\frac{B}{\rho\lambda^2}\Big(1 - \frac{\lambda^2 x^2}{6} + \frac{7\lambda^4 x^4}{360} + O(x^4)\Big)\left(1 + \lambda^2\frac{x^2}{2} + \lambda^4\frac{x^4}{24} + O(x^6)\right) + c &= \\ -\frac{B}{\rho}x^2\text{Var}(\tilde{x}) + 2\frac{B}{\rho\lambda^2}\Big(1 - \frac{\lambda^2 x^2}{6} + \frac{7\lambda^4 x^4}{360} + O(x^4)\Big)\left(1 + \frac{\lambda^2 x^2}{2}\mathbb{E}[\tilde{x}^2] + \frac{\lambda^4 x^4}{24}\mathbb{E}[\tilde{x}^4] + O(x^6)\right), \end{aligned}$$

collecting the terms with constants, $2\frac{B}{\rho\lambda^2} = 2\frac{B}{\rho\lambda^2}$ so they cancel. Collecting terms with x^2:

$$\begin{aligned} -\frac{B}{\rho}x^2 + 2\frac{B}{\rho\lambda^2}(\lambda^2\frac{x^2}{2} - \frac{\lambda^2x^2}{6}) &= -\frac{B}{\rho}x^2\text{Var}(\tilde{x}) + 2\frac{B}{\rho\lambda^2}\left(\frac{\lambda^2x^2}{2}\mathbb{E}[\tilde{x}^2] - \frac{\lambda^2x^2}{6}\right) \\ -\frac{\lambda^2x^2}{6}2\frac{B}{\rho\lambda^2} &= -\frac{\lambda^2x^2}{6}2\frac{B}{\rho\lambda^2} \end{aligned}$$

so they also cancel. While only considering terms with x^4, we have:

$$\begin{aligned} c + 2\frac{B}{\rho\lambda^2}(\lambda^4\frac{x^4}{24} - \frac{\lambda^4x^4}{12} + \frac{\lambda^4x^4}{360}) &= 2\frac{B}{\rho\lambda^2}(\frac{\lambda^4x^4}{24}\mathbb{E}[\tilde{x}^4] + \frac{\lambda^4x^4}{360} - \frac{\lambda^4x^4}{12}\mathbb{E}[\tilde{x}^2]) \\ c &= 2\frac{B}{\rho\lambda^2}\left(\frac{\lambda^4x^4}{24}\mathbb{E}[\tilde{x}^4] - \frac{\lambda^4x^4}{12}\mathbb{E}[\tilde{x}^2] + \lambda^4\frac{x^4}{24}\right) \\ c &= x^4\frac{B\lambda^2}{12\rho}\mathbb{E}\left[(1-\tilde{x}^2)^2\right], \end{aligned}$$

so we have that $c = x^4\frac{B\lambda^2}{12\rho}\mathbb{E}\left[(1-\tilde{x}^2)^2\right] + O(x^6)$, and thus

$$\bar{x} = (\frac{6\sigma^2c}{B}\frac{1}{\mathbb{E}\left[(1-\tilde{x}^2)^2\right]})^{1/4} + o(c^{1/4}), \tag{C.1}$$

so the asymptotic behavior is similar to Dixit (1991) with a correction factor $\frac{1}{\mathbb{E}[(1-\tilde{x}^2)^2]}$. Note that since the support of $\tilde{x}^2$ is in $[0,1)$, we have $1 \geqslant 1-\tilde{x}^2 > 0$ so $0 < \mathbb{E}\left[(1-\tilde{x}^2)^2\right] \leqslant 1$, with strict inequality whenever $\tilde{x}^2$ is non-degenerate.

C.3 Proposition 4

C.3.1 Aggregation

I first derive the formulas under the assumption that F is a continuous distribution with density f, and then I show that the formula generalizes to any symmetric F with support in $(-\bar{x}, \bar{x})$. Given $\bar{x}$, the KFE is $0 = \frac{\sigma^2}{2}g_{xx} + Nf(x), \quad x \in (-\bar{x}, \bar{x})$. Integrating twice, the solution satisfies $g(x) = a + bx - \frac{2}{\sigma^2}N\int_{-\bar{x}}^{x}F(z)dz$. We have three boundary conditions, $g(-\bar{x}) = g(\bar{x}) = 0$ and $\int_{-\bar{x}}^{\bar{x}}g(x)dx = 1$, and three unknowns, a, b, N. Using $g(-\bar{x}) = 0$ yields the restriction $a = b\bar{x}$. Using that and that $g(\bar{x}) = 0$, we get $2b = \frac{2}{\sigma^2}N$, so $b = N/\sigma^2$. Note that this implies that:

$$g(x) = b(x+\bar{x}) - \frac{2}{\sigma^2}N\int_{-\bar{x}}^{x}F(z)dz = b\int_{-\bar{x}}^{x}(1-2F(z))dz,$$

so using that the density must integrate to one: $b = 1/\int_{-\bar{x}}^{\bar{x}}\int_{-\bar{x}}^{x}(1-2F(y))dydx$. Computing the integral in the denominator by integration by parts I obtain $\int_{-\bar{x}}^{\bar{x}}\int_{-\bar{x}}^{x}(1-2F(y))dydx = \bar{x}^2 - \int_{-\bar{x}}^{\bar{x}} x^2 dF(x)$, where I used that since f is symmetric then $F(y) - 1/2$ is antisymmetric and thus $\int_{-\bar{x}}^{\bar{x}}(1-2F(y))dy = 0$. This implies that $b = [\bar{x}^2 - \text{Var}(x^*)]^{-1}$ so the main claim follows.

For a general reset distribution F, we can write the stationary KFE in weak form. In particular, the stationary density is defined as the function g such that for any twice differentiable function ϕ,

$$\int_{-\bar{x}}^{\bar{x}} g(x)[\frac{\sigma^2}{2}\phi_{xx}]dx + N\left[\int_{-\bar{x}}^{\bar{x}}\phi dF - \frac{1}{2}[\phi(-\bar{x}) + \phi(\bar{x})]\right] = 0. \tag{C.2}$$

Given our expression for N and g, we can just check that this equation holds. Replacing and integrating by parts twice:

$$\begin{aligned}
\int_{-\bar{x}}^{\bar{x}} g(x)[\frac{\sigma^2}{2}\phi_{xx}]dx &= \int_{-\bar{x}}^{\bar{x}} \frac{\int_{-\bar{x}}^{x}(1-2F(z))dz}{\int_{-\bar{x}}^{\bar{x}}\int_{-\bar{x}}^{x}(1-2F(y))dydx}[\frac{\sigma^2}{2}\phi_{xx}]dx \\
&= \frac{N}{2}\left[\phi_x\int_{-\bar{x}}^{x}(1-2F(z))dz|_{-\bar{x}}^{\bar{x}} - \int_{-\bar{x}}^{\bar{x}}(1-2F(x))\phi_x dx\right] \\
&= -N\left[\int_{-\bar{x}}^{\bar{x}}\phi dF - \frac{1}{2}[\phi(-\bar{x}) + \phi(\bar{x})]\right],
\end{aligned}$$

so (C.2) holds. Since given our assumptions $g(x)$ is always once differentiable at the boundary, we can compute the frequency as $N = \frac{\sigma^2}{2}[g_x(-\bar{x}) - g_x(\bar{x})] = \sigma^2/(\bar{x}^2 - \text{Var}(x^*))$, which is the formula in the main text.

C.3.2 Impulse Response

Given that after resetting the contribution of the firm is zero, we can use the results in Alvarez and Lippi (2022). Their general formula is:

$$G(t; f, \hat{P}) = \int_{-\bar{x}}^{\bar{x}} \mathcal{G}(f)(x,t)[dP(x;0) - d\bar{P}(x)],$$

where $\bar{P}(x)$ is the steady-state distribution, $P(x;0)$ is the new distribution (after the shock), and $\mathcal{G}(f)(x,t) = \mathbb{E}[1_{\{t\leqslant\tau\}}f(x(t))|x(0) = x]$ is the expected value of $f(x)$ for x evaluated t periods ahead, with zero contribution after firms adjust at t. For a small monetary shock, $dP(x;0) - d\bar{P}(x) = (g(x+\delta) - g(x))dx \approx \delta g'(x)dx$, so following the main text, we need

$Y(t) = \int_{-\bar{x}}^{\bar{x}} \mathcal{G}(-x)(x,t)[g'(x)]dx$, since output corresponds to $-x(t)$. Since there are no mass points, Theorem 1 in Alvarez and Lippi (2022) implies:

$$Y(t) = \sum_{n=1}^{\infty} e^{\lambda_n t} \left(\int_{-\bar{x}}^{\bar{x}} \varphi_n(x)(-x)dx \right) B_n.$$

Where $B_n = \left(\int_{-\bar{x}}^{\bar{x}} \varphi_n(x) g'(x) dx \right)$, with $\lambda_n = -[\frac{\sigma^2}{2}(\frac{n\pi}{2\bar{x}})^2]$ and $\varphi_n(x) = \frac{1}{\sqrt{\bar{x}}} \sin\left(\frac{n\pi(x+\bar{x})}{2\bar{x}} \right)$. From their paper, we also have that $\int_{-\bar{x}}^{\bar{x}} (-x) \frac{1}{\sqrt{\bar{x}}} \sin\left(\frac{n\pi(x+\bar{x})}{2\bar{x}} \right) dx = 4\frac{\bar{x}^{3/2}}{n\pi}$ for $n = 2, 4, 6, \ldots$ and zero otherwise. Thus:

$$\begin{aligned} Y(t) &= \sum_{n=1}^{\infty} B_n \exp(-\frac{\sigma^2}{2}(n\pi/2\bar{x})^2 t) \int_{-\bar{x}}^{\bar{x}} (-x) \frac{1}{\sqrt{\bar{x}}} \sin\left(\frac{n\pi(x+\bar{x})}{2\bar{x}} \right) dx \\ &= \sum_{j=1}^{\infty} B_{2j} \frac{2\bar{x}^{3/2}}{j\pi} \exp(-\frac{\sigma^2}{2}(j\pi/\bar{x})^2 t). \end{aligned}$$

To compute B_n, note that, when defined, $g'(x) = \frac{N}{\sigma^2}(1 - 2F(x))$. Thus:

$$\begin{aligned} B_{2j} &= \frac{N\bar{x}}{\sigma^2} \int_{-1}^{1} \frac{1}{\sqrt{\bar{x}}} (1 - 2F(u\bar{x})) \sin\left(j\pi(u+1)\right) du, \\ &= \frac{N\sqrt{\bar{x}}}{\sigma^2} \frac{2}{j\pi} (1 - \int_{-1}^{1} \cos\left(j\pi(u+1)\right) d\tilde{F}(u)) \end{aligned}$$

where I changed variables to $u = x/\bar{x}$ and integrated by parts. Using the expression for the frequency and replacing:

$$Y(t) = \frac{1}{1 - \mathbb{E}[\tilde{x}^2]} \sum_{j=1}^{\infty} \left((1 - \int_{-1}^{1} \cos\left(j\pi(u+1)\right) d\tilde{F}(u)) \right) (\frac{2}{j\pi})^2 \exp(-N(1 - \mathbb{E}[\tilde{x}^2]) \frac{(j\pi)^2}{2} t).$$

Note that if there is perfect reset, $\mathbb{E}[\tilde{x}^2] = 0$, and we recover the formula in Alvarez and Lippi (2022) for the standard menu cost model, $Y(t) = \frac{8}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(1+2k)^2} \exp(-N\frac{((1+2k)\pi)^2}{2} t)$.

Cumulative Impulse Response. From Alvarez et al. (2016a), the cumulative IRF can be computed as $\mathcal{M}'(0) = -\int m_x g dx$, where $m(x) = -\frac{(\bar{x})^2}{3\sigma^2} x(1 - (x/\bar{x})^2)$, so $m'(x) = -\frac{(\bar{x})^2}{3\sigma^2}(1 - 3(x/\bar{x})^2)$ so:

$$\mathcal{M}'(0) = \int_{-\bar{x}}^{\bar{x}} \frac{(\bar{x})^2}{3\sigma^2} (1 - 3(x/\bar{x})^2) g dx = \frac{1}{\sigma^2} \left[\frac{\bar{x}^2}{3} - \int_{-\bar{x}}^{\bar{x}} x^2 g dx \right].$$

Using $\phi = x^4$ in equation (C.2), we get that: $\int_{-\bar{x}}^{\bar{x}} x^2 g dx = \frac{N}{6\sigma^2}\left[\bar{x}^4 - \int_{-\bar{x}}^{\bar{x}} x^4 dF\right]$. Using that $N/\sigma^2 = 1/(\bar{x}^2 - \int_{-\bar{x}}^{\bar{x}} x^2 dF)$ and operating we get

$$\mathcal{M}'(0) = \frac{\bar{x}^2 - \int_{-\bar{x}}^{\bar{x}} x^2 dF}{6\sigma^2} \times \int_{-\bar{x}}^{\bar{x}} (\bar{x}^2 - x^2)^2 dF / \left(\bar{x}^2 - \int_{-\bar{x}}^{\bar{x}} x^2 dF\right)^2$$

but $\frac{\bar{x}^2 - \int_{-\bar{x}}^{\bar{x}} x^2 dF}{6\sigma^2} = 1/(6N)$, and using that $\tau(x) = \frac{\bar{x}^2 - x^2}{\sigma^2}$ we can write:

$$\frac{\int_{-\bar{x}}^{\bar{x}} (\bar{x}^2 - x^2)^2 dF}{\left(\bar{x}^2 - \int_{-\bar{x}}^{\bar{x}} x^2 dF\right)^2} = \frac{\int_{-\bar{x}}^{\bar{x}} ((\bar{x}^2 - x^2)/\sigma^2)^2 dF}{\left(\int_{-\bar{x}}^{\bar{x}} (\bar{x}^2 - x^2)/\sigma^2 dF\right)^2} = \frac{\mathbb{E}_F[\tau(x)^2]}{\mathbb{E}_F[\tau(x)]^2} = 1 + \frac{\mathbb{V}_F[\tau(x)]}{\mathbb{E}_F[\tau(x)]^2}.$$

C.4 Proposition 6

Compare the IRF with strictly imperfect reset, $Y^{imp}(t)$, and the one with perfect reset, $Y^{per}(t)$. As $t \to +\infty$ the behavior is dominated by the first eigenvalue, $\lambda = -\frac{\sigma^2}{(\bar{x})^2}\frac{\pi^2}{2}$. For the comparative static exercise, the bands widen so λ falls in magnitude. This implies the second part of the proposition: as $t \to +\infty$, $Y^{imp}(t) - Y^{per}(t) > 0$, and since $B_j > 0$, $\frac{Y^{per}(t)}{Y^{imp}(t)} \to 0$. To show the first part, we need to study the behavior close to $t = 0$. The following proposition describes the asymptotic behavior of $Y(t)$ close to $t = 0$:

Proposition 12. ***Response near*** $t = 0$.*For* $t \downarrow 0$ $Y(t)$ *satisfies:*

$$Y(t) = 1 - 2\sqrt{\frac{2Nt}{\pi(1 - \mathbb{E}[\tilde{x}^2])}} + o(\sqrt{t}),$$

Proof. See Online Appendix OA.C.2. □

Given this, we note that since $\sqrt{\frac{2N}{\pi(1-\mathbb{E}[\tilde{x}^2])}} > \sqrt{\frac{2N}{\pi}}$, we have $Y^{imp}(t) - Y^{per}(t) < 0$ for all $t \in (0, \varepsilon)$ for a fixed N. Since the frequency increases, the comparison continues to hold. By continuity, Bolzano's theorem implies that there is at least one crossing, with $Y^{imp}(t)$ below $Y^{per}(t)$ before the crossing and above afterwards. This implies that there exists a t_1 such that $Y^{imp}(t) < Y^{per}(t)$ for all $t \in (0, t_1)$.

C.5 Proposition 5

If c is small, then we can obtain the inaction region from the approximation derived in Proposition 3. In particular from C.1, since $[\mathbb{E}[(1 - \tilde{x}^2)^2]]^{-1} \geqslant 1$, with equality only if $\tilde{x} = 0$

almost surely, the size of price changes increases. The frequency is:

$$N = \frac{\sigma^2}{\bar{x}^2(1-\mathbb{E}[\tilde{x}^2])} = \sqrt{\frac{\sigma^2 B}{6c}}\sqrt{\frac{\mathbb{E}[(1-\tilde{x}^2)^2]}{(1-\mathbb{E}[\tilde{x}^2])^2}}(1+o(1)).$$

Due to Jensen's inequality, if we let $Y = 1-\tilde{x}^2$, then $\mathbb{E}[(1-\tilde{x}^2)^2] \geqslant (\mathbb{E}_F[1-\tilde{x}^2])^2$, so N weakly increases (strictly if $1-x^2$ is not degenerate). For non-neutrality:

$$\mathcal{M}'(0) = \frac{1}{6N}\frac{\int_{-\bar{x}}^{\bar{x}}(\bar{x}^2-x^2)^2 dF}{\left(\bar{x}^2-\int_{-\bar{x}}^{\bar{x}}x^2 dF\right)^2} = \sqrt{\frac{c}{6B\sigma^2}}\sqrt{\frac{\mathbb{E}[(1-\tilde{x}^2)^2]}{\mathbb{E}[(1-\tilde{x}^2)]^2}}(1+o(1)),$$

which for the same reason as above, increases with imperfect reset since $\frac{\mathbb{E}[(1-\tilde{x}^2)^2]}{\mathbb{E}[(1-\tilde{x}^2)]^2} \geqslant 1$. As the proof shows, imperfect reset increases the frequency of price adjustment with the square root of $\frac{\mathbb{E}[(1-\tilde{x}^2)^2]}{\mathbb{E}[(1-\tilde{x}^2)]^2} = 1+\frac{\mathbb{V}_F(\tau(x))}{\mathbb{E}_F[\tau(x)]^2}$, and this force decreases non-neutrality. However, the effect of dispersion increases non-neutrality by a factor of $\frac{\mathbb{E}[(1-\tilde{x}^2)^2]}{\mathbb{E}[(1-\tilde{x}^2)]^2}$, so this effect weakly dominates the increase in average frequency. The key for this result is that inaction bands endogenously expand. Intuitively, firms now landing close to 0 will take longer to adjust for the first time after the shock, and this effect dominates the higher adjustment frequency (which is driven by firms landing closer to the inaction bands).

C.6 Proposition 9

I show that the CIR can diverge using the formulae in Proposition 11. This follows from taking the limit to $\beta_1 \to -\frac{1}{2}$ and choosing the value of $\text{Kurt}(x^*)$ appropriately. We need the constraint that $\text{Kurt}(\Delta p) > 2$ because, as we take the limit to $\sigma^2 \to 0$, price changes are equal to $\Delta p = x_2^* - x_1^*$, i.e., the difference between two independent draws from the reset price distribution. It turns out that the minimum value for the kurtosis of such a random variable is 2, so if $\text{Kurt}(\Delta p) < 2$, we cannot make σ^2 arbitrarily small. This follows from:

$$\text{Kurt}(x_2^* - x_1^*) = \frac{\mathbb{E}[(x^*)^4] + 6Var(x^*)^2 + \mathbb{E}[(x^*)^4]}{\text{Var}(x_1^* - x_2^*)^2} = \frac{1}{2}(\text{Kurt}(x^*)+3),$$

I show in Online Appendix OA.B.7 that if $\text{Kurt}(\Delta p) > 2$, we can always find a distribution for x^* that can generate that kurtosis even if $\sigma^2 = 0$. Online Appendix OA.B.7 also constructs specific examples for each limit case ($CIR \to \pm\infty$). In both cases, I set $\sigma^2 = 0$. For

$CIR = +\infty$, the idea is to have a set of firms that never adjusts prices: all price changes are driven by other firms. After the shock, these firms have their prices misalisgned forever, and thus the CIR diverges. For the other case, the example features a mass of firms very close to a discontinuity in Λ. When there is a shock, this mass of firms moves into a region with much larger adjustment probability. This triggers a large amount of price increases relative to the size of the shock, and thus prices overshoot forever.

C.7 Proposition 11

The proof follows from a sequence of Lemmas. I first present them in order to understand the basic argument, and later on I present the proofs.

Lemma 1. ***CIR.*** *Assume $\sigma^2 > 0$. The CIR $\mathcal{M}'(0)$ satisfies $\mathcal{M}'(0) = \dfrac{Var(\check{x}) + \bar{M}}{\sigma^2}$, where $-Var(x^*) \leqslant \bar{M} \leqslant 0$. The lower bound is attained with Calvo ($\Lambda(y) = N, \bar{y} = +\infty$).*

Proof. See Appendix C.7. □

This result has several implications. First, note that if reset gaps are deterministic, then $\bar{M} = 0$ and we obtain $\mathcal{M}'(0) = \text{Var}(\check{x})/\sigma^2$. In that case, price changes are solely driven by σ^2. Thus, if the variance of gaps is large relative to the size of the underlying shocks, it must be that pricing frictions are more severe, and thus non-neutrality is higher. Once we add imperfect reset, there is still a connection between the CIR and $\text{Var}(\check{x})$, but it is not direct: there is the additional $\bar{M}$ term, which depends on all objects $(\sigma^2, n, \bar{y}, \Lambda, F)$ in a nontrivial way, but can be bounded. Intuitively, high $\text{Var}(x^*)$ increases $\text{Var}(\check{x})$ regardless of the size of pricing frictions, so we need to add the nonpositive correction $\bar{M}$ to correct for this distortion. It is also interesting to note that the lower bound on non-neutrality is attained when hazards are completely flat, as in Calvo. This is in line with Proposition 8: the interaction between imperfect reset and state dependence increases non-neutrality.

From Lemma 1 it is clear that if we aspire to at least bound non-neutrality, we need to measure $\sigma^2, \text{Var}(\check{x})$ and $\text{Var}(x^*)$. I show how to do that in a sequence of Lemmas below. The first lemma relates σ^2 to the frequency and variance of price changes, and $\text{Var}(x^*)$.

Lemma 2. ***Variance of Idiosyncratic Shocks.*** *The variance of price changes satisfies $Var(\Delta p) = 2\,Var(x^*) + \frac{\sigma^2}{N}$ and thus $\sigma^2 = N\,Var(\Delta p)(1 - 2\frac{Var(x^*)}{Var(\Delta p)})$.*

Proof. See Appendix C.7. □

This lemma shows that, once reset is imperfect, there are two drivers of price changes: the usual idiosyncratic shocks σ^2, and also variation in reset gaps, $\text{Var}(x^*)$. For a given variance of price changes, this automatically implies that the inferred variance of idiosyncratic shocks would be lower in the presence of imperfect reset. Turning to $\text{Var}(\check{x})$, the following lemma links $\text{Var}(\check{x})$ to observable moments (most notably, the kurtosis of price changes), but it also depends on $\text{Var}(x^*), \text{Kurt}(x^*)$.

Lemma 3. ***Variance of Price Gaps.*** *If $\sigma^2 > 0$, the stationary variance of gaps is:*

$$Var(\check{x}) = \frac{Kurt(\Delta p)\, Var(\Delta p)}{6(1 - 2\frac{Var(x^*)}{Var(\Delta p)})} - Var(x^*)\left[1 + \frac{1}{3}\frac{Var(x^*)(Kurt(x^*)+3)}{Var(\Delta p)\left(1 - 2\frac{Var(x^*)}{Var(\Delta p)}\right)}\right].$$

Proof. See Appendix C.7. □

Lemmas 2 and 3 show that σ^2 and $\text{Var}(\check{x})$ are measurable up to some key moments of the distribution of reset gaps: $\text{Var}(x^*)$ and $\text{Kurt}(x^*)$. Using Proposition 10, we can measure these moments directly by means of β_1 and β_2. Replacing the relevant terms in the bounds in Lemma 1 and operating:

$$\begin{aligned}
\frac{\text{Var}(\check{x})}{\sigma^2} &= \frac{1}{N}\left[\frac{\text{Kurt}(\Delta p)}{6(1+2\beta_1)^2} + \frac{\beta_1}{1+2\beta_1} - \frac{\beta_2(\text{Kurt}(\Delta p)-1)+4\beta_1^2}{3(1+2\beta_1)^2}\right] \\
&= \frac{1}{N}\frac{(1-2\beta_2)\text{Kurt}(\Delta p) + 2\beta_1(2\beta_1+3)+2\beta_2}{6(1+2\beta_1)^2} = \frac{U}{N}, \qquad \text{and} \\
\frac{\text{Var}(\check{x}) - \text{Var}(x^*)}{\sigma^2} &= \frac{U}{N} + \frac{\beta_1}{N(1+2\beta_1)} = \frac{L}{N}.
\end{aligned}$$

Therefore $L/N \leqslant CIR \leqslant U/N$, as claimed.

C.7.1 Proving the Lemmas

Throughout the proofs, I let x be a random variable with distribution equal to that of a price gap conditional on a finished spell (but before reset), x^* be a random variable with distribution F (i.e., that of reset gaps), and $\check{x}$ be a random variable with distribution equal to the price gaps of ongoing spells. I first introduce a lemma that will be of use in several proofs. Furthermore, I assume that $\text{supp}(y^*) \subset [0, \bar{y}]$.[22]

[22] If we draw x^* such that we leave the inaction region, the firm immediately pays the cost of reset and draws again, but this is not counted as an observed price change.

Lemma 4. ***Link between ongoing and completed distributions****. For any even positive integer $m = 2, 4, \ldots$ such that these moments exist:*

$$\mathbb{E}[x^m] = \mathbb{E}[(x^*)^m] + m(m-1)\frac{\sigma^2}{2N}\mathbb{E}[(\check{x})^{m-2}].$$

Proof. Consider the process $z_{i,t} = x_{i,t}^m$. Then by Itô's lemma, $dx_{i,t}^m = \frac{\sigma^2}{2}m(m-1)x_{i,t}^{m-2}dt + \sigma m x_{i,t}^{m-1}dW_{i,t}$. Integrating from 0 to the (random) duration until the next price change τ,

$$(x_i(\tau))^m = (x_i(0))^m + \frac{\sigma^2}{2}m(m-1)\int_0^\tau (x_i(s))^{m-2}ds + \sigma m\int_0^\tau (x_i(s))^{m-1}dW_{i,s}.$$

Taking expectations and using that $N = 1/\mathbb{E}[\tau]$ and the martingale property we get:

$$\begin{aligned}\mathbb{E}[(x_i(\tau))^m] &= \mathbb{E}[(x_i(0))^m] + \frac{\sigma^2}{2N}m(m-1)\mathbb{E}[\int_0^\tau (x_i(s))^{m-2}\frac{1}{\mathbb{E}[\tau]}ds] \\ \mathbb{E}[x^m] &= \mathbb{E}[(x^*)^m] + \frac{\sigma^2}{2N}m(m-1)\mathbb{E}[(\check{x})^{m-2}].\end{aligned}$$

□

C.7.2 Proposition 10

In order to see why, let $\Delta p_{i,n-1}$ and $\Delta p_{i,n}$ be two consecutive price changes for the same product of the same firm, where $\Delta\tau$ is the duration of the spell. Using the definition of price changes and the evolution of x:

$$\Delta p_{i,n-1} = x_{i,n-1}^* - x_{i,\tau_{i,n-1}} \quad \text{and} \quad \Delta p_{i,n} = x_{i,n}^* - \underbrace{(x_{i,n-1}^* + \sigma(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}))}_{x_{i,\tau_{i,n}}},$$

where $W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}$ is a Brownian increment, and $x_{i,\tau_{i,n-1}}$ denotes the value of x right *before* reset. The covariance between consecutive price changes is:

$$\text{Cov}(\Delta p_{i,n-1}, \Delta p_{i,n}) = \text{Cov}(x_{i,n-1}^* - x_{i,\tau_{i,n-1}}, x_{i,n}^* - (x_{i,n-1}^* + \sigma(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}))) = -\text{Var}(x^*)$$

since $\Delta p_{i,n}$ is independent of $x_{i,\tau_{i,n-1}}$, $x_{i,n}^*$ is independent of $x_{i,n-1}^*$, and $W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}$ is uncorrelated with $x_{i,n-1}^*$ (*not* independent). The lack of independence is because the duration of the spell $\Delta\tau_{i,n}$ depends on the initial value of x for that spell. If $|x_{i,n-1}^*|$ is large and there is state dependence, then the duration $\Delta\tau_{i,n}$ is expected to be smaller, which affects the

distribution of $W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}$. However, lack of independence is not an issue since:

$$\begin{aligned}\text{Cov}(x^*_{i,n-1}, W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}) &= \mathbb{E}[x^*_{i,n-1}(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})] - \mathbb{E}[x^*_{i,n-1}]\mathbb{E}[W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}] \\ &= \mathbb{E}[x^*_{i,n-1}\mathbb{E}[(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})|\mathcal{F}_{\tau_{i,n-1}}]] = 0,\end{aligned}$$

where in the first line I used that $\mathbb{E}[x^*_{i,n-1}] = 0$ and in the second line I used that since $W_{i,t}$ is a martingale, by the optional stopping theorem the stopped version also is, and thus $\mathbb{E}[W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}|\mathcal{F}_{\tau_{i,n-1}}] = 0$. Thus, the regression coefficient of predictability across successive price spells is:

$$\beta_1 = \frac{\text{Cov}(\Delta p_{i,n-1}, \Delta p_{i,n})}{\text{Var}(\Delta p_{i,n-1})} = -\frac{\text{Var}(x^*)}{\text{Var}(\Delta p)} = -\frac{\text{Var}(x^*)}{2Var(x^*) + \frac{\sigma^2}{N}},$$

so it is directly informative about the share of variance that is explained by reset gaps. This implies that we can recover σ^2 from $\text{Var}(\Delta p), N, \beta_1$ as $NVar(\Delta p)(1 + 2\beta_1) = \sigma^2$, and since $\beta_1 < 0$, $NVar(\Delta p) > \sigma^2$. Note furthermore that $\beta_1 > -1/2$ so $1 + 2\beta_1 > 0$.

Consider now a regression of squared price changes on squared price changes. In that case we need to compute:

$$\begin{aligned}\text{Cov}((\Delta p_{i,n-1})^2, (\Delta p_{i,n})^2) &= \text{Cov}((x^*_{i,n-1} - x_{i,\tau_{i,n-1}})^2, (x^*_{i,n} - x^*_{i,n-1} - \sigma(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}))^2) \\ &= \text{Cov}((x^*_{i,n-1})^2 - 2x^*_{i,n-1}x_{i,\tau_{i,n-1}} + (x_{i,\tau_{i,n-1}})^2, \\ &\quad (x^*_{i,n} - x^*_{i,n-1})^2 - 2(x^*_{i,n} - x^*_{i,n-1})\sigma(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}) + \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2).\end{aligned}$$

Let's work term by term: $\text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2)$ survives due to the lack of independence between $x^*_{i,n-1}$ and $W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}$. Then since $\mathbb{E}[x^*_{i,n-1}x_{i,\tau_{i,n-1}}] = 0$ and since $x_{i,\tau_{i,n-1}}$ is independent of $x^*_{i,n-1}, W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}}$, we have $\text{Cov}(x^*_{i,n-1}x_{i,\tau_{i,n-1}}, (W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2) = \mathbb{E}[x^*_{i,n-1}x_{i,\tau_{i,n-1}}(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2] = \mathbb{E}[x_{i,\tau_{i,n-1}}]\mathbb{E}[x^*_{i,n-1}(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2] = 0$. A similar argument shows that $\text{Cov}(-2x^*_{i,n-1}x_{i,\tau_{i,n-1}}, (x^*_{i,n} - x^*_{i,n-1})^2 + 2(x^*_{i,n} - x^*_{i,n-1})\sigma(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})) = 0$. The variable $x_{i,\tau_{i,n-1}}$ is independent of the future variables in this covariance so that term drops from the covariance. Thus, we end up with

$$\begin{aligned}\text{Cov}((\Delta p_{i,n-1})^2, (\Delta p_{i,n})^2) &= \text{Cov}((x^*_{i,n-1})^2, (x^*_{i,n} - x^*_{i,n-1})^2) + \text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2) \\ &= \text{Cov}((x^*_{i,n-1})^2, (x^*_{i,n})^2 - 2x^*_{i,n}x^*_{i,n-1} + (x^*_{i,n-1})^2) \\ &\quad + \text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2) \\ &= \text{Cov}((x^*_{i,n-1})^2, (x^*_{i,n-1})^2) + \text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2) \\ &= \text{Var}((x^*_{i,n-1})^2) + \text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2) \\ &= \mathbb{E}[(x^*_{i,n-1})^4] - \mathbb{E}[(x^*_{i,n-1})^2]^2 + \text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2) \\ &= \text{Var}(x^*)^2[\text{Kurt}(x^*) - 1] + \text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2).\end{aligned}$$

So we need to measure the term $\text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2)$. In order to account for the extra term, note that:

$$\text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2) \quad = \quad \sigma^2\text{Cov}((x^*_{i,n-1})^2, [(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2 - \Delta\tau_{i,n}] + \Delta\tau_{i,n}).$$

Define $\mathcal{W}_{i,n,s} = (W_{\tau_{i,n-1}+s} - W_{\tau_{i,n-1}})^2 - s$. Note that $d\mathcal{W}_s = d(W_{\tau_{i,n-1}+s} - W_{\tau_{i,n-1}})^2 - ds = 2(W_{\tau_{i,n-1}+s} - W_{\tau_{i,n-1}})dW_{\tau_{i,n-1}+s} + ds - ds = 2(W_{\tau_{i,n-1}+s} - W_{\tau_{i,n-1}})dW_{\tau_{i,n-1}+s}$ so it is a local martingale, and since it is square-integrable, it is also a martingale. Therefore its stopped version is also a martingale. Therefore:

$$\begin{aligned}
\text{Cov}((x^*_{i,n-1})^2, (W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2 - \Delta\tau_{i,n}) &= \text{Cov}((x^*_{i,n-1})^2, \mathcal{W}_{i,n,\Delta\tau_{i,n}}) \\
&= \mathbb{E}[(x^*_{i,n-1})^2\mathcal{W}_{i,n,\Delta\tau_{i,n}}] - \mathbb{E}[(x^*_{i,n-1})^2]\mathbb{E}[\mathcal{W}_{i,n,\Delta\tau_{i,n}}] \\
&= \mathbb{E}[(x^*_{i,n-1})^2\mathbb{E}[\mathcal{W}_{i,n,\Delta\tau_{i,n}}|\mathcal{F}_{\tau_{i,n-1}}]] - \mathbb{E}[(x^*_{i,n-1})^2]\mathbb{E}[\mathcal{W}_{i,n,\Delta\tau_{i,n}}] \\
&= 0
\end{aligned}$$

because of the definition of $\mathcal{W}_{i,n,s}$, the fact that Brownian increments are independent, and the martingale property, $\mathbb{E}[\mathcal{W}_{i,n,\Delta\tau_{i,n}}|\mathcal{F}_{\tau_{i,n-1}}] = \mathbb{E}[\mathcal{W}_{i,n,\Delta\tau_{i,n}}] = 0$. Therefore,

$$\text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2) \quad = \quad \sigma^2\text{Cov}((x^*_{i,n-1})^2, \Delta\tau_{i,n}).$$

Note that the RHS can be measured using the following regression:

$$\Delta\tau_{i,n} \quad = \quad \alpha_3 + \tilde{\beta}_3(\Delta p_{i,n-1})^2 + \text{Error},$$

i.e., we try to predict the duration of a pricing spell using the squared size of the previous price change (i.e., right before that spell started). Note that:

$$\begin{aligned}
\text{Cov}((\Delta p_{i,n-1})^2, \Delta\tau_{i,n}) &= \text{Cov}((x^*_{i,n-1} - x_{i,\tau_{i,n-1}})^2, \Delta\tau_{i,n}) \\
&= \text{Cov}((x^*_{i,n-1})^2 - 2x^*_{i,n-1}x_{i,\tau_{i,n-1}} + (x_{i,\tau_{i,n-1}})^2, \Delta\tau_{i,n}),
\end{aligned}$$

due to independence, $\text{Cov}(x^2_{i,\tau_{i,n-1}}, \Delta\tau_{i,n}) = 0$ (since all that matters for $\Delta\tau_{i,n}$ is the initial condition, which under our assumptions is independent of the value of x before reset). Note also that since $\mathbb{E}[x^*_{i,n-1}x_{i,\tau_{i,n-1}}] = 0$, we have:

$$\begin{aligned}
\text{Cov}(x^*_{i,n-1}x_{i,\tau_{i,n-1}}, \Delta\tau_{i,n}) &= \mathbb{E}[x^*_{i,n-1}x_{i,\tau_{i,n-1}}\Delta\tau_{i,n}] \\
&= \mathbb{E}[x_{i,\tau_{i,n-1}}]\mathbb{E}[x^*_{i,n-1}\Delta\tau_{i,n}] = 0,
\end{aligned}$$

since $\mathbb{E}[x_{i,\tau_{i,n-1}}] = 0$ and $x_{i,\tau_{i,n-1}}$ is independent of $x^*_{i,n-1}$ and $\Delta\tau_{i,n}$. This implies that:

$$\tilde{\beta}_3 \quad = \quad \frac{\text{Cov}((\Delta p_{i,n-1})^2, \Delta\tau_{i,n})}{\text{Var}((\Delta p_{i,n})^2)} = \frac{\text{Cov}((x^*_{i,n-1})^2, \Delta\tau_{i,n})}{\text{Var}(\Delta p)^2[\text{Kurt}(\Delta p) - 1]},$$

and thus we can measure the extra term directly with this moment. Completing the derivation for $\tilde{\beta}_2$, we have that $\text{Var}((\Delta p)^2) = \text{Var}(\Delta p)^2[\text{Kurt}(\Delta p) - 1]$ so the regression coefficient of a predictability regression on squared price changes is:

$$\begin{aligned}
\tilde{\beta}_2 &= \frac{\text{Cov}((\Delta p_{i,n-1})^2, (\Delta p_{i,n})^2)}{\text{Var}((\Delta p)^2)} \\
&= \frac{\text{Var}(x^*)^2[\text{Kurt}(x^*) - 1] + \text{Cov}((x^*_{i,n-1})^2, \sigma^2(W_{i,\tau_{i,n}} - W_{i,\tau_{i,n-1}})^2)}{\text{Var}(\Delta p)^2[\text{Kurt}(\Delta p) - 1]} \\
&= (\beta_1)^2 \frac{\text{Kurt}(x^*) - 1}{\text{Kurt}(\Delta p) - 1} + \tilde{\beta}_3 N \text{Var}(\Delta p)(1 + 2\beta_1).
\end{aligned}$$

C.7.3 Lemma 2

Since price changes have distribution $\Delta p = x^* - x$ and x^* and x are independent,

$$\text{Var}(\Delta p) \quad = \quad \text{Var}(x^*) + \text{Var}(x) = 2\text{Var}(x^*) + \frac{\sigma^2}{N}$$

using Lemma 4 for $m = 2$. From here, some rearranging yields $N\text{Var}(\Delta p)(1 - 2\frac{\text{Var}(x^*)}{\text{Var}(\Delta p)}) = \sigma^2$, and using Proposition 10 we replace $\beta_1 = -\frac{\text{Var}(x^*)}{\text{Var}(\Delta p)}$ for the result.

C.7.4 Lemma 3

The kurtosis is defined as $\text{Kurt}(\Delta p) = \frac{\mathbb{E}[(\Delta p)^4]}{\text{Var}(\Delta p)^2}$. But $\mathbb{E}[(\Delta p)^4] = \mathbb{E}[(x^* - x)^4] = \mathbb{E}[(x^*)^4] + 6\mathbb{E}[(x^*)^2]\mathbb{E}[x^2] + \mathbb{E}[x^4]$. Using Lemma 4 for $m = 2$ and 4 to substitute for $\mathbb{E}[x^2]$ and $\mathbb{E}[x^4]$, using $\mathbb{E}[(x^*)^2] = \text{Var}(x^*)$, and dividing by $\text{Var}(\Delta p)^2$ yields the result.

C.7.5 Lemma 1

We first note that the function $m = -\mathbb{E}\left[\int_0^\tau \sum_{i=1}^n x_i(t)dt | x(0) = x\right]$ does not depend on F^y, since it describes what happens for a given initial point until reset. As described in Alvarez et al. (2016a), the function m only depends on $z = \sum_{i=1}^n x_i$ and on $y = \sum_{i=1}^n x_i^2$. Furthermore,

between resets, these variables follow the processes:

$$dy = n\sigma^2 dt + 2\sigma\sqrt{y}dW^a \quad \text{and} \quad dz = \sqrt{n}\sigma\left[\frac{z}{\sqrt{ny}}dW^a + \sqrt{1-(\frac{z}{\sqrt{ny}})^2}dW^b\right],$$

where W^a and W^b are independent standard Brownian motions. The function m solves the following PDE $\Lambda(y)m = -z + n\sigma^2 m_y + \frac{1}{2}[4\sigma^2 y m_{yy} + m_{zz}(n\sigma^2) + 4\sigma^2 z m_{zy}]$. We can guess and verify that the solution has a form $m(z,y) = zk(y)$ where k solves $\Lambda(y)k = -1 + \sigma^2(n+2)k_y + 2\sigma^2 y k_{yy}$. As pointed out in Alvarez et al. (2016a), $k_n(y)$ is actually minus the expected time until the next change starting at y for an economy with $n+2$ products. We now need to derive the stationary density g using the KFE. Here is where the biggest difference lies, since we have mass entering at different points. We can write the KFE heuristically as $\Lambda(y)g = -n\sigma^2 g_y + \frac{1}{2}\sigma^2\frac{\partial^2}{\partial y^2}[4yg] + N(dF^y)$ where dF^y is the "density" of the reset value of y. This is correct if F^y admits a smooth density, but this may not be the case (especially since the benchmark case with no mistakes has a mass point at $x = 0$). More generally, we can write the Kolmogorov Forward Equation in weak form as:

$$\int_0^{\bar{y}}(-n\sigma^2 g_y + \frac{1}{2}\sigma^2\frac{\partial^2}{\partial y^2}[4yg] - \Lambda(y)g)\phi(y)dy + N\int_0^{\bar{y}}\phi(y)dF^y - \phi(\bar{y})(N - \int \Lambda g dy) = 0, \quad \text{(C.3)}$$

for any smooth function $\phi(y)$. There is some distribution of z conditional on y, $q(z|y)$ that we characterize later on. Here we see that except for the case in which $n = 1$, the generators for g and m are different. Letting $h(z,y) = q(z|y)g(y)$ be the joint stationary density of (z,y), the IRF for a shock δ is $\mathcal{M}(\delta) = \frac{1}{n}\int\int m(z - n\delta, y - 2z\delta + n\delta^2)h(z,y)dzdy$. Differentiating, evaluating at $\delta = 0$ and using that $m = zk(y)$,

$$\begin{aligned}
\mathcal{M}'(0) &= \frac{1}{n}\int\int\left[-nk(y) + zk'(y)(-2z)\right]h(z,y)dzdy \\
&= -\frac{1}{n}\int\int\left[nk(y) + 2z^2k'(y)\right]h(z,y)dzdy \\
&= -\frac{1}{n}\left[\int nk(y)g(y)dy + 2\int[\int z^2 q(z|y)dz]k'(y)g(y)dy\right].
\end{aligned}$$

So to continue to make progress we need to compute $\int z^2 q(z|y)dz = \mathbb{E}[(\sum_{i=1}^n x_i)^2 | \sum_{i=1}^n (x_i)^2 = y]$. To do so, note that $\mathbb{E}[(\sum_{i=1}^n x_i)^2 | \sum_{i=1}^n (x_i)^2 = y] = \mathbb{E}[\sum_{i=1}^n (x_i)^2 + \sum_i \sum_{j\neq i} x_i x_j | \sum_{i=1}^n (x_i)^2 = y] = y + \mathbb{E}[\sum_i \sum_{j\neq i} x_i x_j | \sum_{i=1}^n (x_i)^2 = y]$, but by symmetry of the initial condition and absence of drift, for each of the terms there is equal chance that x_j has either sign, so $\mathbb{E}[x_i x_j | \sum_{i=1}^n (x_i)^2 = y] = 0$ for all $i \neq j$ and thus $\int z^2 q(z|y)dz = y$. Therefore we get

$\mathcal{M}'(0) = -\left[\int [k(y) + \frac{2}{n} y k'(y)] g(y) dy\right].$

Let F^y be the distribution of $y^* = \sum_{i=1}^n (x_i^*)^2$ induced by the individual reset-gap distribution F. Using (C.3) with $\phi = k(y)y$ we obtain:

$$\begin{aligned}
\int_0^{\bar{y}} (-n\sigma^2 g' + \frac{1}{2}\sigma^2 \frac{\partial^2}{\partial y^2}[4yg]) k(y) y dy + N \int_0^{\bar{y}} k(y) y dF^y &= \int_0^{\bar{y}} \Lambda(y) g(y) k(y) y dy \\
\int_0^{\bar{y}} (-n\sigma^2 g' + \frac{1}{2}\sigma^2 \frac{\partial^2}{\partial y^2}[4yg]) k(y) y dy + N \int_0^{\bar{y}} k(y) y dF^y &= \int_0^{\bar{y}} [-1 + \sigma^2(n+2)k' + 2\sigma^2 y k''] g(y) y dy,
\end{aligned}$$

so

$$\int_0^{\bar{y}} [N k(y) y dF^y + g y dy] = \sigma^2 \int_0^{\bar{y}} \left[g((n+2)k' + \frac{1}{2}(4y)k'') - k(-ng' + \frac{1}{2}(4yg)'') \right] y dy.$$

but

$$\begin{aligned}
[g((n+2)k' + \frac{1}{2}(4y)k'') - k(-ng' + \frac{1}{2}(4yg)'')] &= 2k'g + n(k'g + g'k) + \frac{1}{2}[(4yg)k'' - (4yg)''k] \\
&= 2k'g + n(kg)' - \frac{1}{2}[k(4yg)' - k'(4yg)]'.
\end{aligned}$$

Replacing that expression, operating, and integrating by parts:

$$\begin{aligned}
\int_0^{\bar{y}} [N k(y) y dF^y + g y dy] &= \sigma^2 \left(\int_0^{\bar{y}} [2k'g + n(kg)' - \frac{1}{2}[k(4yg)' - k'(4yg)]'] y dy \right) \\
&= \sigma^2 \left(\int_0^{\bar{y}} [2k'gy] dy + n \int_0^{\bar{y}} (kg)' y dy - \frac{1}{2} \int_0^{\bar{y}} [k(4yg)' - k'(4yg)]'] y dy \right) \\
&= \sigma^2 \left(\int_0^{\bar{y}} [2k'gy] dy + n \left[kgy|_0^{\bar{y}} - \int_0^{\bar{y}} kg dy \right] \right. \\
&\quad \left. -2 \left([k(yg)' - k'(yg)] y|_0^{\bar{y}} - \int_0^{\bar{y}} [k(yg)' - k'(yg)] dy \right) \right) \\
&= \sigma^2 \left(\int_0^{\bar{y}} [2k'gy] dy + n \left[- \int_0^{\bar{y}} kg dy \right] + 2 \left(\int_0^{\bar{y}} [k(yg)' - k'(yg)] dy \right) \right) \\
&= \sigma^2 \left(2 \int_0^{\bar{y}} [k'gy] dy - n \left[\int_0^{\bar{y}} kg dy \right] + 2 \left(\int_0^{\bar{y}} [k(yg)' - k'(yg)] dy \right) \right),
\end{aligned}$$

where the boundary conditions vanish since $y = 0$ and $g(\bar{y}) = k(\bar{y}) = 0$ (and g, k are well behaved close to the boundaries). We then have

$$\int_0^{\bar{y}} [k(yg)' - k'(yg)] dy = k(yg)|_0^{\bar{y}} - 2 \int_0^{\bar{y}} k'gy dy = -2 \int_0^{\bar{y}} k'gy dy,$$

and thus

$$\begin{aligned}\int_0^{\bar{y}}[Nk(y)ydF^y + gydy] &= \sigma^2\left(2\int_0^{\bar{y}}[k'gy]dy - n\left[\int_0^{\bar{y}}kgdy\right] - 4\int_0^{\bar{y}}k'gydy\right)\\ &= -\sigma^2\left(n\left[\int_0^{\bar{y}}kgdy\right] + 2\int_0^{\bar{y}}k'gydy\right)\\ \int_0^{\bar{y}}[Nk(y)ydF^y + gydy] &= \sigma^2 n\mathcal{M}'(0),\end{aligned}$$

since $\int_0^{\bar{y}} ygdy = \mathbb{E}[\sum_{i=1}^n (x_i)^2] = n\text{Var}(\check{x})$ (i.e., n times the variance of ongoing spells) we can divide by n to get

$$\sigma^2\mathcal{M}'(0) = \text{Var}(\check{x}) + \frac{N}{n}\int_0^{\bar{y}} k(y)ydF^y,$$

so letting $\bar{M} = \frac{N}{n}\int_0^{\bar{y}} k(y)ydF^y$, we get the first part of the proposition. Our task is now to bound the second term.

Proving the Bound. First, note that as mentioned in Alvarez et al. (2016a), $k(y) = -\mathcal{T}_{n+2}(y)$ where $\mathcal{T}$ is the expected time until the next price adjustment in an economy with $n+2$ products if we start at $y(0) = y$ (but with the same $\bar{y}$). Note that by definition $k(y)$ is always negative, and y is always positive, so $\int_0^{\bar{y}} k(y)ydF^y/n \leqslant 0$. To prove the bound in the other direction we need to show that $-\frac{\text{Var}(x^*)}{N} \leqslant \int_0^{\bar{y}} k(y)ydF^y/n$. Note that if we have a pure Calvo model and $k(y) = -\frac{1}{N}$, then the above holds with equality since $\int_0^{\infty} k(y)ydF^y/n = -\frac{1}{N}\mathbb{E}[\sum_{i=1}^n (x_i^*)^2]/n$ by definition. Therefore, we need to show that any state dependence in the hazard makes this expectation smaller in magnitude. Note that:

$$\frac{1}{n}\int_0^{\bar{y}} k(y)ydF^y = \mathbb{E}_{y^*}[k(y^*)y^*]/n = \mathbb{E}_{y^*}[k(y^*)]\mathbb{E}_{y^*}[y^*]/n + \text{Cov}(k(y^*), y^*)/n,$$

where I use the notation $\mathbb{E}_{y^*}[w(y^*)] = \int_0^{\bar{y}} w(y)dF^y(y)$. Since $k(y) = -\mathcal{T}_{n+2}(y)$ and the expected time is decreasing in y, then k is increasing, so $\text{Cov}(k(y^*), y^*) \geqslant 0$. Furthermore, $\mathbb{E}_{y^*}[y^*]/n = \text{Var}(x^*)$, so $\int_0^{\bar{y}} k(y)ydF^y/n \geqslant \text{Var}(x^*)\mathbb{E}_{y^*}[k(y^*)]$. So if we prove that $\mathbb{E}_{y^*}[k(y^*)] \geqslant -\frac{1}{N}$ we are done. Note that $-\frac{1}{N} = -\mathbb{E}_{y^*}[\mathcal{T}_n(y^*)]$, so showing that inequality is equivalent to showing that $\mathbb{E}_{y^*}[\mathcal{T}_{n+2}(y^*)] \leqslant \mathbb{E}_{y^*}[\mathcal{T}_n(y^*)]$. Let $y_j(t)$ be a process with:

$$y_j(t) = y_j(0) + \sigma^2 b_j t + 2\sigma\int_0^t \sqrt{y_j(s)}dW.$$

Note therefore that $dy_j = \sigma^2 b_j dt + 2\sigma\sqrt{y_j}dW$. Here $y(0)$ is drawn according to y^*. Condition on an initial point $y(0) \geqslant 0$. I now show that $y_{n+2} \geqslant y_n$ pathwise, by invoking Theorem 1.1 in Ikeda and Watanabe (1977). Note that y_j is the solution of an SDE of the form $dX = \sigma^2 b_j dt + \sigma(x)dW$, with $b_{n+2} = n+2 \geqslant n = b_n$ and $\sigma(x) = 2\sigma\sqrt{x}$. Note furthermore that $|\sigma(x) - \sigma(y)| = 2\sigma|\sqrt{x} - \sqrt{y}| \leqslant 2\sigma\sqrt{|x-y|} = 2\sigma\rho(|x-y|)$, and thus for any $\epsilon > 0$, $\int_0^\epsilon \rho(\xi)^{-2}d\xi = (2\sigma)^{-2}\int_0^\epsilon \xi^{-1}d\xi = +\infty$. This implies that all conditions of the theorem stated above hold, and therefore, conditioning on $y_n(0) = y_{n+2}(0)$, $y_{n+2} \geqslant y_n$ pathwise.

Since the hazard is non-decreasing in y, comparing the processes y_n and y_{n+2} conditioning on the same initial condition and path of Brownian shocks, we must have $\int_0^t \Lambda(y_{n+2}(s))ds \geqslant \int_0^t \Lambda(y_n(s))ds$ so $\mathbb{E}[e^{-\int_0^t \Lambda(y_{n+2}(s))ds} - e^{-\int_0^t \Lambda(y_n(s))ds}|y_{n+2}(0) = y_n(0) = y^*, \{W\}] \leqslant 0$ pathwise (i.e., for any given realization of Brownian shocks), and thus:

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[\int_0^\infty e^{-\int_0^t \Lambda(y_{n+2}(s))ds}dt|y_{n+2}(0) = y^*, \{W\}]] &\leqslant \mathbb{E}[\mathbb{E}[\int_0^\infty e^{-\int_0^t \Lambda(y_n(s))ds}dt|y_n(0) = y^*, \{W\}]] \\
\mathbb{E}[\mathbb{E}[\int_0^\tau dt|y_{n+2}(0) = y^*, \{W\}]] &\leqslant \mathbb{E}[\mathbb{E}[\int_0^\tau dt|y_n(0) = y^*, \{W\}]] \\
\mathbb{E}_{y^*}[\mathcal{T}_{n+2}(y^*)] &\leqslant \mathbb{E}_{y^*}[\mathcal{T}_n(y^*)].
\end{aligned}$$

As shown above, $\mathbb{E}_{y^*}[\mathcal{T}_{n+2}(y^*)] \leqslant \mathbb{E}_{y^*}[\mathcal{T}_n(y^*)]$ is equivalent to $\mathbb{E}_{y^*}[k(y^*)] \geqslant -\frac{1}{N}$, and thus:

$$-\frac{\mathrm{Var}(x^*)}{N} \leqslant \mathrm{Var}(x^*)\mathbb{E}_{y^*}[k(y^*)] \leqslant \int_0^{\bar{y}} k(y)ydF^y/n.$$